PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

# Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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# Abstract

The rapid advancement of digital technologies has transformed modern manufacturing by enabling organizations to improve production monitoring, operational efficiency, data management, and data-driven decision-making. Despite these developments, some food manufacturing operations continue to rely on manual and paper-based production recording methods, which may result in delayed reporting, human recording errors, inaccurate computations, inconsistent documentation, data redundancy, and limited visibility of production performance. At AlDi Foods Inc., production personnel manually recorded information concerning raw materials received, raw material usage, extraction outputs, packed yields, production slip making it difficult to monitor production activities, evaluate production performance, retrieve historical records, and generate timely production reports. These challenges established the need for a centralized and accessible digital production monitoring solution capable of organizing production information and providing timely and reliable analytical results.

This study developed a web-based production monitoring system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* The system was developed to digitize the existing production recording process by providing a centralized platform for recording, processing, monitoring, and reporting production data. It included functions for user authentication, production batch management, raw materials received, raw material recording, extraction output recording, packed yield recording, production slip, automated yield percentage computation, extraction efficiency monitoring, production stability classification, dashboard visualization, historical record management, and automated report generation. The system calculated production indicators based on recorded raw material quantities, extraction outputs, and packed yields and classified production stability using predefined organizational thresholds. It also generated weekly, monthly, quarterly, and yearly reports to support the monitoring of raw material utilization and production performance. Through these functions, the system addressed the limitations associated with manual recording and computation while improving the accessibility and organization of production information.

The study employed a Developmental Research Design and used the Agile Development Model to guide the analysis, planning, design, development, testing, and evaluation of the system. System requirements were obtained through interviews with management, production supervisors, and production personnel, direct observation of existing production workflows, and review of relevant company production documents, records, logbooks, and reports. The system was designed using a three-tier architecture consisting of the Client Layer, Application Layer, and Database Layer. HTML, CSS, and JavaScript were used for the user interface and client-side functionality, PHP was used for server-side processing, MySQL was used for centralized database management, and Chart.js was used to present production analytics through graphical visualizations. The system focused on manually encoded production data and was designed as a cost-effective digital solution without requiring advanced Internet of Things (IoT), industrial automation, robotics, or machine-learning infrastructure.

The developed system was tested and evaluated based on functionality, usability, accuracy, efficiency, reliability, and user satisfaction. The evaluation assessed the system's ability to perform its intended functions, accurately process production data, provide accessible production information, maintain reliable records, and support users in monitoring production performance. The results demonstrated that the system provided a centralized approach to production data recording and processing while supporting automated calculations, production analytics, historical record retrieval, and periodic report generation. By replacing fragmented manual documentation with an organized digital platform, the system improved the availability and presentation of production information and supported more timely monitoring and decision-making.

The study concluded that the developed web-based production monitoring system provided a practical, accessible, and cost-effective solution for digitizing production operations at AlDi Foods Inc. The system supported the monitoring of yield performance, extraction efficiency, production stability, and raw material utilization while reducing reliance on manual calculations and paper-based records. It also demonstrated the applicability of a web-based information system in supporting the digital transformation of food manufacturing operations through

centralized data management and automated production analytics. The study recommended continuous system maintenance and future enhancement of analytical and reporting capabilities based on the evolving operational requirements of AlDi Foods Inc.

Index Terms— production monitoring, yield analytics, extraction efficiency, production stability, raw material utilization, web-based system, food manufacturing, digital production management.

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# CHAPTER I

# INTRODUCTION

## 1.1 Background of the Study

In today's increasingly competitive and technology-driven business environment, manufacturing organizations continuously seek ways to improve operational efficiency, minimize material losses, maintain product quality, and support timely decision-making. The food manufacturing industry is particularly dependent on efficient production processes because the quality and quantity of finished products are influenced by the proper utilization of raw materials, equipment performance, production procedures, and workforce activities. As organizations generate increasing amounts of production information, the ability to accurately record, organize, process, and interpret such information has become an important part of effective production management. Digital technologies provide opportunities for organizations to transition from fragmented and manually maintained records toward centralized information systems that can improve the accessibility and reliability of operational data.

Digital transformation in manufacturing involves the application of digital technologies to improve existing business and production processes. One important aspect of this transformation is the digitization of production records and the use of data analytics to support operational monitoring. Digital production systems can organize information from different stages of a production process, automate repetitive calculations, present performance indicators, and provide historical records that can be reviewed when needed. Through centralized data management and graphical visualization, production personnel and managers can obtain a clearer understanding of production performance and identify areas that require attention. These capabilities are particularly useful in production environments where decisions depend on timely information concerning material consumption, output, yield, and process efficiency.

Despite the increasing availability of digital technologies, some small and medium-sized enterprises (SMEs) continue to use manual methods for recording and managing production information. Manual records may include logbooks, paper forms, spreadsheets, and other independently maintained documents. Although these methods can provide basic documentation of production activities, they may become difficult to manage as the volume of production data increases. Manual data entry can also introduce human errors, inconsistent records, duplicate information, and inaccuracies in calculations. In addition, manually recorded information may not be immediately available to management, resulting in delays between the completion of production activities and the analysis of production performance. These limitations can make it difficult for organizations to identify production losses, monitor resource utilization, and respond promptly to operational concerns.

The availability of timely production information is particularly important in food manufacturing because production performance is closely related to the utilization of raw materials and the quantity of usable output obtained from

them. In juice production, for example, raw materials undergo extraction processes in which the amount of extracted product is compared with the quantity of materials used. Yield is therefore an important production indicator because it provides information about how effectively raw materials are converted into usable output. Monitoring yield over different production batches can help production personnel identify variations in performance and determine whether production results remain within established organizational expectations.

Extraction efficiency is another important indicator in the production process because it provides information regarding the effectiveness of extracting usable output from the available raw materials. Variations in extraction performance may be associated with factors such as the condition of equipment, characteristics of raw materials, production procedures, and operator practices. However, when production information is recorded manually and calculations are performed separately, monitoring these indicators consistently can become time-consuming and susceptible to errors. Without a centralized mechanism for processing and comparing production records, identifying changes in yield and extraction performance across different batches can also be difficult.

Production stability is likewise relevant to the monitoring of manufacturing operations. A production process may produce varying results from one batch to another due to changes in raw material quantities, extraction output, packed yield, or other operational conditions. Establishing predefined organizational thresholds provides a basis for classifying whether recorded production performance falls within an expected range. However, manually determining production status for multiple records can increase the workload of production personnel and may delay the identification of batches that require further review. A digital system can assist by automatically processing recorded values and presenting the resulting production status in a more understandable format.

At AlDi Foods Inc., production personnel rely on existing manual methods to record important production information, including raw material usage, extraction outputs, and packed yield. These records serve as a source of information for monitoring production activities; however, the manual process creates challenges in consolidating, calculating, retrieving, and analyzing production data. Information recorded in separate logbooks, spreadsheets, or documents may require additional manual processing before it can be used for production analysis. Consequently, management may experience difficulty obtaining an immediate overview of production performance, comparing historical records, monitoring raw material utilization, and determining extraction efficiency across production batches.

The limitations of the existing process also affect the timely identification of production-related concerns. When yield and extraction efficiency calculations are performed manually or after production activities have been completed, potential variations in production performance may not be recognized immediately. Delayed access to production information can limit the opportunity for production personnel and management to investigate possible causes of low yield, inefficient extraction, excessive material usage, or inconsistent packed output. Furthermore, the absence of a centralized historical database makes it more difficult to retrieve previous production records and use them as references for monitoring production trends.

Another challenge involves the preparation of periodic production reports. Weekly, monthly, quarterly, and yearly production information may need to be consolidated from individual records before management can review raw material utilization and extraction performance. When this process is performed manually, preparing reports can consume additional time and increase the possibility of computation or transcription errors. A centralized digital platform can address this limitation by storing production records in a structured database and generating summarized information based on selected reporting periods. Such functionality can reduce repetitive manual work while making historical production information more accessible.

In response to these challenges, the digitization of production recording provides a practical approach for improving the management of production information at AlDi Foods Inc. Rather than requiring complex industrial automation infrastructure, a web-based production monitoring system can provide a centralized environment where authorized users can encode production data, process production measurements, view analytical results, and retrieve historical

records. The system can transform manually recorded production information into useful indicators through automated calculations and visualizations. This approach is appropriate for the scope of the study because it focuses on improving the existing information-management process using accessible and cost-effective technologies.

Therefore, this study proposes the development of *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* The proposed system is designed to centralize production data and digitize the recording and monitoring activities currently performed through manual methods. It provides modules for user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, yield analysis, extraction efficiency monitoring, production stability classification, dashboard visualization, historical record management, and automated reporting. The system also provides automated computation of yield percentage and extraction efficiency based on recorded production values, while production stability is determined using predefined organizational thresholds.

The proposed system further provides graphical dashboards and historical production records to improve the visibility of production performance. Through the centralized database, authorized users can access organized production information and review previous production batches without relying exclusively on separate physical or manually maintained records. The reporting functions are intended to support the generation of weekly, monthly, quarterly, and yearly summaries of raw material utilization and extraction performance. These features are designed to reduce manual calculations, improve the organization of production records, and provide production personnel and management with more timely information for operational monitoring.

The development of the system follows a Developmental Research Design and the Agile Development Model, allowing the system to undergo systematic planning, analysis, design, development, testing, and evaluation. System requirements are obtained through interviews with relevant production personnel, direct observation of existing production workflows, and examination of available production records and documents. The system utilizes a three-tier architecture consisting of the Client Layer, Application Layer, and Database Layer. HTML, CSS, and JavaScript are used for the user interface and client-side functionality, PHP is used for server-side processing, MySQL is used for centralized data management, and Chart.js is used to present production information through graphical visualizations.

The proposed system is intended to provide a practical and accessible digital solution that addresses the identified limitations of manual production monitoring at AlDi Foods Inc. By centralizing production records, automating key production calculations, presenting yield and extraction-efficiency analytics, classifying production stability, and generating periodic reports, the system can support more organized production monitoring and improve the availability of information for decision-making. Ultimately, the study seeks to demonstrate how a cost-effective web-based production monitoring platform can contribute to the digital transformation of food manufacturing operations while remaining appropriate to the technological resources, operational requirements, and scope of AlDi Foods Inc.

## 1.2 Statement of the Problem

AlDi Foods Inc. currently relies on manual methods for recording and managing production information, including raw material usage, extraction outputs, and packed yields. Although these records provide documentation of production activities, the existing process requires production personnel to manually encode, calculate, consolidate, and review information from different production records. This process may consume considerable time and increase the possibility of recording, computation, and transcription errors. The absence of a centralized digital system also makes it difficult to consistently organize production information and provide timely access to data required for production monitoring and analysis.

The manual recording process also limits the availability of production information for immediate monitoring. Since production data must be recorded and processed before it can be analyzed, delays may occur between the completion of production activities and the availability of useful production indicators. This limitation can make it difficult for production personnel and management to promptly determine whether production performance is within expected

levels. In particular, variations in raw material utilization, extraction output, and packed yield may not be immediately visible when information is maintained through separate manual records.

Another concern is the manual computation and monitoring of production performance indicators. Yield percentage and extraction efficiency are important measurements for determining how effectively raw materials are converted into production output. When these calculations are performed manually, repeated computation may increase the possibility of errors and require additional time. Furthermore, without an automated mechanism for classifying production performance, identifying whether a production batch meets predefined organizational thresholds may require additional manual review.

The existing process also presents limitations in historical data retrieval and production reporting. Previous production records may need to be manually searched, consolidated, or reviewed before they can be used for comparison and analysis. This can make it difficult to examine production performance across different batches and reporting periods. Similarly, the preparation of weekly, monthly, quarterly, and yearly reports on raw material utilization and extraction performance may require repetitive manual consolidation and calculation. These limitations can reduce the efficiency of production reporting and restrict the availability of organized historical information for management review.

The absence of centralized production analytics further limits the ability of the organization to identify production trends and variations. Without a dashboard that presents production indicators in an organized and understandable manner, users may need to examine individual records to determine production performance. This can make it more difficult to identify changes in yield, extraction efficiency, raw material utilization, and production stability across different production batches. A centralized digital monitoring system can address these limitations by organizing production information in a structured database and transforming recorded values into accessible analytical information.

The problem is therefore not only the continued use of manual production records but also the lack of an integrated system that can connect production recording, automated computation, analytics, historical data management, and reporting in a single platform. The proposed system addresses this gap by providing authorized users with a web-based platform for recording production data and monitoring production performance. It is designed to automate the computation of yield percentage and extraction efficiency, classify production stability using predefined organizational thresholds, visualize production information through dashboards, maintain historical production records, and generate periodic reports.

In line with the objectives of the study, the following questions are formulated:

1. How does the existing manual production recording process affect the accuracy, consistency, timeliness, and reliability of production information at AlDi Foods Inc.?

2. What limitations are encountered in the existing process in terms of recording, organizing, retrieving, and accessing production data?

3. How can a centralized web-based system improve the recording and monitoring of raw material inputs, extraction outputs, and packed yields?

4. How can automated computation of yield percentage and extraction efficiency improve the monitoring and analysis of production performance?

5. How can production stability classification based on predefined organizational thresholds assist in identifying production performance that requires attention?

6. How can dashboard visualization and historical production records improve the accessibility and interpretation of production performance information?

7. How can automated weekly, monthly, quarterly, and yearly reports improve the monitoring of raw material utilization and extraction performance?

8. How can the proposed system support timely and informed managerial decision-making through centralized and accessible production information?

9. To what extent does the developed system satisfy the intended users in terms of functionality, usability, accuracy, efficiency, reliability, and user satisfaction?

## 1.3 Objectives of the Study

### 1.3.1 General Objective

The objective of the study is to design, develop, and evaluate a web-based digital production monitoring system for AlDi Foods Inc. that centralizes raw materials received, raw materials management, juice extraction records, and pack yield; production slips; and automates real-time yield analytics, efficiency tracking, stability classification, and reporting to improve overall operational decision-making.

### 1.3.2 Specific Objectives

Specifically, the study aims to:

1. Analyze and document the existing production process of AlDi Foods Inc., particularly the recording and monitoring of raw material usage, extraction outputs, packed yields, production batches, and production reports, to identify existing limitations and system requirements.

2. Identify the problems associated with the existing manual production recording process, particularly those involving data accuracy, consistency, timeliness, accessibility, computation, record retrieval, and production reporting.

3. Design a user-friendly web-based production monitoring system that provides a centralized platform for recording, processing, monitoring, analyzing, and managing production information.

4. Develop a user authentication and access-control function that allows authorized users to securely access the system and perform production-related activities based on their assigned roles and permissions.

5. Develop a production batch management function that organizes production activities and associates raw material, extraction, and packed-yield information with their corresponding production batches.

6. Develop a raw material management and recording function that allows authorized users to record and monitor the quantities of raw materials used during production.

7. Develop an extraction output recording function that allows users to encode and manage the quantity of output obtained during the extraction process.

8. Develop a packed yield recording function that allows users to record the quantity of finished or packed product associated with each production batch.

9. Implement automated yield percentage computation using the appropriate recorded production inputs and outputs to reduce repetitive manual calculations and improve the consistency of production measurements.

10. Implement automated extraction efficiency computation and monitoring to provide users with a standardized indicator for evaluating extraction performance based on the production data recorded in the system.

11. Implement production stability classification using predefined organizational thresholds to help users determine whether recorded production performance falls within the expected operating range.

12. Develop a centralized production dashboard that presents relevant production indicators and performance information through tables, summary indicators, and graphical visualizations.

13. Provide real-time access to encoded production information and analytics so that authorized users can view updated production records and calculated indicators without relying solely on separate manual documents.

14. Develop historical production record management that allows authorized users to retrieve, review, and examine previous production batch information for monitoring, comparison, and reference purposes.

15. Develop automated production reporting functions capable of generating weekly, monthly, quarterly, and yearly summaries related to raw material utilization, extraction performance, and other relevant production indicators.

16. Centralize production information in a structured database to improve the organization, storage, retrieval, and management of production records.

17. Improve data consistency and reduce manual recording and computation errors by standardizing data entry fields and automating selected production calculations and reporting processes.

18. Improve the accessibility and timeliness of production information by providing authorized users with a centralized platform through which production records and analytical results can be accessed and reviewed.

19. Support the identification of production variations and potential yield losses by presenting yield, extraction efficiency, raw material utilization, and production stability information in an organized analytical format.

20. Support data-driven operational decision-making by providing management and production personnel with timely, organized, and understandable production information through dashboards, historical records, and automated reports.

21. Evaluate the developed system in terms of functionality, usability, accuracy, efficiency, reliability, and user satisfaction to determine whether the system meets its intended technical and operational requirements.

Determine the practical applicability and acceptability of the developed system as a digital production monitoring solution for improving production data management and monitoring within AlDi Foods Inc.

## 1.4 Scope and Delimitation

This study focuses on the design, development, testing, and evaluation of a web-based digital production monitoring system for AlDi Foods Inc. The system is intended to address the limitations of the company's existing manual production recording process by providing a centralized platform for recording, organizing, processing, monitoring, analyzing, and reporting production information. The primary focus of the study is the monitoring of production activities related to raw materials received, raw material usage, extraction output, packed yield, production slip, yield percentage, extraction efficiency, production stability, and raw material utilization within the company's juice extraction operations.

The proposed system covers the digital recording of production information that is currently documented through manual production records. Authorized users are provided with appropriate system functions for entering and managing production data, including raw materials received, raw material quantities, extraction outputs, packed yields, production slip and production batch information. The system organizes these records within a centralized database to improve data accessibility, consistency, retrieval, and management. The digitization of these records is intended to reduce dependence on separate paper-based or manually maintained production documents while providing a more organized approach to production information management.

User authentication and access control, which restrict system functions to authorized users, define the system's scope. The system also includes production batch management for organizing production records according to individual batches. The raw materials received will allow users to encode the materials they received and the quantity of each material. The raw material management and recording function allows users to encode information regarding materials used during production. The extraction output function records the output obtained during the extraction process. The packed yield function records the resulting packed production output associated with each batch, and the production slip will allow the users to input information about the product item and the accountable one in the products that they will deliver to each supplier.

The system also covers the automated processing of selected production indicators. Based on the production information entered by authorized users, the system computes yield percentage to provide an indicator of the relationship between production input and output. The system also computes and presents extraction efficiency to assist in monitoring the effectiveness of the extraction process. These automated calculations are intended to reduce repetitive manual computation and provide more consistent production measurements.

Another component within the scope of the study is production stability classification. The system uses predefined organizational thresholds to classify recorded production performance according to the established criteria. This functionality provides users with an additional indicator for identifying production results that may require further review. The system does not independently determine the causes of production variations; rather, it presents calculated indicators and classifications based on the production data and thresholds entered or established for the system.

The study also includes a centralized production dashboard that provides summary information and graphical visualizations of relevant production indicators. The dashboard is designed to present production information in an understandable format, allowing authorized users to monitor raw materials received, raw material utilization, yield performance, extraction efficiency, production slip, production stability, and other available production information. The visualization component is intended to improve the accessibility and interpretation of production data without requiring users to manually consolidate individual production records.

The system further includes historical production record management, allowing authorized users to retrieve and review previously recorded production batches. Historical records provide a reference for examining production performance across different batches and reporting periods. This function supports the organization's ability to review previous production information and compare production indicators over time based on the data available in the system.

Automated reporting is also included within the scope of the study. The system is designed to generate production summaries for weekly, monthly, quarterly, and yearly periods. These reports focus primarily on raw material utilization, extraction performance, yield-related information, and other production indicators supported by the system. The reporting functionality is intended to reduce the time required to manually consolidate production records and provide management with organized information for production monitoring and operational review.

The proposed system is developed as a web-based prototype intended specifically for the operational context of AlDi Foods Inc. The system follows a three-tier architecture consisting of the Client Layer, Application Layer, and Database Layer. The development utilizes HTML, CSS, and JavaScript for the user interface and client-side functionality, PHP for server-side processing, MySQL for centralized database management, and Chart.js for graphical visualization of production analytics. The system is developed and tested within the defined technical environment established by the study.

Production information within the prototype is manually encoded by authorized users. The system provides real-time access to the information after it has been entered and processed, but it does not automatically collect production measurements directly from machines, equipment, sensors, or other physical devices. Therefore, the term real-time monitoring in this study refers primarily to the immediate availability, processing, visualization, and retrieval of newly encoded production information within the web-based system.

The study includes system testing and evaluation to determine whether the developed system meets its intended requirements. The evaluation considers functionality, usability, accuracy, efficiency, reliability, and user satisfaction. Testing is conducted using the intended system functions and relevant production data to determine whether the system can properly record, process, calculate, display, store, retrieve, and report production information. User evaluation is also used to determine the perceived usability and practical acceptability of the system within the identified organizational setting.

The study is delimited to production monitoring and analysis related to the juice extraction operations of AlDi Foods Inc. It does not cover the complete management of all company operations. Business functions such as inventory management beyond production-related raw material monitoring, sales and marketing, accounting, payroll, human resource management, purchasing, supply chain management, customer relationship management, and other enterprise-level business processes are excluded from the system.

The study also does not include full-scale industrial automation, robotics, advanced machine learning, predictive maintenance, automated machine control, or sophisticated Industrial Internet of Things (IIoT) infrastructure. The proposed system does not directly control production machinery or automatically adjust production processes. It is primarily an information-management and production-monitoring solution that processes data entered by authorized users. Any possible future integration with sensors, automated equipment, or other advanced technologies is outside the current implementation and may be considered only as a potential future enhancement.

The study is also limited to the production indicators and calculations defined within the system. Although the system can present yield, extraction efficiency, production stability, raw material utilization, and related production information, it does not automatically determine all possible causes of production losses or inefficiencies. Factors such as raw material quality, equipment condition, operator practices, environmental conditions, and other operational variables may influence production performance, but the system does not independently diagnose these factors. Users may use the available production records and analytics as references for further operational investigation.

The system is developed and evaluated within a controlled environment and within the operational requirements identified during the study. Consequently, the results of system testing and user evaluation represent the conditions, data, processes, and users involved in the study and may not represent every possible production condition that could

occur during long-term or large-scale implementation. Changes in production procedures, organizational policies, system requirements, or operational conditions may require future modifications to the system.

The findings of the study are therefore primarily applicable to the production monitoring requirements of AlDi Foods Inc. and should not be assumed to be directly generalizable to all food manufacturing companies or manufacturing environments. Other organizations may have different production processes, performance indicators, reporting requirements, organizational structures, and technological resources that would require corresponding system modifications.

Finally, the study does not include a comprehensive financial feasibility analysis, full commercial deployment, long-term maintenance assessment, enterprise-wide implementation, or measurement of long-term organizational benefits beyond the defined system evaluation. The study focuses primarily on determining whether the developed web-based system can provide a practical, accessible, and organized approach to production data recording, yield analytics, extraction efficiency monitoring, production stability classification, historical record management, and automated reporting within the defined scope of AlDi Foods Inc.

## 1.5 Significance of the Study

This study is significant because it proposes a web-based digital production monitoring system that addresses the limitations of manual production recording and provides a centralized approach to managing, processing, analyzing, and reporting production information. The system is designed to support the monitoring of raw materials received, raw material usage, extraction output, packed yield, production slip, yield percentage, extraction efficiency, production stability, historical production records, and periodic production reports.

The study may provide practical benefits to the following:

**AlDi Foods Inc.-** The proposed system may provide AlDi Foods Inc. with a centralized and organized platform for managing production information. By digitizing the recording of raw material quantities, extraction outputs, and packed yields, the system can reduce the organization's reliance on separate manual records and repetitive manual calculations. The automated computation of yield percentage and extraction efficiency can provide more consistent production measurements, while the production stability classification can assist users in identifying production results that require further review.

The centralized dashboard and historical records may also improve the visibility and accessibility of production information. Management and authorized personnel can review production performance using organized data rather than relying solely on manually consolidated records. Furthermore, automated weekly, monthly, quarterly, and yearly reports may reduce the time required to prepare production summaries and support more systematic monitoring of raw material utilization and extraction performance.

Overall, the system may contribute to improved production information management, reduced manual workload, better data accessibility, and more organized production monitoring. The study does not directly measure long-term profitability, but the improved availability of production information may provide management with useful information for identifying operational concerns and supporting informed decisions.

**Production Managers**- Production Managers may benefit from having timely access to centralized production information. Instead of manually reviewing separate production records, they can use the dashboard to view relevant indicators such as raw material utilization, yield percentage, extraction efficiency, and production stability. This can make it easier to monitor production performance and review variations between production batches.

The historical production records may also assist managers in comparing current production information with previous batches and reporting periods. Automated reports can further support periodic production reviews by providing organized summaries of production information. Through these capabilities, the system may help

managers and supervisors obtain relevant information more efficiently and support more timely operational decision-making.

**Production Employees and Personnel-** Production employees may benefit from a more structured and organized method of recording production information. The system provides designated functions for entering raw material quantities, extraction outputs, packed yields, and production batch information, reducing the need to repeatedly perform manual calculations and consolidate information from separate records.

Automated calculations can also reduce the possibility of arithmetic errors associated with manually computing yield percentage and extraction efficiency. By providing a standardized digital process for recording production data, the system may improve consistency in data entry and reduce repetitive administrative tasks. This allows production personnel to focus more on their assigned production activities while maintaining properly organized production records.

**Management and Organizational Decision-Makers–**Management and organizational decision-makers may benefit from the availability of consolidated production information and analytical results. The system transforms recorded production data into measurable indicators and graphical visualizations that can be reviewed through the dashboard. These information resources may assist management in understanding production performance, identifying variations in production results, reviewing raw material utilization, and examining extraction efficiency.

The automated reporting function may also support management in reviewing production performance across different periods. Weekly, monthly, quarterly, and yearly summaries can provide organized information for operational discussions, production planning, and performance monitoring. The system therefore serves as an information-support tool rather than replacing managerial judgment or decision-making.

**BSIT Students-** The study may benefit Bachelor of Science in Information Technology (BSIT) students by providing a practical example of how information technology concepts can be applied to an actual organizational problem. The development of the proposed system demonstrates the application of web development, database management, software engineering, system analysis and design, user interface development, data processing, and data visualization.

The project may also help students understand how requirements can be gathered through interviews, observation, and document review and then translated into system functions. The use of technologies such as HTML, CSS, JavaScript, PHP, MySQL, Chart.js, and XAMPP provides an example of how different development technologies can be integrated to create a functional information system. The study may therefore serve as an academic and practical reference for students developing similar information-management applications.

**Future Researchers-** The study may serve as a reference for future researchers who intend to develop or investigate digital production monitoring systems, manufacturing information systems, production analytics, yield monitoring, extraction efficiency, or related areas of information technology. The system architecture, development approach, production indicators, evaluation criteria, and identified limitations may provide a foundation for further research and system enhancement.

Future researchers may also build upon the study by investigating additional analytical methods, expanded production indicators, predictive analytics, broader production management functions, or integration with other technologies. Potential future developments may include automated data acquisition, sensor-based monitoring, advanced analytics, or integration with other organizational systems. Such developments, however, are outside the scope of the present study and may be considered in subsequent research.

**Academic Institution-** The study may also contribute to the academic institution by demonstrating the practical application of information technology in addressing an actual production and business-process

problem. It provides an example of how academic knowledge in system development, database management, data analytics, and software engineering can be applied to support organizational digital transformation.

The completed study may also serve as supplementary academic material for future students and researchers who are interested in developing web-based information systems for manufacturing, food production, and other operational environments.

**Production and Food Manufacturing Sector** -Although the system is specifically designed for AlDi Foods Inc., the study may provide a useful example of how small and medium-sized food manufacturing operations can begin digitizing production information using a centralized and relatively accessible web-based platform. The approach demonstrates that production monitoring does not necessarily require complex industrial automation infrastructure when the immediate objective is to organize production records, automate calculations, provide analytics, and improve access to operational information.

## 1.6 Definition of Terms

**Analytics** – The process of analyzing data to extract meaningful insights and support decision-making.

**Centralized Database** – A digital storage system where production data is organized and managed in a single location for easier access and monitoring.

**Dashboard** – A graphical user interface that displays real-time production information, analytics, and system reports.

**Digitizing Production** – The conversion of manual production records and monitoring processes into digital format using information technology.

**Efficiency** – A measure of how effectively resources are utilized to produce desired outputs.

**Extraction –** The process of obtaining juice or liquid product from raw materials during production.

**Manufacturing Industry-** The study may contribute to the advancement of digital transformation practices in manufacturing by demonstrating the benefits of real-time monitoring systems in improving efficiency and reducing production waste.

**Monitoring–** The continuous observation, tracking, and evaluation of production processes and operational performance.

**Real-Time –** Immediate processing, updating, and availability of data without significant delay.

**Yield –** The ratio of output produced compared to input materials, usually expressed as a percentage.

**Automated Yield Calculation** – The automated mathematical process of computing the ratio of usable output to raw material input without manual intervention.

**Extraction Efficiency** – A quantitative measurement comparing the actual amount of liquid extracted to the theoretical maximum yield possible from a given batch of raw materials.

**Manual Recording Method** – The traditional, paper-based practice of logging operational data, material usage, and production metrics by hand.

**Operational Visibility** – The degree to which real-time production activities, resource flows, and machine performances are accessible and transparent to managers and operators.

**Production Monitoring System** – A specialized digital application designed to continuously record, track, and evaluate production workflows, inputs, and outputs.

**Prototype System** – A preliminary, functional version of software developed to demonstrate core capabilities, validate system design, and test features in a controlled environment.

**Raw Material Input** – The baseline quantity of unprocessed agricultural goods or ingredients introduced into the extraction machinery at the start of a production run.

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# CHAPTER II

# REVIEW OF RELATED LITERATURE AND STUDIES

## 2.1 Foreign Studies

A study [1] by Verhoef et al. (2021) analyzed the impact of transitioning from legacy paper-based logs to centralized digital dashboards in manufacturing. The researchers highlighted that manual data silos create "blind spots" or information gaps, where production anomalies—such as waste hotspots or mechanical inefficiencies—are often only discovered long after a batch is completed. By implementing a digitized analytics system, the facility achieved a more responsive production environment, allowing managers to make data-driven decisions based on live efficiency metrics rather than historical, error-prone logs. This study [1] serves as a foundational reference for your claim that digitizing the production floor is a prerequisite for Industry 4.0 and is essential for reducing financial losses caused by avoidable process inefficiencies.

A study [2] by Makhmudah et al. (2025) explored the implementation of a digital production reporting system in a manufacturing environment. Using a qualitative case study approach, the researchers replaced manual reporting processes with a system integrated with sensors, automated data transmission, and dashboard visualization. The results showed improvements in data accuracy, operational efficiency, and decision-making. This study [2] is relevant to the present research as it demonstrates the effectiveness of digital systems in improving production monitoring and data management.

A study [3] by Mansour and Al-Hamdani (2024) examined the application of internal Key Performance Indicators (KPIs) in evaluating production efficiency within the food industry, with a specific focus on non-conforming products identified during manufacturing operations. The researchers proposed a structured performance measurement system centered on the internal parts per million (INTppm) indicator, which quantifies defective units detected within production orders relative to the total number of items produced. This approach allows organizations to assess production quality at the operational level rather than relying solely on end-product inspection, thereby enabling immediate, data-driven decision-making. By continuously tracking these metrics, the authors emphasized that food production performance depends heavily on the continuous monitoring of quality-related deviations to maintain process stability. In their case analysis involving food-related processing lines, researchers [3] monitored various parameters across manual and automated operations, including quality control checks, labeling procedures, measurement systems, and line adjustments. The findings revealed that internal defect rates varied significantly depending on the production stage, with technical problems—such as mechanical errors, clamping inconsistencies, tool wear, and alignment issues—serving as primary drivers of elevated INTppm values.

Additionally, inadequate maintenance and delayed equipment replacements further exacerbated production instability. Consequently, the internal defect rate proved to be a critical indicator of both product quality and machine reliability. To mitigate these operational inefficiencies, the researchers [3] highlighted the necessity of corrective and preventive measures, such as enhancing automated process controls, executing random quality checks during shifts, and updating employee training programs. The researchers also recommended integrating skill evaluation

checkpoints and fostering workforce communication to minimize human errors alongside technical updates. Cultivating this long-term data collection approach allows manufacturing organizations to transition from reactive problem-solving to proactive process optimization, shifting focus toward controllable internal variables like equipment conditions and operator behavior. Ultimately, it was concluded **[3]** that utilizing internal KPIs like INTppm offers a reliable methodology for optimizing food industry operations by linking real-time defect rates directly to strategic business outcomes like waste reduction and cost efficiency. This literature is highly relevant to the present study as it provides a robust structural foundation for utilizing internal performance indicators to target production inefficiencies. Furthermore, it directly supports the proposed framework of developing a data-driven, real-time tracking system for AlDi Foods Inc. to monitor extraction efficiency and establish long-term process optimization.

A comprehensive review **[4]** conducted by **Aftab Siddique, Ashish Gupta, Jason T. Sawyer, Tung-Shi Huang**, and Amit Morey (2025), entitled " Big Data Analytics in the Food Industry: A State-of-the-Art Literature Review", examined the increasing influence of artificial intelligence (AI), machine learning (ML), and big data analytics on the modernization of food industry operations. Published in "npj Science of Food", the study explored numerous scholarly works related to digital transformation and data-driven technologies across food production, processing, quality assurance, supply chain management, and food safety. The researchers emphasized that the food industry is undergoing a significant technological shift as organizations increasingly adopt digital systems capable of collecting, storing, processing, and analyzing large volumes of operational data. As production processes grow more complex and consumer demands evolve, the deployment of advanced analytics has become a pivotal strategy for improving efficiency, productivity, and competitiveness. According to Siddique et al. (2025), modern food manufacturing facilities generate substantial amounts of data from various stages of production, including raw material handling, processing, packaging, storage, distribution, and quality control **[4]**. Traditionally, much of this data remained underutilized due to legacy constraints in collection and analytical methods. However, advancements in digital technologies enable organizations to transform raw production data into actionable insights to support real-time decision-making and operational improvements.

The study explained that big data analytics provides organizations with the capacity to identify patterns, detect hidden inefficiencies, forecast outcomes, and monitor performance far more accurately than conventional, manual approaches. The review highlighted that AI and ML technologies have become essential tools for enhancing food production systems by processing large datasets to generate insights that are difficult to obtain through manual analysis. AI-based models learn from historical and real-time data, making it possible to predict trends, isolate operational bottlenecks, and optimize processing parameters. Through continuous data analysis, organizations can achieve tighter process control, minimize material waste, and enhance overall manufacturing efficiency. A major focus of the study involved the application of predictive analytics and artificial neural networks (ANNs) in food processing operations, particularly in activities such as drying, baking, fermentation, extrusion, freezing, and quality assessment. By utilizing historical operational data, these predictive systems can anticipate final product characteristics, estimate exact processing times, model complex multi-variable relationships, and dynamically identify optimal operating conditions. Furthermore, researchers **[4]** underscored the importance of real-time monitoring systems over traditional monitoring methods, which heavily rely on manual recording procedures that are time-consuming and prone to human error. In contrast, digital monitoring systems continuously collect and analyze operational data, allowing managers and operators to observe production performance dynamically. This real-time monitoring provides immediate visibility into production loops and enables organizations to respond swiftly to operational anomalies. The authors further discussed the integration of computer vision systems and image-processing technologies to automatically inspect products during production.

This automated visual intelligence reduces the need for manual inspection while increasing consistency and accuracy in food grading, defect detection, and quality evaluation. In addition to production monitoring and quality control, the review explored the role of big data analytics in food safety management, supply chain optimization, and resource sustainability. Advanced analytical systems can identify contamination risks, track disease outbreaks, and improve overall traceability to mitigate economic losses associated with product recalls. From an operational standpoint,

AI-powered forecasting models analyze historical sales data and consumer behavior to predict future demand, enabling manufacturers to optimize inventory levels and schedule production efficiently. It was also noted [4] that data-driven systems support corporate sustainability goals by optimizing the use of raw materials, energy, and water, thereby reducing environmental impacts.

Despite challenges such as data quality, system integration, and workforce readiness, the authors concluded that the future of the food industry depends on intelligent systems capable of transforming raw metrics into actionable knowledge [4]. This extensive review is highly relevant to the present capstone project, "Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.", because it provides strong theoretical and empirical validation for the implementation of digital monitoring and analytics systems within food processing environments. The findings directly support the technical architecture of a platform capable of monitoring production yield and extraction efficiency in real time. By establishing a system for continuous data collection, management can move away from latency-heavy, manual logs to instantly identify process variations, evaluate extraction performance, and detect production losses. Consequently, this literature [4] serves as a foundational benchmark for the proposed system, demonstrating that real-time data integration is critical to modernizing waste management and achieving operational excellence.

A study [5] by **Rivadeneira et al. (2023)**, published in *Food Research*, examined the production of pectin from "saba" banana peel waste using ultrasound-assisted extraction and evaluated its performance as an emulsifying agent compared with commercial low-methoxyl pectin (LMP). The study emphasized the potential of agricultural waste valorization by converting banana peels into functional food ingredients suitable for emulsion-based systems. Findings revealed that the extracted pectin had lower equivalent weight and a reduced degree of esterification than commercial LMP, indicating differences in molecular structure that influence its functional behavior in food applications. Although UEP showed lower purity, it contained relatively higher amounts of protein and ash, which contributed positively to its interfacial and emulsifying properties.

Structural and physical analyses further confirmed these differences, where FTIR results indicated variations in esterified carboxyl groups between UEP and LMP. Microscopic observation showed that emulsions stabilized by UEP formed smaller and more evenly distributed oil droplets compared to those stabilized by commercial pectin. This droplet size reduction contributed to improved emulsion stability, as it enhanced interfacial coverage and minimized phase separation. The study also reported that changes in formulation factors such as pH level, oil concentration, and ionic strength significantly affected emulsion behavior, with UEP consistently showing stronger adaptability under varying conditions.

In terms of functional performance, emulsions containing UEP exhibited higher viscosity and better resistance to breakdown compared to those stabilized by LMP [5]. The study also found that UEP influenced lipid digestion, as it reduced free fatty acid release during in vitro digestion, suggesting a potential role in controlling lipid bioaccessibility. Overall, the researchers concluded **[5]** that ultrasound-extracted pectin from banana peel waste is a promising sustainable alternative to commercial pectin, offering improved emulsifying stability and functional benefits for food formulation and processing systems.

A study [6] by **Sabih, Farid, Ejaz, Husam, Khan, and Farooq (2023)** investigated the use of computer vision and machine learning technologies to automate the measurement of raw material flow rates on a conveyor belt system used in soda ash manufacturing. The researchers recognized that many industrial processes still depend on manual monitoring and measurements, which can lead to errors, inefficiencies, and delays in decision-making. To address this challenge, an automated framework was proposed [6] capable of estimating the flow rate of raw materials transported on a conveyor belt without requiring continuous human intervention.

The study was conducted in a soda ash manufacturing plant where limestone and coke are transported through a conveyor belt system. These materials are critical inputs in the production process because maintaining the correct ratio between limestone and coke is essential for generating carbon dioxide gas with the required purity level.

Traditionally, operators manually conducted experiments and observations to determine the flow rate of materials and adjust conveyor settings accordingly. However, this approach was time-consuming and susceptible to inaccuracies. The researchers sought to improve this process through the implementation of an intelligent monitoring system.

To collect data, a high-resolution camera was installed above the conveyor belt to continuously capture video footage of the moving materials. The data collection process spanned twenty-four weeks and included videos recorded under various environmental conditions such as changes in lighting, dust levels, and weather conditions. Since the conveyor belt was located outdoors, these environmental factors introduced additional challenges that had to be addressed by the proposed system. The researchers designed their framework to be robust enough to operate effectively despite these changing conditions.

The methodology involved several stages of image processing and machine learning **[6]**. First, a region of interest was extracted from each video frame so that only the conveyor belt area containing the raw materials would be analyzed. This step reduced computational requirements and eliminated unnecessary background information. The images were then preprocessed using sharpening and filtering techniques to improve image quality and reduce the effects of dust, noise, and poor lighting. Bilateral filtering was specifically selected because it reduced image noise while preserving important edge details needed for accurate material detection.

After preprocessing, the researchers focused on segmenting the raw materials from the conveyor belt background. Several segmentation approaches were evaluated, including deep learning-based methods, watershed segmentation, frame differencing, and background subtraction. However, because of limited labeled data and highly variable environmental conditions, some methods produced inconsistent results. To overcome these limitations, a voting-based segmentation approach was developed combining frame differencing and the BackgroundSubtractorCNT algorithm. By retaining only the pixels identified by both techniques, the system was able to isolate the most reliable regions representing the raw materials. This method improved segmentation accuracy and reduced the influence of environmental noise and lighting variations.

Once the materials were successfully segmented, the system extracted various image features to support flow-rate estimation **[6]**. The researchers used moment-based features to describe the shape, size, position, and distribution of the segmented materials. Additional contour-based features were extracted to measure the area occupied by limestone and coke on the conveyor belt. K-means clustering was also applied to separate different image regions based on color information. These features were aggregated over specific time windows to accommodate the variable lengths of the video recordings and ensure consistent input dimensions for machine learning models.

To improve model performance and reduce computational complexity, feature selection techniques were employed, particularly Recursive Feature Elimination with Cross-Validation (RFECV). This method identified the most significant features while eliminating less relevant variables. The selected features were then used to train and evaluate several machine learning algorithms, including Decision Tree, Random Forest, XGBoost, Gradient Boosting, Bagging Regressor, Bayesian Ridge, Gamma Regressor, RANSAC, and Theil-Sen Regressor. Both linear and non-linear models were assessed to determine the most effective approach for predicting material flow rates.

The performance of the models was evaluated using several regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). They also utilized leave-one-out validation and k-fold cross-validation to ensure the reliability of the results. The findings revealed that model performance was significantly influenced by the choice of window size, aggregation method, and feature selection strategy. While feature selection improved some non-linear models, certain linear models achieved better results when all extracted features were retained.

Among the evaluated algorithms, the Bagging Regressor demonstrated the best overall performance in terms of prediction accuracy **[6]**. The final model was able to estimate material flow rates with a percentage error as low as seven percent, indicating a high level of reliability. The study showed that combining computer vision techniques

with machine learning algorithms can effectively automate flow-rate monitoring while minimizing the need for manual measurements. Furthermore, the proposed system successfully operated in challenging outdoor environments where variations in lighting, dust, and weather conditions could otherwise compromise accuracy **[6]**.

It was concluded **[6]** that the integration of computer vision and machine learning provides a practical and cost-effective solution for industrial process monitoring. By using only a standard RGB camera and advanced data analytics techniques, the system was able to deliver accurate real-time estimates of material flow rates. This capability supports more informed operational decisions, enhances process efficiency, and reduces the likelihood of human error in industrial settings [6].

A study **[7]** by Sharma et al. (2021) examined the growing role of artificial intelligence (AI) and big data analytics (BDA) in transforming the food industry toward more sustainable, efficient, and intelligent operations. The researchers conducted a systematic literature review, gathering and analyzing previous studies from reputable academic databases, including Scopus, ScienceDirect, Wiley Online Library, Taylor & Francis, and Web of Science. Their objective was to investigate how AI and BDA technologies are being utilized across different stages of the food industry, from agricultural production and food processing to logistics, quality assurance, supply chain management, and distribution. The review emphasized that the increasing demand for food, combined with the need for higher product quality, food safety, and sustainable resource management, has encouraged food manufacturers to adopt digital technologies that can optimize production processes and improve operational performance.

The authors explained that artificial intelligence serves as a powerful computational technology capable of analyzing complex production data, recognizing patterns, making predictions, and supporting intelligent decision-making. AI encompasses various technologies such as machine learning, artificial neural networks (ANNs), computer vision, robotics, fuzzy logic, genetic algorithms, and expert systems. These technologies enable food manufacturers to automate repetitive tasks, monitor production processes, detect defects, forecast production outcomes, and improve quality control. Meanwhile, big data analytics complements AI by collecting, processing, and analyzing large volumes of structured and unstructured data generated throughout production operations. Through advanced data analysis, manufacturers can identify operational trends, predict equipment failures, optimize production schedules, reduce waste, and improve resource utilization. Together, AI and BDA create an intelligent production environment that supports real-time monitoring and continuous process improvement.

The review highlighted numerous applications of AI within food production systems. Machine learning algorithms are widely used for demand forecasting, production planning, yield prediction, and quality classification. Artificial neural networks assist manufacturers in modeling complex production processes and predicting product quality based on multiple process variables. Computer vision systems perform automated inspection by identifying defects, grading food products according to size, shape, color, and texture, and ensuring compliance with quality standards. Robotics contributes to automating labor-intensive operations such as food handling, sorting, packaging, and processing while maintaining product consistency and reducing human error. The authors also discussed intelligent systems such as electronic noses, hyperspectral imaging, infrared sensing, and machine vision technologies that improve food safety, contamination detection, and product evaluation with greater accuracy than traditional manual inspection methods. These technologies reduce processing time while maintaining high levels of precision and reliability.

It was further explained **[7]** that big data analytics strengthens decision-making by converting massive amounts of operational data into meaningful insights. The researchers described three primary categories of analytics used in food manufacturing. **Descriptive analytics** helps identify production patterns, operational trends, and performance indicators by analyzing historical production data. **Predictive analytics** estimates future outcomes such as production demand, equipment performance, product quality, weather impacts, customer behavior, and inventory requirements using statistical models and machine learning algorithms. **Prescriptive analytics** goes one step further by recommending the most effective actions based on optimization models and simulation techniques, allowing managers to make informed decisions regarding production scheduling, inventory management, logistics, and

resource allocation. These capabilities enable manufacturers to respond more effectively to production challenges while minimizing operational risks and improving efficiency.

Another important contribution of the study is its discussion of real-time monitoring technologies integrated with AI and BDA. Modern food manufacturing facilities generate enormous amounts of production data through sensors, Internet of Things (IoT) devices, radio-frequency identification (RFID), automated equipment, and intelligent monitoring systems. These technologies continuously collect information regarding machine performance, production output, environmental conditions, inventory levels, and product quality. AI algorithms analyze these data streams in real time, enabling immediate detection of production abnormalities, equipment malfunctions, or quality deviations. Instead of relying on delayed manual reporting, manufacturers receive instant information that supports faster corrective actions and more effective production management. Real-time visibility also improves traceability throughout the production process, helping organizations maintain compliance with food safety regulations while reducing waste and operational costs.

The researchers also emphasized the growing role of AI and big data analytics across various sectors of the food industry. In dairy processing, AI models predict milk yield, monitor animal health, and estimate product shelf life. In meat processing, machine learning, computer vision, and electronic nose technologies assess meat freshness, identify spoilage, and automate quality grading. For fruits and vegetables, intelligent refrigeration systems monitor product aging and shelf life, while machine vision systems automate sorting and grading based on physical characteristics. AI applications have also been successfully implemented in bakery operations through intelligent oven control systems that optimize baking conditions and improve product consistency. Similarly, beverage manufacturers utilize AI technologies to evaluate flavor, aroma, appearance, and consumer preferences, while robotics assists in packaging and production automation. These examples demonstrate the versatility of AI technologies in addressing operational challenges across different food manufacturing environments.

Despite these advantages, several challenges were acknowledged associated with implementing AI and BDA in the food industry **[7]**. The adoption of intelligent technologies requires substantial financial investment in digital infrastructure, sensors, automated equipment, and information systems. Organizations also need skilled personnel capable of operating AI-based technologies, interpreting analytical results, and maintaining digital production systems. Additionally, concerns regarding cybersecurity, data privacy, workforce adaptation, and organizational readiness remain significant barriers to successful implementation. However, the researchers argued that the long-term benefits—including improved operational efficiency, enhanced product quality, reduced costs, faster decision-making, better traceability, and increased production sustainability—far outweigh these implementation challenges. As digital technologies continue to evolve, their adoption is expected to become increasingly essential for maintaining competitiveness within the global food industry.

## 2.2 Local Studies

A study **[8]** by Bagaforo and Albason (2024), presented at the 19th European Conference on Innovation and Entrepreneurship (ECIE), investigated the digital transformation journeys of Micro, Small, and Medium Enterprises (MSMEs) in Iloilo Province, identifying the specific enablers and barriers local businesses face when adopting new technologies. The research utilized a descriptive-correlational approach to evaluate how local infrastructure and workforce "readiness" influence the transition from traditional to digitized operations. Their findings highlighted that while digital tools significantly enhance local business competitiveness and operational efficiency, success is often hindered by technical literacy gaps and varying levels of infrastructural support **[8]**. This study provides critical contextual evidence for AlDi Foods Inc. regarding the feasibility of implementing automated systems within the local economic landscape, directly supporting the Significance of the Study by illustrating how digital integration optimizes resource management and regional market positioning.

A study **[9]** by De Leon et al. (2017), archived in the Central Philippine University (CPU) Institutional Repository, focused on the design and development of an online ordering system for Iloilo Supermart using the V-Model Software Development Life Cycle (SDLC). The researchers implemented a secure, centralized database and a

user-friendly management dashboard to streamline operations for the Iloilo-based food retailer, effectively mitigating risks associated with physical data damage and manual record-keeping **[9]**.

A study conducted **[10** by Palero et al. (2026), titled *"Green Extraction and Sustainability Assessment of Total Carotenoids from Carrot (Daucus carota) Peel Waste using Soybean Oil,"* explored a sustainable and environmentally responsible approach for recovering valuable bioactive compounds from agricultural waste materials. Published in the Davao Research Journal, the study primarily addressed the growing concern over the environmental and health hazards associated with conventional solvent-based extraction methods, particularly those involving organic solvents such as hexane. These traditional solvents, while efficient in extracting carotenoids, are widely criticized due to their toxicity, flammability, and environmental persistence. In response to these limitations, the researchers investigated soybean oil as a safer, biodegradable, and food-grade alternative solvent capable of efficiently extracting carotenoids from carrot peel waste while aligning with green chemistry and circular economy principles.

Carotenoids are naturally occurring pigments responsible for the orange, yellow, and red coloration in many fruits and vegetables, and they are widely recognized for their antioxidant, anti-inflammatory, and health-promoting properties. Because of their high commercial value and wide applicability in food, pharmaceutical, and cosmetic industries, there is increasing global interest in improving their extraction from natural sources, especially from agricultural by-products that are often discarded. Carrot peels, which are generated in large quantities during food processing, represent a significant yet underutilized source of carotenoids **[10]**. The study emphasized that valorizing such waste materials not only contributes to resource efficiency but also reduces environmental pollution and supports sustainable production systems.

To achieve its objectives, the researchers employed a two-factor, three-level full factorial experimental design to systematically evaluate and optimize the extraction process. The independent variables considered were the liquid-to-solid (L/S) ratio and extraction temperature, both of which are known to significantly influence mass transfer and solute diffusion during extraction processes. The dependent variable was the total carotenoid content (TCC), expressed in micrograms of β-carotene per gram of dried carrot peel. The experimental procedure involved freeze-drying carrot peels to preserve bioactive compounds, followed by pulverization to increase surface area and enhance solvent penetration. The prepared samples were then mixed with soybean oil under controlled conditions and subjected to agitation using a shaking water bath. This setup ensured consistent contact between the solvent and plant material, facilitating efficient extraction of carotenoids into the oil phase.

Quantitative analysis of carotenoid content was performed using UV-Visible spectrophotometry at 450 nm, applying the Beer-Lambert law for concentration determination. The study also implemented rigorous statistical tools to evaluate the reliability and validity of the experimental model. These included analysis of variance (ANOVA), regression modeling, and diagnostic tests such as the Shapiro-Wilk test for normality, Durbin-Watson test for autocorrelation, Cook's distance for identifying influential outliers, and Variance Inflation Factor (VIF) for detecting multicollinearity. Additionally, model performance was assessed using $R^2$, adjusted $R^2$, and predicted $R^2$ values to ensure both accuracy and predictive capability.

The experimental results demonstrated that both the liquid-to-solid ratio and temperature significantly influenced carotenoid extraction efficiency, although their effects were not equal in magnitude. The highest observed total carotenoid content was 84.22 ± 5.56 µg β-carotene/g of dried peel, achieved at a liquid-to-solid ratio of 40 mL/g and a temperature of 45 °C **[10]**. This result indicated that increasing the solvent proportion relative to the biomass significantly enhances extraction efficiency by improving concentration gradients and promoting better diffusion of carotenoids from the plant matrix into the oil solvent. In contrast, temperature exhibited a positive but comparatively weaker effect on extraction performance, suggesting that thermal enhancement plays a supporting role rather than a dominant one in the process.

Statistical analysis confirmed that the overall model was highly significant ($p < 0.0001$), with a strong coefficient of

determination ($R^2$ = 0.95), indicating that the model effectively explained most of the variability in carotenoid recovery **[10]**. The adjusted $R^2$ (0.94) and predicted $R^2$ (0.93) values further supported the robustness and predictive reliability of the model. However, a statistically significant lack of fit was also observed, suggesting that certain complex interactions or nonlinear behaviors within the system were not fully captured by the model. This indicates that while the model is strong in predictive capability, there remains room for refinement to improve its descriptive accuracy.

Through response surface methodology optimization, the study identified the most favorable extraction conditions as a liquid-to-solid ratio of 40 mL/g and a temperature of 30 °C. Under these conditions, the predicted carotenoid yield was 77.49 ± 13.13 µg β-carotene/g of dried peel. To validate the model, experimental runs were conducted under the same optimized conditions, yielding an actual average value of 76.04 ± 3.71 µg β-carotene/g. The small deviation between predicted and experimental values, reflected in a low relative error, confirmed the reliability and practical applicability of the developed model. This close agreement demonstrates that the optimization approach used in the study is effective for predicting carotenoid extraction performance under varying process conditions.

Further analysis revealed that the liquid-to-solid ratio was the most influential factor affecting carotenoid recovery. This can be attributed to its direct effect on mass transfer dynamics. A higher solvent volume increases the concentration gradient between the solid matrix and the liquid phase, thereby enhancing diffusion rates and preventing early saturation of the solvent. This allows more carotenoids to be released from the plant material into the soybean oil. While temperature also contributes by improving solubility and weakening cellular structures, its effect becomes limited once the system approaches equilibrium or when saturation conditions are reached. In oil-based extraction systems, this suggests that solvent quantity plays a more critical role than thermal energy in determining overall extraction efficiency.

The study also compared its findings with previous research involving other biomass sources such as pumpkin, peach, and carrot byproducts. These comparisons showed consistent trends, where higher solvent-to-solid ratios generally resulted in improved carotenoid recovery. However, excessive temperature increases were often associated with reduced efficiency due to possible degradation of heat-sensitive carotenoid compounds. This reinforces the idea that moderate temperature conditions combined with optimized solvent ratios are ideal for maximizing yield in green extraction systems.

In addition to extraction efficiency, the study conducted a sustainability evaluation using the Path2Green assessment framework, which measures the environmental performance of extraction processes across 12 standardized principles. These principles include biomass sourcing, transport, pretreatment, solvent selection, scaling, purification, energy use, and waste management. The overall Path2Green score obtained was 0.593 out of a possible maximum of 1.00, indicating a moderately sustainable and environmentally friendly process **[10]**.

The sustainability assessment revealed that the process performed strongly in several key areas. High scores were recorded for biomass utilization, as the study effectively repurposed agricultural waste rather than relying on freshly cultivated crops. Similarly, solvent selection scored highly due to the use of soybean oil, which is biodegradable, non-toxic, non-volatile, and safe for food-related applications. The process also demonstrated strong performance in scalability, purification, post-treatment, application potential, and waste management, as the extracted product could be directly utilized without requiring extensive purification or solvent removal.

However, certain limitations were identified in the sustainability analysis. Energy consumption received a low score due to the use of freeze-drying, grinding, and controlled temperature water bath systems, all of which require significant energy input. Pretreatment processes also contributed to reduced sustainability performance because mechanical processing steps were necessary to prepare the biomass for extraction. Additionally, yield efficiency was identified as an area for improvement, indicating that the extraction process was not fully exhaustive and that some carotenoids likely remained unextracted in the residual biomass. The study also noted the absence of a closed-loop extraction system, which limited its circular economy potential.

Despite these limitations, the study highlighted that soybean oil-based extraction remains a promising green alternative to conventional solvent systems. The integration of environmentally friendly solvents with waste valorization strategies demonstrates a strong alignment with sustainable development goals and green chemistry principles. Furthermore, the extracted carotenoid-rich oil can potentially be applied directly in food formulations, cosmetics, pharmaceuticals, and nutraceutical products without requiring further purification, thereby reducing processing costs and environmental burden.

In conclusion, Palero et al. (2026) **[10]** successfully demonstrated that soybean oil is an effective and sustainable solvent for carotenoid extraction from carrot peel waste. The optimized extraction conditions provided high yields with strong model reliability, while the sustainability assessment confirmed its moderate environmental performance. Although improvements are still needed in energy efficiency, extraction completeness, and system integration, the study provides strong evidence that agricultural waste can be effectively transformed into high-value bioactive compounds using green extraction technologies. This research contributes significantly to the growing field of sustainable bioprocessing and supports the development of eco-friendly industrial practices that prioritize both efficiency and environmental responsibility.

A study **[11]** by **Manalo (2021)** discussed the role of Digital Agriculture in supporting the modernization and industrialization goals of the Philippine agricultural sector through the adoption of Information and Communications Technologies (ICTs). The study highlighted how digital technologies have become important tools in improving agricultural productivity, efficiency, and access to information. Various ICT-based innovations have already been utilized in agriculture, including web-based disease diagnosis systems, crop monitoring sensors, mobile applications, and digital trading platforms. In the Philippines, DA-PhilRice developed several ICT-driven applications such as the Leaf Color Chart (LCC) app, e-binhi, e-damuhan, and AgriDoc, which provide farmers with timely and relevant agricultural information. These technologies were designed to support better crop management, improve decision-making, and enhance overall agricultural operations. The study emphasized that digital technologies can help farmers access critical information related to crop production, nutrient management, pest control, and market opportunities, enabling them to make informed decisions and improve farm performance. Through these innovations, Digital Agriculture has been recognized as a significant strategy in transforming traditional farming practices into more efficient, data-driven, and technology-enabled systems.

The study further examined the level of ICT adoption among Filipino farmers and identified several challenges that hinder the effective use of digital technologies. Based on the results of the 2016–2017 Rice-Based Farm Households Survey, approximately 93 percent of rice-farming households had access to mobile phones, smartphones, and internet services. Similarly, a large percentage of farmers expressed willingness to receive agricultural information through text messaging and internet-based platforms. Despite this high level of access, actual utilization of ICTs for farming activities remained relatively low. The survey revealed that only 31 percent of farmers used text messaging as a source of agricultural information, while only 21 percent actively searched for farming-related information online. These findings suggest that access to technology alone does not guarantee its effective use. The study argued that several factors contribute to this gap, including low levels of digital literacy, limited exposure to technology, lack of confidence in using digital tools, and ICT anxiety among farmers.

ICT anxiety was identified as one of the major barriers to technology adoption in agriculture **[11]**. ICT anxiety refers to the discomfort, fear, or uncertainty experienced by individuals when interacting with computers, mobile devices, or other digital technologies. This challenge is particularly common among older populations, including many Filipino farmers. Since a significant portion of the agricultural workforce in the Philippines consists of aging farmers, the adoption of new technologies often becomes difficult. The study explained that limited experience with digital tools and insufficient technical knowledge can discourage farmers from utilizing ICT-based solutions, even when such technologies are readily available. As a result, many farmers continue to rely on traditional methods of obtaining information, such as consulting fellow farmers, agricultural extension workers, and local community networks. This preference for interpersonal communication demonstrates that technological innovations must be accompanied by strategies that address behavioral and educational barriers to ensure successful implementation.

To address these challenges, the study introduced the concept of infomediaries as a means of bridging the gap between technology and farmers. Infomediaries are individuals who facilitate access to information by helping others use digital tools and communication technologies. The study described infomediaries as important agents in reducing the digital divide, particularly among populations that have limited technological skills. The digital divide refers to the disparity between individuals who can effectively access and utilize digital technologies and those who cannot. This divide is influenced by various factors, including educational attainment, socioeconomic status, technological literacy, and access to resources. By mobilizing infomediaries, agricultural organizations can help ensure that the benefits of digital technologies reach a wider population of farmers, including those who may be hesitant or unable to use ICTs independently.

The study highlighted the implementation of the Infomediary Campaign by DA-PhilRice from 2012 to 2017 as an innovative approach to promoting digital inclusion in agriculture. The campaign engaged high school students from rice-farming communities and encouraged them to serve as infomediaries for their parents and local farmers. These students accessed agricultural information through digital platforms and shared the information with farmers who lacked the skills or confidence to use ICT tools themselves. Through this initiative, young people became active participants in agricultural development while helping bridge the information gap within their communities. The campaign demonstrated how youth engagement can contribute to the dissemination of agricultural knowledge and encourage greater acceptance of digital technologies among farmers.

Results of the Infomediary Campaign showed positive outcomes in promoting information access and technology adoption. The study reported that more than 4,000 student-infomediaries submitted nearly 20,000 inquiries to the PhilRice Text Center between 2012 and 2016. These inquiries covered a wide range of agricultural topics, including rice varieties, pest management, nutrient management, crop establishment, and general farming practices. Students not only gathered information but also actively shared their knowledge with their parents, relatives, and other members of the farming community. The study observed that information dissemination often extended beyond the immediate recipients, creating a multiplier effect where agricultural knowledge was continuously shared among farmers. This process contributed to improved awareness of modern farming techniques and increased utilization of available information resources.

Furthermore, the study found evidence that the campaign positively influenced farmers' information-seeking behavior. Farmers who were exposed to the initiative became more likely to use text messaging services and digital communication platforms to obtain agricultural information. In some communities, the habit of consulting digital information sources continued even after the campaign ended. These findings suggest that targeted interventions and support systems can help overcome barriers to technology adoption and encourage the long-term use of ICT-based agricultural solutions. The study emphasized that successful digital transformation requires not only technological infrastructure but also educational programs, capacity-building initiatives, and community support mechanisms that enable users to maximize the benefits of digital tools.

The study concluded that Digital Agriculture can only achieve its full potential if efforts are made to ensure inclusivity and accessibility among all stakeholders. While digital technologies offer significant opportunities for improving productivity, information management, and decision-making, their effectiveness depends largely on the willingness and ability of users to adopt them. Initiatives such as the Infomediary Campaign demonstrate that human-centered approaches can play a critical role in facilitating technology adoption and reducing barriers associated with digital literacy and ICT anxiety. The study recommended expanding infomediary programs, strengthening ICT training initiatives, integrating agriculture-related digital skills into educational curricula, and providing continuous support to both farmers and agricultural extension workers.

This literature is relevant to the present study as it demonstrates the significant role of digital technologies in improving operational efficiency, information management, and data-driven decision-making. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights how digital solutions can enhance monitoring processes, facilitate timely access to information, and support more effective

management practices. The findings support the development of digital production systems by emphasizing the value of technology in collecting, analyzing, and utilizing data to improve productivity and operational performance. Furthermore, the study reinforces the importance of user adoption and accessibility in ensuring the successful implementation of digital innovations within agricultural and food-processing industries

A study **[12]** by **Barroga, Rola, Depositario, Digal, and Pabuayon (2019**) investigated the determinants of the extent of technological innovation adoption among micro, small, and medium food processing enterprises (MSMFEs) in the Davao Region, Philippines. The research focused on enterprises that were beneficiaries of the Department of Science and Technology–Small Enterprises Technology Upgrading Program (DOST-SETUP), a government initiative that provides technological support, upgraded equipment, and technical assistance to improve productivity and competitiveness in the food processing sector. The study aimed to determine not only whether MSMFEs adopt technological innovations but also the degree or extent to which these innovations are implemented within their operations.

The researchers utilized a census of 52 MSMFEs in Davao Region and employed both quantitative and qualitative methods to analyze the data. Specifically, an ordered logistic regression model was used to identify the factors influencing the level of technological innovation adoption. The study also used index construction to measure the extent of adoption across multiple technological interventions, including food safety training, Good Manufacturing Practices (GMP) assessment, plant layout improvement, packaging and labeling, laboratory testing, product development, energy audit, cleaner production technology (CPT) consultancy, MPEX consultancy, and technology training. These innovations represent a comprehensive package of technical and managerial upgrades introduced through the SETUP program.

Findings of the study showed that MSMFEs in the Davao Region generally exhibited varying levels of technological innovation adoption, ranging from non-adoption to full adoption. A small portion of the enterprises failed to adopt any of the technological innovations provided, while others demonstrated low to moderate adoption levels. However, the majority of MSMFEs were classified as highly adopters, indicating that most enterprises were able to integrate several SETUP innovations into their operations. Only a smaller segment of the sample achieved full adoption, meaning they implemented nearly all of the technological innovations offered by the program.

Among the specific innovations introduced by SETUP, food safety training emerged as the most widely adopted intervention, followed by GMP assessment and MPEX consultancy. These innovations were particularly important for MSMFEs because they directly address compliance with food safety standards and operational efficiency. On the other hand, innovations such as product development and cleaner production technology consultancy were among the least adopted. The study noted that more advanced or resource-demanding innovations tended to have lower adoption rates, especially among smaller enterprises with limited capacity and technical expertise.

The study further highlighted that the adoption of technological innovations does not only depend on whether enterprises accept or reject a particular technology, but also on the depth and breadth of implementation across various business functions. This means that even if MSMFEs adopt certain innovations, the level of integration into their production processes, management systems, and product development activities may differ significantly. Based on this, the researchers categorized MSMFEs into five groups according to their extent of adoption: did not adopt, less adopted, moderately adopted, highly adopted, and fully adopted.

In terms of distribution, most medium-sized enterprises were found to have lower levels of adoption compared to micro and small enterprises. Micro enterprises, despite their limited resources, showed a relatively higher tendency toward adopting multiple technological innovations. Small enterprises were mostly classified as moderate to high adopters, depending on their exposure to export markets and access to external support services. The study suggests that enterprise size alone does not fully determine innovation adoption, but interacts with other organizational and contextual factors.

One of the key findings of the study is that the combination of food safety training, GMP assessment, and MPEX consultancy was the most commonly adopted set of innovations. These three are closely interconnected, as MPEX consultancy often recommends improvements related to GMP compliance and food safety practices. This indicates that MSMFEs tend to adopt innovations in clusters rather than individually, especially when the technologies are complementary and collectively improve production efficiency and compliance with regulatory standards.

The study also revealed that full adoption of technological innovations was relatively limited, with only about 21% of MSMFEs achieving this level **[12]**. Enterprises under this category were able to implement most or all of the SETUP-supported innovations, demonstrating strong organizational capacity and readiness for technological upgrading. Micro enterprises in this category adopted nearly all available innovations, while small enterprises typically adopted a slightly smaller set, often excluding more advanced innovations such as laboratory analysis or product development.

Highly adopted MSMFEs represented a significant portion of the sample. These enterprises generally implemented key foundational innovations such as GMP assessment and food safety training, along with selected productivity-enhancing interventions. However, they still lacked full integration of more advanced technological services. This suggests that while these enterprises are relatively progressive in adopting innovation, constraints such as cost, technical expertise, and organizational capacity still limit full adoption.

Moderately adopted MSMFEs showed partial integration of SETUP innovations, often focusing on basic operational improvements. These enterprises typically adopted essential interventions such as food safety training and plant layout improvements but did not fully engage in more complex innovations like laboratory analysis or product development. Their level of adoption indicates an intermediate stage of technological readiness, where awareness and initial implementation exist but full assimilation remains limited.

Less-adopted MSMFEs were characterized by minimal engagement with SETUP innovations, typically adopting only a few interventions such as technology training or food safety training. In this category, medium enterprises were more dominant, suggesting that some established firms may perceive less need for external technological assistance due to existing internal capabilities. This group highlights that adoption is not always higher among larger firms, especially when they already possess in-house technical expertise.

At the lowest level, the study identified a very small number of enterprises that did not adopt any technological innovations at all. These cases were linked to external shocks such as natural disasters and management issues, which disrupted operations and prevented the proper utilization of SETUP assistance. These findings emphasize that external environmental factors and governance issues can also significantly affect innovation adoption outcomes.

The study further examined the determinants of technological innovation adoption using an ordered logistic regression model. The results identified four statistically significant factors influencing the extent of adoption: sex of the owner, educational attainment, business scale, and type of market **[12]**. These variables were found to consistently influence whether MSMFEs adopt innovations at a higher or lower level.

In terms of sex of the owner, the study found that male-owned enterprises were more likely to achieve higher levels of technological innovation adoption compared to female-owned enterprises. Although female entrepreneurs represented a larger proportion of MSMFE owners in the sample, male owners were more likely to engage in extensive adoption. This was attributed to differences in mobility, access to training opportunities, and exposure to external information networks, which may influence innovation-related decision-making.

Educational attainment was also found to have a positive relationship with the extent of adoption. MSMFE owners with higher educational levels were more likely to adopt a wider range of technological innovations. This is because education enhances the ability to understand technical information, evaluate innovation benefits, and manage the risks associated with technological change. As a result, more educated owners tend to be more open to upgrading their production systems and integrating new technologies into their operations.

Business scale, on the other hand, showed a negative relationship with adoption extent, indicating that micro enterprises were more likely to adopt innovations compared to small and medium enterprises. This finding suggests that smaller firms are more flexible and can implement changes more quickly due to simpler organizational structures and centralized decision-making. In contrast, larger firms may experience more complex internal processes that slow down innovation adoption.

The type of market was also identified as a significant determinant. MSMFEs that cater to both domestic and export markets were more likely to adopt technological innovations at higher levels. This is because export-oriented firms are required to comply with stricter international standards, which encourages them to adopt advanced technologies and quality control systems to remain competitive in global markets.

Other variables such as age, years of experience, organizational type, and sources of information were found to be statistically insignificant in influencing the extent of technological innovation adoption. This suggests that while these factors may have some descriptive influence, they do not significantly determine the level of adoption when analyzed together with stronger predictors such as education, gender, business scale, and market orientation.

Overall, the study concluded that technological innovation adoption among MSMFEs in the Davao Region is a multi-dimensional process influenced by both individual and organizational factors. The findings highlight that innovation adoption is not simply a matter of access to technology, but also depends on the characteristics of decision-makers and the market environment in which enterprises operate.

The study is relevant to the present research because it provides empirical evidence on how personal demographics and organizational characteristics influence the extent of technological innovation adoption in food processing enterprises. It also highlights the importance of government support programs such as DOST-SETUP in promoting innovation among MSMEs, particularly in developing regions.

A study **[13]** by **Ang, Amongo, Paras Jr., Suministrado, and Peralta (2022),** entitled *Development of Banana Peduncle Juice Extractor for Ethanol and Fiber Production*, focused on the conceptualization, design, fabrication, and performance evaluation of a specialized mechanical system intended for the efficient processing of banana (Musa acuminata C.) peduncles. The research was motivated by the increasing concern regarding agricultural waste management in the Philippines, particularly within large-scale banana plantations where peduncles are typically discarded despite their significant potential as raw materials for bioethanol production and natural fiber utilization. The authors emphasized that although banana peduncles contain valuable lignocellulosic components and fermentable sugars, their utilization remains limited due to the lack of appropriate extraction technology that can simultaneously maximize juice recovery while preserving fiber integrity. Conventional extraction machines, originally designed for other agricultural commodities such as sugarcane, were found to be inefficient when applied to banana peduncles due to differences in material structure, density, and mechanical resistance. This gap in technology prompted the development of a dedicated juice extraction system tailored specifically for banana peduncle processing.

The developed machine was designed using engineering principles that prioritized functionality, manufacturability, safety, and cost-effectiveness. The researchers deliberately utilized locally available materials in order to ensure affordability and ease of replication, particularly for potential use in rural agricultural communities and small-scale processing facilities. The system employed a crushing roll mechanism as the primary extraction component, consisting of a main roll and multiple feeder rolls arranged in a configuration that enables controlled compression of the peduncle material. This configuration was intended to optimize juice release while minimizing excessive mechanical damage to the fiber structure, thereby allowing the biomass to be utilized for both ethanol fermentation and fiber recovery applications. The structural design also incorporated protective enclosures and adjustable components to enhance operational safety and accommodate varying operator conditions. In addition, the transmission system was engineered using a combination of engine-driven pulleys, gear reducers, and chain-sprocket

assemblies, allowing controlled variation of rotational speed to evaluate performance under different operational settings.

To assess the effectiveness of the developed machine, the researchers employed a systematic experimental design using a two-factor, three-level full factorial approach. The two independent variables considered were roll rotational speed (50 rpm, 70 rpm, and 90 rpm) and crushing load (0.8 kN, 1.2 kN, and 1.6 kN). A total of 27 experimental runs were conducted to determine the influence of these variables on machine performance. Each trial utilized approximately 40 kilograms of freshly harvested banana peduncle samples, which were processed individually to ensure accurate measurement of performance indicators. The study followed standardized testing procedures based on established agricultural machinery evaluation protocols, specifically PAES 235:2008 for multi-crop juice extractors and PAES 229:2005 for fiber decorticators. These standards ensured that the evaluation process was consistent, reliable, and comparable to other agricultural machinery studies.

The performance of the machine was evaluated using multiple response variables, including input capacity, juice extraction rate, juice recovery, extraction losses, extraction efficiency, fuel consumption, fiber extraction rate, fiber recovery, and fiber tensile strength. In addition, the °Brix level of the extracted juice was measured to determine sugar concentration, which is an important factor in ethanol production. The inclusion of both mechanical and biochemical indicators allowed for a comprehensive assessment of the system, covering not only productivity but also product quality. The tensile strength of the extracted fiber was also analyzed using a universal testing machine following ASTM D3822-01 standards to determine whether mechanical processing had adversely affected fiber integrity.

Results of the study showed that both roll rotational speed and crushing load significantly influenced machine performance, although their effects varied depending on the specific output variable. Higher roll rotational speeds generally resulted in increased input capacity, juice extraction rate, and fiber extraction rate due to reduced processing time and faster material throughput within the crushing chamber. Conversely, lower rotational speeds tended to reduce throughput but improved certain recovery metrics due to longer retention time within the extraction system. Crushing load, on the other hand, exhibited a dual effect depending on the response variable. Lower crushing loads facilitated easier passage of peduncle material through the rolls, thereby increasing processing speed, while higher crushing loads enhanced the mechanical compression of the material, resulting in improved juice recovery and extraction efficiency.

The interaction between roll speed and crushing load revealed important trade-offs in system performance. For instance, although higher crushing loads improved juice recovery, they also increased resistance within the system, which reduced processing speed and increased fuel consumption. Similarly, higher rotational speeds improved throughput but contributed to higher extraction losses, as some juice was not fully recovered before the material exited the machine. These findings highlight the complexity of optimizing mechanical systems for biomass processing, where improvements in one performance aspect may negatively affect another. As such, the study emphasized the importance of multi-response optimization techniques to determine the most balanced operating conditions.

Using response surface methodology (RSM), the researchers performed a multi-objective optimization analysis to identify the most desirable combination of operating parameters. The optimization process considered multiple performance goals simultaneously, including maximizing input capacity, juice recovery, extraction efficiency, fiber extraction rate, and fiber recovery, while minimizing extraction losses and fuel consumption. Based on desirability function analysis, the optimal operating condition was identified at approximately 71.378 rpm and 0.8 kN crushing load. At this optimal setting, the machine achieved an input capacity of about 307.24 kg/h, a juice extraction rate of 82.92 kg/h, juice recovery of 26.08%, extraction efficiency of 18.46%, fiber extraction rate of 54.83 kg/h, fiber recovery of 42.29%, and fuel consumption of 0.391 L/h **[13]**. These results demonstrate that the system can achieve a balanced level of efficiency while maintaining both juice and fiber output quality when operated under optimized conditions.

Further analysis revealed that neither roll rotational speed nor crushing load significantly affected the °Brix content of the extracted juice or the tensile strength of the recovered fiber. This indicates that while the mechanical parameters influence the quantity of juice and fiber produced, they do not significantly alter the chemical composition of the juice or the structural strength of the fiber. This finding is particularly important because it suggests that the quality of the end products remains stable regardless of operational variations, making the system more reliable for industrial applications where consistency is essential.

In terms of fermentation performance, the extracted juice was subjected to ethanol production analysis using Saccharomyces cerevisiae as the fermentation agent. The results showed that the fermented slurry contained approximately 10–11% ethanol by volume across all treatment combinations. Statistical analysis indicated no significant differences in ethanol content among the different mechanical settings, reinforcing the observation that sugar concentration, rather than mechanical extraction conditions, is the primary determinant of ethanol yield. This further supports the conclusion that the machine primarily influences extraction efficiency and throughput rather than biochemical conversion outcomes.

In conclusion, the study demonstrated that the developed banana peduncle juice extractor is an effective machine for converting agricultural waste into valuable products such as bioethanol feedstock and natural fiber **[13]**. The system successfully addressed the limitations of conventional extraction machines by introducing a design specifically optimized for banana peduncle characteristics. The findings highlight the importance of mechanical optimization in improving productivity while maintaining product quality. Moreover, the study underscores the potential of banana peduncle as a sustainable raw material for bioenergy and fiber-based industries, contributing to waste reduction, renewable energy development, and value-added agricultural processing. The authors further recommended future improvements such as modifying roll geometry, improving transmission efficiency, experimenting with alternative crushing mechanisms, and testing additional banana varieties to enhance the versatility and efficiency of the system.

A s**tudy [14] by Jiang and Po (2023**) investigated the influence of human resource practices on employee retention in the food manufacturing industry in the Philippines and examined the moderating role of the work environment. The researchers focused on five key human resource practices: training and development, employee participation in decision-making, compensation, job security, and performance appraisal. Grounded in Social Exchange Theory, the study proposed that employees are more likely to remain with an organization when they perceive that the organization provides adequate support, fair treatment, and opportunities for growth. In return, employees reciprocate through commitment, loyalty, and long-term retention.

The study employed a quantitative research design using a cross-sectional survey method. Data were collected from 206 employees working in various food manufacturing companies in the Philippines through both online and offline questionnaires. The researchers used a five-point Likert scale to measure employee perceptions regarding human resource practices, work environment, and employee retention. Statistical analyses, including regression and moderation analysis, were conducted using JAMOVI software to determine the relationships among the variables **[14]**.

The findings revealed that employee participation in decision-making, compensation, job security, and performance appraisal significantly and positively influenced employee retention. Employees who were given opportunities to participate in organizational decisions tended to develop a stronger sense of belonging and commitment to their companies. Likewise, fair compensation and benefits were found to be major factors that encouraged employees to remain in their organizations. The results further indicated that job security contributed positively to retention because employees preferred stable employment arrangements that reduced uncertainty regarding their future careers. Performance appraisal was also shown to strengthen employee retention by providing feedback, recognition, and opportunities for professional improvement.

Among the identified human resource practices, compensation emerged as the strongest predictor of employee retention. The researchers found that employees were more likely to remain in organizations when they perceived

that they were fairly compensated for their work and contributions. This finding emphasizes the importance of providing competitive salaries and benefits in order to attract and retain skilled employees in the food manufacturing sector. The study concluded that compensation plays a critical role in maintaining workforce stability and reducing employee turnover.

In contrast, the study found that training and development did not significantly influence employee retention. This result differed from several previous studies that suggested training opportunities increase employee loyalty and commitment. The researchers proposed that training programs may not automatically lead to retention unless employees perceive them as valuable, relevant, and beneficial to their career advancement. This finding highlights the need for organizations to carefully evaluate the effectiveness of their training and development initiatives and ensure that such programs align with employee needs and organizational objectives.

Another significant contribution of the study was its examination of the work environment as a moderating variable **[14]**. The findings showed that the work environment strengthened the relationship between employee participation in decision-making, compensation, job security, performance appraisal, and employee retention. Employees working in supportive environments were more likely to respond positively to human resource practices and demonstrate stronger intentions to remain in the organization. The study emphasized that a positive work environment includes both physical and non-physical factors, such as workplace safety, adequate facilities, healthy interpersonal relationships, effective communication, and a supportive organizational culture.

The researchers further explained that improvements in the physical work environment, including proper ventilation, safe production equipment, reduced workplace noise, and clean facilities, can increase employee satisfaction and productivity. At the same time, non-physical aspects such as teamwork, leadership support, organizational structure, and employee relationships contribute significantly to employee motivation and retention. These findings suggest that organizations should not only focus on implementing effective human resource policies but also invest in creating a work environment that promotes employee well-being and engagement.

The study has important implications for food manufacturing companies in the Philippines. It suggests that organizations should prioritize compensation systems, employee involvement, job security measures, and performance evaluation mechanisms to improve retention rates. Furthermore, managers should recognize the importance of maintaining a positive work environment to maximize the effectiveness of these human resource practices. By doing so, organizations can reduce employee turnover, improve workforce stability, and enhance overall organizational performance **[14]**.

This literature is relevant to the present study because it demonstrates how organizational practices, performance monitoring, and workplace conditions influence efficiency, productivity, and long-term organizational success in the food manufacturing industry. The findings support the development of digital systems that provide real-time monitoring, performance tracking, and data-driven decision-making capabilities. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights the importance of accurate information, continuous monitoring, and effective management practices in improving operational performance and achieving organizational goals.

## 2.3 Related Literature

A study **[15]** by Somsen, Capelle, and Tramper (2004) introduced Production Yield Analysis (PYA), a systematic method used to evaluate raw material efficiency in food processing. The study developed a “Yield Index” to distinguish between necessary losses and avoidable losses, helping identify inefficiencies in production systems. The authors emphasized that relying solely on historical benchmarks may conceal actual waste, highlighting the importance of model-based analysis. This literature is relevant to the present study as it supports the development of a digital system capable of identifying and analyzing production losses in real time.

A study [16] by **Charbonnier, Béranger, and Cognon (2005)** investigated the application of trend extraction and analysis techniques for complex system monitoring and decision support in industrial and medical environments. The researchers focused on developing an online monitoring methodology capable of analyzing continuously changing data from highly complex systems such as industrial automation processes, food manufacturing operations, and intensive care monitoring systems. According to the researchers, modern industrial environments generate large volumes of operational data that are difficult for human operators to interpret manually, especially in situations where process conditions rapidly change and immediate decisions are required. The study emphasized that traditional monitoring approaches often rely heavily on manual observation and operator expertise, which may result in delayed responses, reduced efficiency, and increased risk of operational errors during abnormal conditions.

To address these challenges, the researchers proposed a real-time trend extraction methodology designed to analyze univariate time-series data using semi-qualitative representations. The system utilized three primary trend primitives: **Increasing, Decreasing, and Steady.** These primitives were used to represent operational behavior over specific time intervals, allowing users to interpret process conditions more effectively. According to the study, converting numerical signals into symbolic trend representations helps reduce cognitive overload and improves operator understanding of complex operational situations.

The methodology involved several stages. First, continuously generated process data were segmented into linear intervals. These intervals were then classified into temporal patterns such as increasing trends, decreasing trends, stable conditions, transient variations, and step changes. After classification, the detected temporal patterns were transformed into semi-quantitative episodes describing the behavior of operational variables over time. The researchers explained that this approach enabled the monitoring system to identify abnormal process behavior without requiring highly complex mathematical models of the entire industrial system.

The study also explored the application of trend extraction for predictive maintenance and operational supervision in industrial systems. The researchers explained that highly automated production systems often operate under demanding conditions to maximize productivity and profitability, increasing the likelihood of equipment degradation and operational faults. By continuously monitoring operational variables and analyzing abnormal trend patterns, the proposed methodology could assist in identifying equipment deterioration before critical failures occurred. The researchers concluded that predictive monitoring capabilities support preventive maintenance strategies, reduce operational downtime, and improve overall equipment reliability [16].

A study [17] by **Davies, Fried, and Gather (2003)** entitled “Robust Signal Extraction for On-line Monitoring Data” examined statistical methods for extracting meaningful information from continuously generated time-series data in real-time monitoring environments. The researchers focused on developing and evaluating robust techniques that can effectively separate true underlying signals from noise, outliers, and irregular fluctuations. The study was motivated by the increasing demand for reliable monitoring systems in critical environments such as intensive care units and industrial process control systems, where fast and accurate interpretation of data is essential for decision-making.

The researchers explained that modern monitoring systems generate large volumes of data due to continuous recording of physiological or operational variables. In healthcare settings, for example, patient data such as heart rate, blood pressure, and oxygen levels are recorded at very short intervals, producing complex datasets that must be analyzed immediately. Similarly, industrial systems also generate high-frequency operational data that reflect machine performance, process efficiency, and production status. However, these datasets are often affected by noise, measurement errors, and external disturbances, which make it difficult to identify meaningful patterns using traditional analysis methods.

According to the study, one of the major challenges in online monitoring is distinguishing clinically or operationally relevant changes from irrelevant noise and outliers. The researchers emphasized that false alarms in monitoring systems are a significant problem, particularly in intensive care environments where excessive alarms may lead to

alarm fatigue among medical staff. This reduces trust in automated systems and may cause important warnings to be ignored. In industrial environments, inaccurate signal interpretation may also lead to unnecessary maintenance actions or failure to detect real system faults. Because of these issues, there is a strong need for robust statistical methods that can reliably extract true signals from noisy data in real time.

Another important contribution of the study is its discussion on the trade-off between robustness and computational efficiency. The researchers emphasized that while highly robust methods provide better resistance to data contamination, they may not always be suitable for real-time applications due to processing delays. Therefore, selecting an appropriate signal extraction method requires balancing accuracy, stability, and computational cost depending on the system requirements.

In conclusion, the study by Davies, Fried, and Gather (2003) **[17]** provides valuable insights into the importance of robust statistical methods in real-time monitoring systems. Its emphasis on noise reduction, trend extraction, and reliable signal interpretation makes it highly applicable to modern industrial systems, particularly in the context of automated food production and digital manufacturing technologies.

A study **[18]** conducted by **Charmine Sheena R. Saflor and Marvin I. Noroña (2021)** explored the implementation of Total Productive Maintenance (TPM) within the Philippine rice production sector. The research focused on addressing the increasing operational and maintenance challenges experienced by farmers and agricultural cooperatives due to the growing use of mechanized farming equipment. The authors emphasized that many rice production operators in the Philippines still rely on reactive maintenance practices, where machines are repaired only after breakdowns occur. This approach often results in frequent equipment failures, increased maintenance expenses, production delays, and reduced overall productivity. The study highlighted that inadequate maintenance management can shorten the lifespan of agricultural machinery and negatively affect the efficiency of rice production operations.

The study is relevant to the present research because it demonstrates the importance of monitoring equipment performance and maintenance activities to improve operational efficiency and reduce production losses. It supports the idea that implementing a digital and data-driven monitoring system can help organizations identify inefficiencies, manage maintenance schedules, and enhance productivity in real time. Similar to the proposed study on real-time yield analytics and extraction efficiency monitoring, the research emphasizes the value of systematic analysis and continuous monitoring in optimizing production processes, minimizing waste, and improving decision-making within industrial and agricultural operations.

A study **[19]** conducted by **Desai Shivani Sarjerao (2020),** Design, Development and Performance Evaluation of Juice Extractor, presented the development of a hydraulically operated juice extraction machine intended for processing pomegranate arils and bottle gourd. The experimental evaluation focused on three main independent variables: size reduction (shred size), applied hydraulic pressure, and filter bag pore size. These were tested in different combinations to determine their effects on several dependent performance indicators, namely juice yield, juice extraction efficiency, extraction losses, machine capacity, and sedimentation percentage. A series of experimental trials were conducted for both pomegranate arils and bottle gourd, with repeated measurements used to ensure reliability of results. Statistical analysis, including ANOVA, was applied to determine the significance of main effects and interaction effects among the variables.

In terms of juice extraction efficiency, the study reported maximum values of 94.30% for pomegranate and 80.27% for bottle gourd under optimal operating conditions. The results also showed a consistent decrease in extraction losses as pressure increased and shred size decreased, demonstrating improved material utilization and reduced wastage. Although most interaction effects between variables were statistically non-significant, individual factors such as pressure and filter bag pore size were found to have a significant impact on extraction performance.

Overall, the study emphasizes that mechanical design parameters and operating conditions play a critical role in determining the efficiency of juice extraction systems. It further suggests that optimizing these variables can

significantly enhance juice recovery, reduce processing losses, and improve machine productivity. This research is relevant to the present study as it provides an engineering-based framework for evaluating extraction systems and supports the development of optimized, efficiency-driven food processing technologies in agricultural production systems.

A study [20] by **Bartoszewicz and Wdowicz (2019)** presented an automated reporting approach that integrates the SAP ERP Production Planning (PP) module with Microsoft Excel and Visual Basic for Applications (VBA). The researchers observed that while SAP ERP provides extensive support for production planning and operations, its standard reporting features may not always satisfy the analytical and reporting needs of production managers. To address this limitation, they developed the **Follow the Plan (FTP) Report**, an application that automatically retrieves production data from SAP ERP, processes the information, and generates reports comparing planned production activities with their actual execution. This automation reduces the need for manual data handling while improving the speed and consistency of production reporting.

The proposed system gathers production information from the **COOISPI** and **MB51** transactions within SAP ERP. These transactions contain data on production orders, work centers, planned schedules, completed orders, and material movements. After extracting the required information, the application organizes and analyzes the data in Microsoft Excel, allowing users to determine whether production orders were completed as planned. The system also identifies production deviations, such as orders that were not started, orders produced beyond planned quantities, and production results that fall outside the accepted tolerance. In addition, users can record standardized root causes, including raw material shortages, equipment failures, low machine performance, quality issues, and inaccurate system data, to support further investigation and continuous process improvement.

The researchers found that automating the reporting process significantly improved operational efficiency. Preparing production compliance reports manually previously required approximately two hours because users had to retrieve, organize, and validate data from multiple SAP transactions. With the FTP Report application, the same process could be completed in about five minutes, while also reducing the possibility of human error. The study concluded that integrating ERP systems with automated reporting tools enhances production monitoring, supports faster managerial decision-making, and provides more reliable information for evaluating production performance. This literature is relevant to the present study because it demonstrates how digital technologies can automate production monitoring and reporting, supporting real-time analysis and more effective evaluation of production efficiency, which aligns with the objectives of developing a real-time yield analytics and extraction efficiency monitoring system for AlDi Foods Inc.

## 2.4 Synthesis of the Review Literature

The reviewed foreign and local literature consistently demonstrates that production systems in agriculture and food manufacturing are undergoing a significant transformation from traditional manual operations toward digitally enabled, data-driven, and automated environments. This transformation is largely driven by rapid advancements in Information and Communications Technology (ICT), artificial intelligence (AI), machine learning, big data analytics, and real-time monitoring systems. Collectively, these technologies enhance operational efficiency, improve production accuracy, and support faster, evidence-based decision-making. Across the studies reviewed, there is a clear consensus that digital integration is no longer optional but a necessary component for modern, competitive, and efficient production systems, particularly in industries involving food processing and extraction operations.

One of the most consistent findings across the literature is the growing importance of real-time monitoring systems and centralized digital platforms in production environments. Studies such as Verhoef et al. (2021), Makhmudah et al. (2025), and Bartoszewicz and Wdowicz (2019) emphasize that traditional paper-based recording systems and manual reporting processes are highly inefficient due to delays, data inconsistencies, and lack of immediate accessibility. These limitations prevent production managers from identifying inefficiencies, equipment malfunctions, and yield losses at the moment they occur. In contrast, digital dashboards, ERP-integrated systems, and automated

reporting tools provide real-time visibility of production activities, allowing faster response to operational issues. This shift highlights that real-time data accessibility is a critical requirement for effective production monitoring systems such as the proposed yield and extraction efficiency monitoring system for AlDi Foods Inc.

Another major finding is the increasing role of data analytics, artificial intelligence, and intelligent computing systems in optimizing production processes. Studies by Siddique et al. (2025) and Sharma et al. (2021) show that AI-powered systems are capable of processing large and complex datasets to identify operational patterns, predict production outcomes, and optimize resource allocation. These systems support predictive, descriptive, and prescriptive analytics, enabling manufacturers to minimize waste, improve productivity, and enhance decision-making accuracy. Furthermore, technologies such as machine learning and computer vision are increasingly used to automate inspection, monitor production flow, and detect irregularities, reducing reliance on manual observation and improving operational reliability in industrial environments.

The literature also highlights the critical importance of production efficiency metrics, yield analysis, and performance evaluation systems in modern manufacturing. Mansour and Al-Hamdani (2024) introduced internal key performance indicators (KPIs) as tools for identifying production inefficiencies and defects, while Somsen et al. (2004) emphasized Production Yield Analysis (PYA) as a structured method for distinguishing between avoidable and unavoidable production losses. Similarly, Dora et al. (2020) found that most food losses occur during the production stage due to machine inefficiencies, human errors, poor process control, and inadequate operational management. These findings strongly indicate that continuous monitoring of yield and efficiency is essential in minimizing production losses, improving resource utilization, and ensuring consistent operational performance—core objectives of real-time production monitoring systems.

In food extraction and processing systems, multiple studies such as Desai (2020), Mphahlele et al. (2016), Aviara et al. (2020), and Ang et al. (2022) demonstrate that mechanical and operational parameters significantly influence extraction yield, product quality, and production efficiency. Variables such as pressure, rotational speed, feed condition, and extraction technique directly affect both the quantity and quality of output. In addition, Mphahlele et al. (2016) further explain that extraction methods not only influence yield but also affect chemical composition, antioxidant content, aroma compounds, and overall product quality. These findings reinforce the idea that monitoring extraction processes is necessary not only for improving efficiency but also for ensuring product consistency and quality control in food manufacturing systems.

The reviewed literature also shows a growing adoption of automation technologies and intelligent monitoring systems in industrial environments. Sabih et al. (2023) demonstrated that computer vision and machine learning can effectively automate material flow monitoring, improving accuracy while reducing manual intervention. Similarly, ERP-based systems studied by Bartoszewicz and Wdowicz (2019) show that automated reporting systems significantly enhance production tracking efficiency, reduce human error, and support faster managerial decision-making. These studies collectively indicate a global shift toward integrated digital production systems that enable real-time monitoring, automated computation, and improved operational transparency.

Local studies further confirm the relevance and urgency of digital transformation in the Philippine context, particularly in agriculture and food manufacturing industries. Bagaforo and Albason (2024) and Barroga et al. (2019) found that while MSMEs benefit from technological adoption, they continue to face challenges such as limited technical expertise, financial constraints, and inadequate infrastructure. De Leon et al. (2017) emphasized that centralized digital systems significantly improve data reliability, production tracking, and operational efficiency in local enterprises. Meanwhile, Manalo (2021) highlighted that although ICT access is widespread among Filipino farmers, actual utilization remains low due to digital literacy gaps, ICT anxiety, and lack of technical support. These findings suggest that the success of digital systems depends not only on availability but also on usability, accessibility, and user capability.

Collectively, the literature establishes that digital transformation, real-time monitoring, and data-driven analytics are

essential components of modern production systems. This is particularly significant in food processing and extraction industries where efficiency, yield optimization, and waste reduction directly affect productivity and profitability. The findings strongly support the development of integrated systems that provide real-time data capture, automated yield computation, and extraction efficiency monitoring—such as the proposed system for AlDi Foods Inc.—to enhance operational performance and support data-driven decision-making in production environments.

## Research Gaps Identified

Despite the extensive body of literature on digital transformation, production optimization, and intelligent manufacturing systems, several important gaps remain evident, particularly when applied to food processing and juice extraction operations.

First, there is a noticeable lack of research and system development focused specifically on real-time monitoring systems designed for juice extraction and similar food processing operations. Most existing studies concentrate on broader industrial manufacturing environments, agricultural ICT applications, or general automation systems. While these studies provide valuable insights into production optimization, they do not sufficiently address the unique requirements of extraction-based processes, where continuous measurement of input materials, juice yield, and extraction efficiency is critical. As a result, there is limited exploration of systems tailored to monitor and evaluate performance in real-time within juice production environments such as AlDi Foods Inc.

Second, the current literature often treats key production functions—such as production monitoring, yield analysis, quality control, and reporting—as separate and disconnected systems. This fragmented approach creates inefficiencies in data handling and delays in decision-making, as information must be processed across multiple platforms or manual procedures. There is a clear absence of a unified and integrated system that consolidates these functions into a single real-time platform capable of automatically computing, analyzing, and presenting production performance indicators in a cohesive manner.

Third, many of the existing advanced solutions in production monitoring rely heavily on high-cost and technically complex technologies, including IoT-based sensor networks, machine learning algorithms, cloud computing infrastructures, and industrial automation systems. While these technologies are highly effective in large-scale manufacturing environments, they are often impractical for small and medium-sized enterprises (SMEs) due to financial constraints, lack of technical expertise, and limited infrastructure support. This creates a technological gap where SMEs in the food processing sector are unable to fully benefit from advanced production monitoring innovations.

Fourth, although local studies in the Philippines confirm the advantages of digital transformation in agriculture and food manufacturing, there is still a lack of localized, real-time production monitoring systems specifically developed for juice extraction industries. Most available systems or studies focus on agricultural production, MSME business processes, or general ICT adoption rather than actual operational monitoring within food processing plants. This gap is particularly evident in companies such as AlDi Foods Inc., where production tracking is still largely dependent on manual or semi-digital recording systems.

Fifth, while modern production environments generate large volumes of operational data, there remains a significant gap in transforming raw production data into real-time, actionable insights. Many systems are capable of collecting data but lack intelligent mechanisms for immediate interpretation, such as automated yield computation, extraction efficiency evaluation, and instant performance feedback. This limitation reduces the usefulness of collected data and prevents production managers from making timely, data-driven decisions during ongoing operations.

Lastly, there is still limited research addressing the systematic transition from traditional manual recording systems to fully digital, real-time production monitoring systems, particularly in food processing industries. Many facilities continue to rely on paper-based logs, spreadsheets, or semi-manual recording methods, which are prone to human error, data loss, and delayed reporting. The literature lacks practical frameworks and implemented systems that guide

SMEs in shifting toward fully integrated digital production monitoring environments without requiring high-cost industrial infrastructure.

Overall, these gaps highlight the need for a cost-effective, integrated, and real-time production monitoring system that is specifically designed for juice extraction processes. The proposed study, *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."*, directly addresses these gaps by developing a centralized system that enables real-time tracking of production input, output, yield computation, and extraction efficiency analysis, thereby supporting improved operational efficiency and data-driven decision-making in food processing operations.

## How the Current Project Addresses the Gaps

The proposed study, "Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.", directly addresses the research gaps identified in the reviewed literature through the development of a centralized, digital, and real-time production monitoring system specifically designed for juice extraction operations. The system aims to replace traditional paper-based production recording with a more efficient digital platform that enables accurate data collection, automated computation, and timely access to production information. By providing production managers with organized and up-to-date operational data, the system supports more effective production management and data-driven decision-making.

First, the proposed system specifically addresses the lack of real-time monitoring solutions developed for juice extraction and food processing operations. While many existing studies focus on general manufacturing automation, agricultural information systems, or industrial monitoring applications, few concentrate on the unique production requirements of juice extraction. The proposed system is designed to monitor critical production variables such as raw material input, extracted juice output, packed yield, yield percentage, and extraction efficiency. These indicators are essential in evaluating production performance and identifying areas where operational improvements can be made. By focusing on extraction-specific performance measures, the study contributes practical knowledge to food processing research while addressing the operational needs of AlDi Foods Inc.

Second, the proposed system addresses the fragmented nature of existing production monitoring approaches by integrating multiple production management functions into a single centralized platform. Instead of maintaining separate records for production monitoring, yield computation, efficiency analysis, and report generation, the system consolidates these processes into one digital environment. Production data entered into the system are automatically processed to generate yield percentages, extraction efficiency values, and production summaries. This integration reduces redundant work, minimizes inconsistencies caused by manual computation, and provides production managers with a comprehensive view of operational performance through a single interface.

Third, the proposed study offers a cost-effective and practical digital solution that is suitable for small and medium-sized food manufacturing enterprises. Unlike many advanced production monitoring systems discussed in the literature that require IoT devices, sophisticated sensors, machine learning algorithms, or expensive industrial automation equipment, the proposed system emphasizes centralized data management, automated calculations, and real-time reporting using readily available digital technologies. This approach makes the system more feasible for organizations with limited financial and technical resources while still providing significant improvements in production monitoring and operational efficiency.

Fourth, the proposed system improves the utilization of production data by transforming manually recorded operational information into real-time analytical insights. Rather than simply storing production records for documentation purposes, the system automatically computes key production indicators and presents them through organized reports and dashboards. This allows production managers to monitor production progress continuously, identify abnormal production trends, detect extraction losses, and evaluate operational performance while production activities are ongoing. As a result, corrective actions can be implemented more quickly, reducing delays associated with end-of-day or end-of-batch manual reporting.

Fifth, the study contributes to the digital transformation of food manufacturing operations within the Philippine context. Although previous local studies have demonstrated the importance of digital technologies in agriculture and manufacturing, relatively few have developed practical production monitoring systems specifically intended for food processing companies. By implementing a real-time yield analytics and extraction efficiency monitoring system at AlDi Foods Inc., the study provides a localized example of how digital technologies can improve production management, support operational transparency, and enhance overall manufacturing efficiency. The proposed system may also serve as a reference model for other small and medium-sized food processing enterprises that seek to modernize their production monitoring practices.

Lastly, the proposed system converts raw production data into meaningful and actionable information that supports continuous process improvement. Through automated computation of yield percentages, extraction efficiency, raw material utilization, and production output, the system enables managers to evaluate production performance objectively using measurable indicators. These real-time analytics facilitate evidence-based decision-making, improve resource utilization, reduce production losses, and enhance overall operational efficiency. By providing timely and accurate production information, the system supports continuous monitoring and process optimization, ultimately contributing to improved productivity, better product consistency, and more effective management of juice extraction operations at AlDi Foods Inc.

Overall, the proposed study responds directly to the limitations identified in the reviewed literature by developing an integrated, practical, and real-time production monitoring system specifically tailored to the operational requirements of juice extraction. Through digitalization, automated yield analytics, extraction efficiency computation, and centralized reporting, the system supports the modernization of production management practices while promoting greater operational efficiency, improved resource utilization, and informed decision-making in food processing environments.

**A study [21] by Mphahlele, Fawole, Mokwena, and Opara (2016)** investigated the influence of different extraction methods on the chemical composition, bioactive compounds, volatile compounds, antioxidant properties, and overall quality of pomegranate (*Punica granatum* L. cv. Wonderful) juice. The researchers emphasized that the quality of fruit juice is not determined solely by the characteristics of the fruit itself but is also greatly affected by the method used during juice extraction. Since consumers increasingly prefer fruit juices that retain their natural nutritional value and health-promoting compounds, identifying extraction methods that maximize juice yield while preserving important chemical and nutritional properties has become an important area of research. The study was conducted to determine how different extraction techniques influence the final composition and quality of pomegranate juice and to provide scientific evidence that could assist food processors in selecting the most appropriate extraction method.

To achieve this objective, the researchers collected commercially harvested pomegranate fruits of the cultivar 'Wonderful' from South Africa and processed them using four different extraction methods. These methods included extracting juice from the arils without crushing the seeds, blending the arils together with the seeds, pressing the whole fruit, and extracting juice from halved fruits using a commercial hand press. The resulting juice samples were analyzed for several quality parameters, including total soluble solids, titratable acidity, pH, juice yield, sugar composition, organic acid content, phenolic compounds, flavonoids, anthocyanins, antioxidant activity, and volatile organic compounds responsible for aroma. Statistical analyses, including analysis of variance and principal component analysis, were also performed to determine whether significant differences existed among the extraction methods and to identify which technique produced the most desirable juice characteristics.

The results demonstrated that the extraction method significantly influenced many of the physicochemical properties of pomegranate juice. Although the total soluble solids remained relatively similar across all extraction methods, noticeable differences were observed in acidity, pH, juice yield, and color. Juice extracted from halved fruits exhibited higher titratable acidity and lower pH, indicating a more acidic product. The researchers explained that lower pH levels may contribute to better microbial stability because acidic environments inhibit the growth of many

spoilage microorganisms. Differences were also observed in juice yield, with extraction methods producing varying amounts of juice depending on how effectively the fruit tissues were disrupted during processing. Juice extracted from halved fruits produced the highest juice yield, while pressing whole fruits resulted in the lowest yield because some juice remained trapped inside the fruit tissues during extraction. These findings demonstrated that extraction efficiency is directly affected by the extraction technique employed during processing.

The researchers also examined the sugar and organic acid composition of the juice produced by each extraction method. Glucose and fructose were identified as the major sugars present in all samples, while citric acid was found to be the predominant organic acid. Although sugar concentrations showed only minor differences among the extraction methods, variations in organic acid content were more pronounced. Juice obtained from the arils contained relatively higher citric acid levels, whereas blending the arils with the seeds produced lower concentrations because the seed components diluted the juice composition. The study emphasized that organic acids play an important role in determining the taste, flavor, and stability of fruit juice, making their accurate evaluation essential for maintaining product quality.

In addition to evaluating basic chemical properties, the researchers investigated the effects of extraction methods on bioactive compounds that contribute to the nutritional value of pomegranate juice. Several flavonoids, including catechin, epicatechin, and rutin, together with gallic acid, were identified in all juice samples. Catechin was the most abundant flavonoid regardless of the extraction method, while gallic acid was the predominant phenolic acid. However, the concentration of these compounds varied depending on the extraction technique. Blending the arils together with the seeds produced higher catechin concentrations because phenolic compounds were also extracted from the seeds. On the other hand, juice extracted from halved fruits contained higher concentrations of several beneficial phenolic compounds because the pressing process allowed compounds from the peel and internal membranes to become incorporated into the juice. These findings demonstrate that the fruit fractions included during extraction significantly influence the nutritional composition of the final product.

The study further analyzed anthocyanins, which are natural pigments responsible for the characteristic red color of pomegranate juice. Eight individual anthocyanin compounds were identified, with cyanidin-3,5-diglucoside representing the dominant pigment in all juice samples. Although the same anthocyanin compounds were detected across all extraction methods, their concentrations differed depending on the extraction technique. Juice extracted from whole fruits exhibited relatively higher total anthocyanin content than several of the other extraction methods. According to the researchers, extraction conditions such as acidity and pressing pressure may influence anthocyanin stability and concentration, ultimately affecting the visual appearance and consumer acceptability of the juice.

The antioxidant properties of the juice were also evaluated using radical scavenging activity and ferric reducing antioxidant power assays. The findings revealed that juice extracted from halved fruits consistently exhibited the highest antioxidant activity among all extraction methods. This increase was closely associated with higher concentrations of total phenolic compounds, tannins, and vitamin C present in the juice. Statistical analysis showed a positive relationship between phenolic content and antioxidant activity, indicating that phenolic compounds are major contributors to the antioxidant capacity of pomegranate juice. These results suggest that extraction methods capable of incorporating beneficial compounds from additional fruit tissues may improve the nutritional value and functional properties of the final product.

Furthermore, the researchers investigated the volatile organic compounds responsible for the aroma and flavor of pomegranate juice. Ten volatile compounds belonging to different chemical groups, including esters, ketones, alcohols, and terpenes, were identified. Ethyl acetate and 3-octanone were the dominant volatile compounds detected in all juice samples. However, their relative concentrations differed according to the extraction method, demonstrating that processing conditions influence aroma development as well as chemical composition. Juice produced by blending arils together with seeds contained the highest overall concentration of volatile compounds, suggesting that including additional fruit tissues during extraction may enhance aroma intensity. Since aroma is one

of the primary quality attributes evaluated by consumers, these findings further highlight the importance of selecting appropriate extraction techniques during juice processing.

To better understand the relationships among the measured quality parameters, the researchers performed principal component analysis. The statistical analysis successfully distinguished the juice samples according to their extraction methods based on differences in chemical composition, antioxidant activity, bioactive compounds, sugars, organic acids, and volatile compounds. This finding confirmed that each extraction technique produced a unique quality profile. The researchers concluded that extraction methods influence multiple quality characteristics simultaneously rather than affecting only a single property, emphasizing the importance of considering overall product quality when selecting processing techniques.

Overall, the study concluded that extraction method is a critical factor influencing juice yield, chemical composition, nutritional value, antioxidant capacity, aroma, and overall product quality. The findings demonstrated that selecting an appropriate extraction technique can improve production efficiency while preserving valuable bioactive compounds that contribute to the health benefits and sensory characteristics of pomegranate juice. The study also showed that monitoring extraction conditions allows food manufacturers to optimize both product quality and processing performance.

This literature is relevant to the present study because it demonstrates that extraction methods directly affect production yield, extraction efficiency, and product quality. The findings support the importance of continuously monitoring production parameters to identify variations that influence output and resource utilization. Likewise, the proposed system, *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.*, aims to provide real-time monitoring and analysis of production yield and extraction efficiency, enabling manufacturers to evaluate processing performance, minimize production losses, improve operational efficiency, and make data-driven decisions that enhance overall productivity and product quality.

A study **[22]** by **Aviara, Lawal, Nyam, and Bamisaye (2020)** focused on the design, construction, and performance evaluation of a multi-fruit juice extractor intended for small- and medium-scale fruit processing operations. The researchers developed the machine to overcome the limitations of traditional juice extraction methods, which are often labor-intensive, time-consuming, unhygienic, and inefficient. The extractor was designed to operate using the combined principles of compression and shear force through a screw conveyor mechanism that gradually squeezes juice from fruits as they move through the extraction chamber. The machine was constructed using locally available materials to ensure affordability, ease of maintenance, and suitability for local fruit juice processors.

To evaluate the performance of the machine, the researchers conducted experimental tests using pineapple, orange, and watermelon in both peeled and unpeeled conditions. The study measured three primary performance indicators: percentage juice yield, extraction efficiency, and extraction loss. Each fruit sample underwent repeated testing to ensure the reliability of the results, and the collected data were analyzed using Analysis of Variance (ANOVA) to determine the effects of fruit type and peel condition on machine performance. The researchers found that both the type of fruit and whether it was peeled significantly influenced the juice extraction results, while variations in feed rate had little effect on overall machine performance.

The findings revealed that peeled fruits generally produced higher juice yields and extraction efficiencies than unpeeled fruits because the removal of the peel reduced resistance during the extraction process. Watermelon recorded the highest juice yield at approximately 89.5% for peeled samples, while peeled pineapple achieved the highest extraction efficiency of about 96.9%. The study also found relatively low extraction losses across all tested fruits, indicating that the machine effectively recovered a large proportion of the available juice. These results demonstrated that the extractor could consistently produce high-quality output while minimizing waste during juice processing.

Furthermore, the researchers concluded that the machine was simple to operate, economically feasible, and capable of supporting small-scale fruit juice production using locally sourced materials. Its efficient design provides an alternative to conventional manual extraction methods by improving productivity, reducing processing time, and increasing juice recovery. The study emphasized that the machine can contribute to reducing post-harvest losses of tropical fruits while promoting value-added processing within local communities.

This literature is relevant to the present study because it highlights the importance of evaluating production performance through measurable indicators such as juice yield, extraction efficiency, and extraction loss. These performance metrics demonstrate how production processes can be monitored to identify inefficiencies and improve operational outcomes. Likewise, the proposed IoT-based production management system aims to digitally record production data, monitor raw material utilization and packed yield, and generate real-time production reports. By providing accurate and organized production information, the system can assist manufacturers in analyzing production efficiency, minimizing losses, and supporting data-driven decision-making throughout the juice production process.

A study [23] by **Dora et al. (2020)** examined the role of sustainable operations in minimizing food loss within the Belgian food processing industry. The researchers recognized that although food waste at the consumer level has been extensively studied, there is limited research focusing on food losses that occur during industrial food processing. To address this gap, they investigated food loss across different production stages in food processing companies to determine where losses most frequently occur and what operational factors contribute to them. The study aimed to identify food loss hotspots and evaluate how sustainable operational practices can help reduce waste and improve production efficiency.

To conduct the research, the authors employed a multiple case study approach involving 47 food processing companies from various sectors in Belgium, including fruit and vegetable processing, dairy, beverages, bakery, chocolate, grain milling, sauces, ready meals, and meat processing. Primary data were collected through a semi-structured questionnaire, company interviews, on-site observations, and documentary analysis. The questionnaire gathered information about company operations, production capacity, food loss at different stages of processing, and the perceived causes of these losses. The collected quantitative data were analyzed using Microsoft Excel and SPSS statistical software, while qualitative observations were used to support and validate the findings.

The findings revealed that food loss accounted for an average of 2.1% of total production among the participating companies. Among the three production stages examined—before production, during production, and after production—the production stage, particularly the processing phase, was identified as the most significant hotspot for food loss. Statistical analysis showed that increases in operational problems during processing were directly associated with higher food loss. The researchers identified several major causes of food loss, including human errors, interrupted production, transportation problems, machine inefficiencies, product changes during production, product defects, packaging errors, poor inventory management, and buyer contract issues. Companies involved in fruit and vegetable processing experienced the highest percentage of food loss, while chocolate and ready-meal manufacturers incurred the greatest financial losses. The study also found that smaller food processing companies generally experienced higher levels of food loss than larger companies.

Based on these findings, the researchers emphasized that sustainable operational practices play a critical role in reducing food loss throughout food processing operations. They recommended improving production planning, inventory management, equipment maintenance, employee training, and quality control procedures to minimize operational inefficiencies. The study also highlighted the importance of implementing management approaches such as Lean Manufacturing, Lean Six Sigma, effective scheduling, and the use of Key Performance Indicators (KPIs) to continuously monitor and reduce food loss. Furthermore, the researchers stressed that systematic measurement of food loss enables companies to identify problem areas, develop targeted improvement strategies, and enhance overall production efficiency. These findings demonstrate that effective operational management not only reduces food waste but also improves productivity, sustainability, and cost efficiency within the food processing industry.

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# CHAPTER III

# METHODOLOGY

## 3.1 Research Design

This study employs a Developmental Research Design as the primary research methodology for the planning, analysis, design, development, testing, implementation, and evaluation of the system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* Developmental research is appropriate for this study because it focuses on the systematic development of a functional technological solution in response to an identified organizational problem. In the context of information technology, this approach allows the researchers to translate actual operational requirements into a web-based system that can be tested and evaluated based on its functionality, usability, accuracy, efficiency, reliability, and user satisfaction.

The research design is directly aligned with the primary problem identified in Chapter 1, wherein production-related information at AlDi Foods Inc. is traditionally recorded through manual processes such as logbooks, paper-based records, spreadsheets, and other manually maintained documents. Production personnel record important information, including raw material input, extraction output, packed yield, and other batch-related information. Although this method allows production activities to be documented, it may result in inconsistent records, human recording errors, delayed information availability, repetitive manual calculations, difficulty in retrieving historical information, and limitations in generating consolidated production reports. These limitations can make it more difficult for production personnel, supervisors, and management to monitor production performance efficiently.

The Developmental Research Design therefore provides a structured approach for developing a digital alternative to the existing manual production monitoring process. Rather than focusing only on describing the existing problem, the study aims to produce and evaluate an actual working system that addresses the identified information-management and monitoring requirements. The proposed system centralizes production information in a web-based platform where authorized users can encode, organize, process, retrieve, and analyze production data within a structured database.

The development of the system begins with problem identification and requirements analysis. The researchers examine the existing production workflow of AlDi Foods Inc. to understand how production information is currently recorded, processed, stored, and reported. Data-gathering activities include interviews with management personnel, production supervisors, and production personnel; direct observation of relevant production procedures; and review of available production records, logbooks, forms, spreadsheets, daily reports, raw material records, extraction records, packed yield records, and other related documents. These activities provide the researchers with information regarding the actual workflow, user responsibilities, data requirements, existing difficulties, and information needed for production monitoring.

The information gathered during this stage serves as the basis for defining the functional and non-functional requirements of the proposed system. Functional requirements describe the operations that the system must perform, including user authentication, user access control, production batch management, raw material recording, extraction output recording, packed yield recording, yield calculation, extraction efficiency calculation, production stability classification, dashboard presentation, historical record retrieval, and automated report generation. Non-functional requirements focus on characteristics such as usability, reliability, accuracy, efficiency, security, accessibility, and responsiveness of the system.

The study also applies an Agile Development Model within the broader Developmental Research Design. Agile development is appropriate because the proposed system requires continuous refinement based on user requirements and feedback. Instead of completing the entire system in a single development cycle, the researchers organize the development process into manageable iterations or sprints. Each iteration focuses on selected system requirements, followed by development, testing, evaluation, debugging, and refinement. Feedback from intended users and project stakeholders is considered throughout the development process to ensure that the system remains aligned with the operational requirements of AlDi Foods Inc.

During the system analysis and design phase, the researchers translate the gathered requirements into technical and functional specifications. Process flows, data structures, database relationships, system architecture, use-case diagrams, data-flow diagrams, entity-relationship diagrams, and user-interface wireframes are prepared to describe how the system will operate. These design artifacts help establish a clear relationship between production activities and the corresponding digital functions of the system. The design also considers the different responsibilities of authorized users, production personnel, supervisors, and administrators.

The proposed system follows a three-tier architecture consisting of the client layer, application layer, and data layer. The client layer provides the interface through which users interact with the system. The application layer processes user requests, performs calculations, applies system rules, and manages system functions. The data layer stores production and user information in a centralized database. This architecture provides an organized structure for separating the user interface, system processing, and data storage components of the application.

The development phase involves the implementation of the approved system design using appropriate web development technologies. The system is developed using HTML, CSS, and JavaScript for the user interface and client-side interaction, PHP for server-side processing and application logic, MySQL for database management, and Chart.js for graphical presentation of production analytics. The development and testing environment uses XAMPP to provide the necessary local web server and database environment. These technologies are selected based on their suitability for developing a practical, maintainable, and cost-effective web-based production monitoring prototype.

The developed system consists of several interconnected modules that correspond to the objectives identified in Chapter 1. The User Authentication and Access Control module manages user accounts and restricts system functions according to authorized access. The Production Batch Management module organizes production activities according to individual batches. The Raw Material Management and Recording module records the quantity and relevant information regarding raw materials used in production. The Extraction Output module records the quantity of product obtained during the extraction process, while the Packed Yield module records the final usable or packed production output.

The system also incorporates automated production calculations to reduce repetitive manual computation. Based on the production data encoded by authorized users, the system calculates yield percentage and extraction efficiency according to the formulas and rules established during the system design. These calculations allow production personnel and supervisors to obtain production indicators without repeatedly performing the calculations manually. The system may also present relevant production loss or variation indicators derived from the recorded values, but these indicators are intended to support monitoring rather than automatically diagnose the underlying causes of production losses.

Another major component is the Production Stability Monitoring function. The system evaluates selected production indicators against predefined thresholds or criteria established for the study and the intended organizational process. Based on these criteria, the system provides a production stability status, such as “Stable” or “Not Stable.” This function is intended to help users identify production records that may require further review. The system does not independently determine the physical or mechanical cause of an unstable production result; instead, it provides a digital indicator based on the encoded production information and established rules.

The proposed system also provides a centralized dashboard and analytics interface. The dashboard presents relevant production information in a summarized and graphical format to make the information easier to interpret. Graphs, tables, indicators, and other visual elements may be used to present yield performance, extraction efficiency, raw material utilization, production stability, and other relevant production information. By consolidating these indicators in one interface, the system reduces the need for users to manually examine multiple records to understand production performance.

The term real-time in this study refers specifically to the immediate processing, updating, and availability of information after production data has been manually entered into the system. When an authorized user records a production batch, raw material quantity, extraction output, or packed yield, the system processes the information and updates the corresponding calculations, dashboard information, and records without requiring a separate manual computation or delayed consolidation. Therefore, real-time monitoring in this study does not mean automatic data collection from industrial machines, sensors, IoT devices, or IIoT infrastructure. The system remains primarily dependent on manually encoded production data provided by authorized users.

The system further provides a historical records management function that allows authorized users to retrieve previously recorded production information. Instead of depending exclusively on physical logbooks or separately maintained spreadsheets, users can search and review stored production records through the system. Historical information may be used to compare production batches, review previous yield performance, examine extraction efficiency, monitor raw material utilization, and identify changes in production performance across different periods.

To support management and operational monitoring, the system includes an automated reporting function. Production information can be summarized according to different reporting periods, including weekly, monthly, quarterly, and yearly reports. These reports may contain information regarding raw material utilization, extraction performance, yield results, and other relevant production indicators. The reporting function is intended to reduce the time required to manually consolidate production information and provide management with organized information for review and operational decision-making.

Data organization and storage are supported through a centralized MySQL database. The database is designed to maintain structured information related to users, production batches, raw materials, extraction records, packed yields, calculations, and historical production information. Proper relationships among the relevant database entities are established during the design process to maintain data consistency and support efficient retrieval. User authentication and access-control mechanisms are also incorporated to help protect production information and restrict system functions to authorized personnel.

The development process includes iterative testing, validation, debugging, and refinement. Each developed module is tested to determine whether it performs according to the identified requirements. Errors or inconsistencies discovered during testing are documented and corrected before the system proceeds to subsequent development iterations. User feedback is also considered during refinement to determine whether system functions, interface layouts, terminology, workflows, and outputs are understandable and appropriate for the intended users.

The researchers conduct functional testing to determine whether the system's individual features operate according to their specified requirements. This includes testing user authentication, production batch creation, raw material

recording, extraction output recording, packed yield recording, automated calculations, production stability classification, dashboard updates, historical record retrieval, and report generation. The researchers also verify whether entered data is properly processed, stored, retrieved, and displayed by the system.

The developed system is subsequently subjected to user evaluation and acceptance assessment. Intended users and appropriate company representatives are involved in evaluating the system based on identified quality criteria. The evaluation considers functionality, usability, accuracy, efficiency, reliability, and user satisfaction. These criteria provide a structured basis for determining whether the developed system is capable of addressing the identified production monitoring requirements and whether it is acceptable and practical for its intended use.

The functionality criterion determines whether the system performs the required functions correctly and completely. Usability examines whether users can understand and operate the system with reasonable ease. Accuracy evaluates whether production information and system-generated calculations correspond to the encoded data and defined formulas. Efficiency considers the system's ability to process and present production information without unnecessary steps or delays. Reliability considers the consistency of system operations and outputs during testing. User satisfaction considers the users' overall assessment of the system based on their experience using its features and outputs.

The research design is also consistent with the scope and delimitations established in Chapter 1. The system focuses on production-related monitoring rather than attempting to replace every business function of AlDi Foods Inc. It does not cover sales, accounting, payroll, human resources, purchasing, customer relationship management, or comprehensive enterprise resource planning. It also does not include full industrial automation, robotics, machine control, predictive maintenance, advanced machine learning, or sophisticated IoT/IIoT infrastructure. The proposed system is primarily a web-based digital production monitoring platform that processes information manually encoded by authorized users.

The methodology also recognizes that the developed system is a prototype designed for the specific operational context of AlDi Foods Inc. Its features, calculations, production indicators, and reporting requirements are based on the information gathered from the organization and the defined objectives of the study. Therefore, the results of the evaluation primarily reflect the system's suitability within the study's intended environment and should not automatically be generalized to all food manufacturing organizations or production facilities with different processes and requirements.

Overall, the application of the Developmental Research Design, supported by Agile development practices, enables the researchers to systematically transform the identified manual production monitoring problems into a functional digital solution. The methodology connects the problem identification in Chapter 1, the related studies and concepts discussed in Chapter 2, the system design and development activities in Chapter 3, the implemented modules and system evaluation presented in Chapter 4, and the conclusions and recommendations discussed in Chapter 5. Through this approach, the study develops a centralized and user-oriented production monitoring system that provides immediate access to manually encoded production information, automated yield and extraction efficiency calculations, production stability indicators, dashboard analytics, historical records, and periodic reports. The resulting system is intended to improve the organization, accessibility, consistency, and timeliness of production information while providing a practical and cost-effective digital approach to production monitoring at AlDi Foods Inc.

## 3.2 System Development Methodology

The study employed the Agile Development Methodology as the primary approach for developing the proposed web-based production monitoring system for AlDi Foods Inc. Agile was selected because it supported an iterative and organized development process in which the system could be developed, tested, reviewed, and improved throughout the different stages of development. The methodology was appropriate for the project because the proposed system consisted of several interconnected modules, including user authentication, production batch

management, raw material recording, extraction monitoring, packed yield monitoring, yield analytics, extraction efficiency monitoring, production stability interpretation, historical data management, dashboard visualization, and automated report generation.

The development process was organized into five major phases: Requirement Analysis, System Design, Development, Testing, and Deployment (Prototype). Each phase contributed to the gradual development of the proposed system and was connected to the objectives, scope, and requirements identified in the study. The iterative nature of Agile also allowed the researchers to review the functionality of the system and make necessary adjustments based on testing results and feedback from the intended users.

## Requirement Analysis

The Requirement Analysis phase focused on identifying and documenting the operational requirements of AlDi Foods Inc. that were relevant to the proposed production monitoring system. The researchers gathered information through interviews with management, production supervisors, and production personnel, as well as through direct observation of existing production processes. Existing production documents, including logbooks, forms, spreadsheets, daily reports, raw material records, extraction output records, and packed yield records, were also considered in determining the information that needed to be recorded and monitored by the system.

During this phase, the researchers examined how production information was currently recorded, calculated, stored, and retrieved. Particular attention was given to the recording of raw material quantities, extraction outputs, and packed yields because these data were necessary for determining production yield and extraction efficiency. The analysis also considered the difficulties associated with manual recording, such as possible encoding errors, inconsistent entries, delayed calculations, difficulty in retrieving previous records, and limited visibility of production performance.

The gathered information was then used to identify the functional requirements of the proposed system. These requirements included user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, automated yield calculation, extraction efficiency monitoring, production stability interpretation, historical record management, dashboard visualization, and automated report generation. The system was designed to allow authorized users to encode production information and immediately process the entered data for monitoring and analysis.

The non-functional requirements were also considered during this phase. These included usability, accuracy, reliability, efficiency, accessibility, and appropriate security of production records. The system was intended to provide a user-friendly interface that production personnel could use without requiring advanced technical knowledge. The requirements identified during this phase served as the basis for the succeeding system design and development activities.

## System Design

The System Design phase translated the requirements identified during the Requirement Analysis phase into the structure and design of the proposed system. This phase focused on designing the system architecture, database structure, user interface, process flow, and interaction between the different system components. The design ensured that the proposed system could properly handle production information from data entry to processing, visualization, historical storage, and report generation.

The proposed system followed a three-tier architecture consisting of the client layer, application layer, and data layer. The client layer represented the user interface through which authorized users interacted with the system. The application layer handled the processing of production information, calculations, validation, and system functions. The data layer was responsible for storing and retrieving production records through the MySQL database.

The database design was developed to organize production information systematically. Relevant entities and relationships were considered to ensure that production batches, raw material records, extraction outputs, packed yields, user accounts, and related historical information could be stored and retrieved properly. The database design also supported the generation of reports and analytics based on recorded production data.

The user interface was designed according to the functions required by the intended users. The interface included pages and modules for user registration and authentication, dashboard monitoring, raw material management, raw material extraction, packed yields, user profile management, administrative functions, batch history, and analytics. The design aimed to provide clear navigation and organized presentation of production information.

System process diagrams, data flow representations, entity-relationship diagrams, use case diagrams, and interface wireframes were also considered during this phase. These design components provided a visual representation of how information would move through the system and how users would interact with the different modules. Automated workflows were incorporated where appropriate, particularly for calculations and the presentation of production analytics.

## Development

The Development phase involved the actual implementation of the system based on the approved requirements and system design. The researchers developed the proposed web-based production monitoring system using HTML, CSS, JavaScript, PHP, MySQL, and Chart.js. These technologies were used to create the user interface, implement system functions, process production data, store records, and present production information through graphical visualizations.

The development process was divided into interconnected modules. The User Authentication and Access Management module was developed to control access to the system and ensure that only authorized users could access appropriate system functions. The Production Management module was developed to organize production batches and their associated production information.

The Raw Material Recording module was developed to allow users to encode the quantity of raw materials used in a production batch. The Extraction Monitoring module was developed to record the extraction output generated during production. The Packed Yield Monitoring module was developed to record the final packed output of the production process.

The system also included Real-Time Yield Analytics, which processed newly entered production data and presented the resulting information without requiring users to perform the calculations manually. In the context of this study, real-time refers to the immediate processing, updating, and display of information after authorized users encode production data. It does not refer to automatic data collection from IoT devices or industrial sensors.

The Extraction Efficiency Monitoring module was developed to support the analysis of extraction performance based on the production data recorded in the system. The Production Stability Interpretation module provided a classification of production performance according to predefined thresholds or criteria established for the study. This allowed the system to indicate whether a recorded production condition was considered stable or not stable based on the defined parameters.

The Historical Data Management module was developed to provide access to previously recorded production batches and related production information. This allowed authorized users to retrieve and review historical records without relying solely on physical logbooks or manually maintained files.

The Dashboard and Data Visualization module was implemented using graphical representations to provide users with a summarized view of production information. Charts and other visual elements were used to present relevant production indicators, including yield and extraction-related information. The Automated Report Generation module

was also developed to organize production information into reports that could support periodic monitoring and review, including weekly, monthly, quarterly, and yearly production summaries.

Throughout development, the modules were implemented incrementally so that individual functions could be checked before being integrated into the complete system. This approach allowed the researchers to identify issues in specific functions and make adjustments before proceeding to the next development activities.

## Testing

The Testing phase was conducted to determine whether the developed system performed according to the identified requirements and specifications. Testing was performed on individual functions as well as on the complete system to identify errors, inconsistencies, calculation problems, interface issues, and integration problems before the prototype was presented for user evaluation.

Unit testing was used to examine individual system functions and modules. Functions such as user authentication, production batch creation, raw material recording, extraction output entry, packed yield entry, automated calculations, and report generation were checked individually to determine whether they produced the expected results.

Integration testing was then conducted to determine whether the different modules worked correctly when connected. For example, the researchers checked whether raw material and production batch information could be properly associated with extraction output and packed yield records. The researchers also examined whether recorded data were correctly processed and reflected in the dashboard, historical records, analytics, and reports.

System testing was conducted on the complete system to evaluate its overall functionality and operation. The researchers verified whether the system could properly support the complete process from user login and production data entry to calculation, visualization, record retrieval, and report generation.

The accuracy of the system's calculations was also checked. Yield calculations, extraction efficiency calculations, and production stability classifications were compared with expected results based on the defined formulas and criteria. The dashboard and historical records were also checked to ensure that the information displayed corresponded to the encoded production records.

The system was further evaluated according to the identified criteria, including functionality, usability, accuracy, efficiency, reliability, and user satisfaction/effectiveness. These criteria were used to determine whether the prototype was capable of supporting the intended production monitoring activities.

User acceptance testing was included as part of the testing process. The intended users, such as production personnel, supervisors, and other appropriate users identified by the organization, were given an opportunity to interact with the prototype and provide feedback regarding its functions, usability, and relevance to their production monitoring activities. The feedback obtained during testing was used to identify areas requiring correction or improvement before prototype deployment.

## Deployment (Prototype)

The Deployment (Prototype) phase involved making the developed system available for controlled use and evaluation within the intended environment of the study. Since the project was developed as a prototype specifically for AlDi Foods Inc., deployment focused on demonstrating and evaluating the system's capabilities rather than implementing a full-scale enterprise production system.

The prototype was prepared for use by authorized users involved in production monitoring and management. Users could access the system to encode production information, including raw material quantities, extraction outputs, and

packed yields. Once the information was entered, the system processed the data and presented the corresponding production indicators, analytics, historical records, and reports.

During prototype deployment, the researchers observed how the intended users interacted with the system and gathered feedback regarding its functionality, usability, accuracy, and relevance to their existing production monitoring activities. Any identified issues or difficulties were documented for possible refinement of the prototype.

The deployment phase also served as the basis for determining whether the proposed system addressed the problems identified during the initial assessment of the manual production recording process. In particular, the prototype was examined in terms of its ability to organize production information in a centralized database, reduce dependence on manual calculations, provide immediate access to newly entered production information, support historical record retrieval, and present production performance through analytics and reports.

The deployment was limited to the scope of the proposed prototype and did not involve full industrial automation, IoT-based production monitoring, robotics, machine control, or large-scale enterprise implementation. The system primarily relied on manually encoded production data from authorized users. Future improvements may be considered based on user feedback and the results of the evaluation conducted during the study.

Overall, the five Agile phases provided a structured process for developing the proposed production monitoring system. Requirement Analysis established the necessary system requirements, System Design translated these requirements into the system architecture and interface, Development implemented the proposed functions, Testing verified the functionality and performance of the system, and Deployment (Prototype) allowed the developed system to be demonstrated and evaluated in a controlled setting. This process ensured that the development of the proposed system remained aligned with the production monitoring needs and objectives identified throughout the study.

## 3.3 System Architecture

The proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* adopts a three-tier system architecture composed of the Client Layer, Application Layer, and Database Layer. The architecture provides a structured framework for organizing the different components and responsibilities of the proposed web-based production monitoring platform. Each layer performs a specific function while maintaining communication with the other layers to support the recording, processing, storage, analysis, and presentation of production information.

The three-tier architecture was selected because it is appropriate for the proposed system's objective of digitizing the production recording process of AlDi Foods Inc. The existing production process involves recording information such as raw material usage, extraction output, and packed yield through manual records. The proposed architecture transforms these activities into an organized digital workflow where production information can be entered through a web-based interface, processed automatically by the application, and stored in a centralized database.

The separation of the system into three layers also improves the organization and maintainability of the proposed platform. The Client Layer focuses on user interaction and presentation, the Application Layer manages system processing and business rules, while the Database Layer manages the storage and retrieval of production information. This separation allows each component to perform its intended responsibility without unnecessarily combining user-interface operations, computations, and data storage within a single component.

Furthermore, the architecture supports the major functions identified in the study, including production batch recording, raw material monitoring, extraction output recording, packed yield recording, yield percentage calculation, extraction efficiency monitoring, production stability classification, dashboard visualization, historical record management, and automated report generation. Through the interaction of the three layers, production information

entered by authorized users can be transformed into useful analytical information for production personnel, supervisors, and management.

## Client Layer

The Client Layer serves as the presentation layer and provides the primary point of interaction between users and the proposed production monitoring system. It contains the web-based interface through which authorized users access the system, enter production information, view processed results, monitor production performance, retrieve historical records, and generate reports. The interface is designed to provide a simple and organized environment so that production personnel can perform their responsibilities without requiring advanced technical knowledge.

The primary users of the Client Layer include production personnel, supervisors, management personnel, and system administrators. Each user is required to log in using valid credentials before accessing the functions assigned to their role. Role-based access ensures that users can only access functions relevant to their responsibilities. For example, production personnel can perform production-related data entry, while authorized supervisors or administrators can access monitoring, historical records, analytics, and administrative functions.

Through the Client Layer, production personnel can digitally record the information required during production operations. These records may include production batch information, production date, raw material quantities, extraction output, packed yield, and other production-related details required by the system. This process replaces repetitive manual recording activities with structured electronic data entry.

The use of standardized input forms also helps improve the consistency of production records. Required fields, input restrictions, and validation messages can guide users when entering information. These interface features help reduce incomplete records, incorrect data formats, and unnecessary repetition during data encoding. Instead of maintaining information across separate logbooks or spreadsheets, users can enter production information directly into the centralized platform.

The Client Layer also provides access to the Production Batch Management Module, where users can create, view, and manage production batch information. Each batch serves as an organized record that connects the raw material inputs, extraction results, packed yield, computed performance indicators, and other relevant production information. This allows related production information to remain associated with the appropriate production activity.

The Raw Material Recording Module enables users to encode the quantity of raw materials used for a production batch. The recorded information becomes an important basis for determining material utilization and evaluating how effectively raw materials are converted into extracted products and packed output.

The Extraction Output Module allows users to record the amount of product obtained during the extraction stage. The recorded extraction output is subsequently processed by the Application Layer to support extraction efficiency analysis. Similarly, the Packed Yield Module allows users to record the final packed output associated with the production batch.

After the required production information has been entered, the Client Layer displays the processed results through the system dashboard. The dashboard provides a visual summary of important production indicators so that users do not need to manually calculate production statistics from individual records. The information may be presented using summary cards, tables, charts, status indicators, and other appropriate visual elements.

The dashboard can display important production indicators such as:

- Raw material consumption
- Extraction output

- Packed yield
- Yield percentage
- Extraction efficiency
- Production stability status
- Historical production trends
- Periodic production summaries

The Real-Time Yield Analytics Module allows users to view updated yield information as newly recorded production data is processed by the system. In this context, real-time monitoring refers to the immediate processing and presentation of newly encoded production information within the system rather than relying on delayed manual calculations or report preparation.

The Client Layer also displays the Production Stability Status generated by the Application Layer. Based on predefined thresholds and rules established for the system, production batches can be classified as Stable or Not Stable. This provides production personnel and supervisors with a quick indication of whether the recorded production performance falls within the expected operational conditions.

Historical information can also be accessed through the Client Layer. Users with appropriate permissions can review previous production batches, examine recorded raw material usage, compare extraction outputs and packed yields, and review previously generated analytical results. This capability addresses the difficulty of retrieving older production information from paper-based records and separate files.

The Client Layer further provides access to periodic reports. Production information stored in the system can be presented through daily, weekly, monthly, quarterly, and yearly summaries depending on the reporting requirements of the organization. These reports provide management with a more organized view of production performance and resource utilization.

To improve usability, the interface follows principles of simplicity, consistency, accessibility, and responsiveness. Navigation elements are organized according to the functions of the system, while related production activities are grouped into appropriate modules. Forms are designed to make data entry straightforward, and dashboard elements are arranged to allow users to identify important production information without navigating through multiple unrelated pages.

Overall, the Client Layer transforms the proposed system into an accessible operational tool for AlDi Foods Inc. It provides the environment where manual production recording activities are converted into structured digital records and where processed production information is presented in a form that can be easily understood by authorized users.

## Application Layer

The Application Layer serves as the central processing component of the proposed production monitoring platform. It acts as the intermediary between the Client Layer and the Database Layer by receiving user requests, validating submitted information, applying system rules, performing production calculations, processing analytical results, and returning processed information to the user interface.

This layer contains the main business logic of the proposed system. Rather than allowing users to manually calculate production indicators, the Application Layer automatically processes the information submitted through the Client Layer. This is particularly important to the study because one of the identified problems in the existing production process is the reliance on manual calculations and separate records.

Before production information is stored, the Application Layer performs validation procedures. These procedures verify whether required information has been provided and whether submitted values follow the expected data format and system rules. Validation helps prevent incomplete or invalid production records from being stored and improves the consistency of the information used for subsequent analysis.

After successful validation, the Application Layer processes the production information according to the system's predefined formulas and business rules. The major automated computations include:

- **Yield percentage calculation**
- **Extraction efficiency calculation**
- **Packed yield processing**
- **Raw material utilization analysis**
- **Production loss monitoring**
- **Historical production comparison**

The Yield Percentage Calculation processes the relevant production values entered by the user and produces the corresponding yield percentage. Automating this calculation eliminates the need for production personnel to repeatedly perform manual computations using calculators or spreadsheets.

The Extraction Efficiency Calculation evaluates the relationship between raw material input and extraction output based on the defined computation used by the system. The resulting value provides an indication of extraction performance for the corresponding production batch.

The Application Layer also processes Raw Material Utilization information. By relating raw material quantities to production outputs, the system provides information that can be used to examine how raw materials are utilized across production batches and reporting periods.

The system can also identify production losses by comparing relevant production quantities and computed results. This allows users to examine differences between input materials, extraction output, and final packed yield. The resulting information can help production personnel and supervisors identify batches that may require further review.

Another major responsibility of the Application Layer is the Production Stability Interpretation Module. This module evaluates computed production indicators against predefined thresholds or standards established for the proposed system. The purpose of this module is not to replace human judgment but to provide a consistent initial classification of production performance based on the rules implemented in the system.

The system classifies production performance into two categories:

- Stable
- Not Stable

A Stable classification indicates that the relevant computed production indicators satisfy the predefined conditions configured in the system. A Not Stable classification indicates that one or more indicators fall outside the predefined conditions and may require further examination by authorized personnel.

This automated classification provides users with a quick interpretation of production performance. Instead of manually reviewing individual calculations for every production batch, supervisors can use the generated status as an initial reference when reviewing production records.

The Application Layer also manages the Real-Time Yield Analytics Module. Whenever new production information is successfully recorded, the application processes the information and updates the corresponding production indicators. This allows newly entered production data to become available for dashboard visualization without requiring users to manually update formulas or prepare separate spreadsheet calculations.

The Extraction Efficiency Monitoring Module is likewise managed within this layer. The module processes extraction-related production data and produces efficiency indicators that can be reviewed for individual production batches or across selected reporting periods. If a batch produces a lower extraction efficiency result than expected, the information can serve as a reference for supervisors when reviewing possible operational factors.

The Application Layer also manages the Automated Report Generation Module. Instead of manually consolidating information from different production records, the system retrieves the required information from the Database Layer and organizes it according to the selected reporting period.

The proposed system can generate reports such as:

- Daily production summaries
- Weekly raw material utilization reports
- Monthly production performance reports
- Quarterly extraction efficiency reports
- Yearly production analytics
- Historical production trend reports

These reports provide organized information that can support production monitoring and management review. The reports may summarize production quantities, raw material usage, yield performance, extraction efficiency, stability classifications, and other relevant indicators stored by the system. In addition to production processing, the Application Layer manages user authentication and role-based access control. When a user submits login credentials, the application verifies the credentials against the information stored in the Database Layer. After successful authentication, the system determines which functions and information the user is authorized to access.

The Application Layer also manages session control, data retrieval, dashboard processing, and communication between the Client Layer and Database Layer. When a user requests historical records, analytics, or reports, the Application Layer retrieves the appropriate information from the database, processes the requested information, and sends the resulting output back to the Client Layer.

Through these responsibilities, the Application Layer serves as the core of the proposed system. It converts raw production entries into meaningful production information while maintaining the business rules, calculations, validation procedures, access controls, and reporting functions required by the system.

## Database Layer

The Database Layer functions as the centralized repository of information generated, entered, processed, and retrieved by the proposed production monitoring platform. It is responsible for storing, organizing, maintaining, and retrieving production information required by the system.

The use of a centralized relational database is important to the proposed system because the current production process involves information that may be recorded through manual documents, logbooks, spreadsheets, and other separate records. By consolidating these records into a centralized database, the proposed system provides a more organized approach to managing production information.

The Database Layer stores different categories of information required by the system, including:

- User roles and permissions
- Production batch records
- Raw material input records
- Extraction output records
- Packed yield information
- Yield percentage results
- Extraction efficiency results
- Production stability classifications
- Historical production records
- Dashboard-related information
- Generated production reports
- System activity information

Production batch records serve as the primary organizational reference for individual production activities. Related raw material, extraction, and packed yield information can be associated with the appropriate batch to maintain consistency among production records.

The database stores Raw Material Input Records, including the quantities entered for each production batch. These records provide the information needed by the Application Layer to process raw material utilization and other production-related calculations.

The database also stores Extraction Output Records and Packed Yield Information. These records allow the system to maintain a historical representation of production outputs and provide the data required for yield and extraction efficiency calculations.

Computed results such as yield percentage, extraction efficiency, and production stability classification can also be associated with the relevant production records. This allows users to retrieve previous analytical results without repeatedly performing calculations for already processed records.

The centralized database also supports historical analysis. Instead of searching through individual paper records or separate spreadsheets, authorized users can retrieve production information through the system based on available search and filtering functions. Historical information can then be used to compare production batches and examine performance over selected periods.

Data integrity is an important consideration in the Database Layer. Production information must remain consistent and properly associated with the correct production batch. The database structure therefore organizes related information into appropriate records and relationships so that the system can retrieve the correct information during analytics and report generation.

Security is also supported through authenticated access and role-based permissions. Users do not receive unrestricted access to all stored information. Instead, the Application Layer determines which operations are available to a particular user based on the user's assigned role.

The centralized database also reduces dependence on physical documents. Problems such as misplaced records, duplicated information, inconsistent entries, and delayed retrieval can be reduced when production information is maintained in a structured digital repository.

Furthermore, the Database Layer serves as the primary source of information for the dashboard, analytics, historical records, and automated reports. When a user requests production information, the Application Layer retrieves the appropriate records from the database and processes them before presenting the results through the Client Layer.

As a result, the Database Layer provides the foundation for the system's centralized information management. It ensures that production records are maintained in an organized repository that can support both daily operational monitoring and longer-term production analysis.

## System Data Flow

The overall operation of the proposed system follows a continuous flow of information between the Client Layer, Application Layer, and Database Layer. The process begins when an authorized user accesses the system through the Client Layer.

First, the user provides valid authentication credentials. The authentication request is transmitted to the Application Layer, where the submitted credentials are processed and verified against the user information stored in the Database Layer. Once authentication is successful, the system grants access according to the user's assigned role.

After accessing the appropriate system functions, production personnel can begin recording production information. The user may enter production batch details, raw material quantities, extraction output, packed yield, production date, and other required information through the appropriate forms in the Client Layer.

The submitted production information is then transmitted to the Application Layer. The Application Layer performs validation procedures to determine whether the required fields have been completed and whether the submitted values follow the defined system rules.

If the submitted information does not satisfy the validation requirements, the system returns an appropriate message to the Client Layer so that the user can correct the information. If the information passes validation, the Application Layer proceeds with the required production processing.

The Application Layer then performs the necessary calculations, including yield percentage, extraction efficiency, raw material utilization, and other applicable production indicators. The system also applies the predefined rules for determining the production stability classification.

After processing, the production information and corresponding results are stored in the Database Layer. The stored information becomes part of the organization's centralized production records and can subsequently be retrieved for historical analysis, dashboard visualization, and report generation.

When users access the dashboard, the Client Layer sends a request to the Application Layer for the required production information. The Application Layer retrieves the relevant records from the Database Layer, processes the information according to the selected dashboard requirements, and returns the resulting data to the Client Layer.

The Client Layer then presents the information through dashboard elements such as summary indicators, tables, charts, and production status classifications. Newly recorded production information can therefore be reflected in the corresponding production indicators after it has been successfully processed and stored.

The same process applies when users request historical information or reports. The system retrieves the appropriate production records from the centralized database, processes and organizes the information according to the requested period or category, and presents the results through the Client Layer.

For example, when a supervisor requests a monthly production report, the Application Layer retrieves the production records corresponding to the selected month from the Database Layer. It then processes the relevant production indicators and organizes them into the required report format before presenting the report through the Client Layer.

The continuous flow can therefore be summarized as:

User → Client Layer → Application Layer → Database Layer → Application Layer → Client Layer → User

This flow ensures that the Client Layer remains focused on user interaction, the Application Layer manages processing and business rules, and the Database Layer manages the storage and retrieval of production information.

Through this architecture, the proposed system supports the central purpose of the capstone: digitizing the production recording and monitoring process of AlDi Foods Inc. Production information that was previously dependent on manual recording can be entered into a centralized web-based system, processed automatically, stored systematically, and presented through real-time production analytics and reports.

The architecture also supports the major objectives of the proposed system by improving the organization of production records, reducing dependence on manual calculations, providing immediate access to production indicators, supporting extraction efficiency monitoring, and enabling historical production analysis. Rather than requiring production personnel and supervisors to manually consolidate information from separate records, the proposed architecture allows production information to move through a structured digital workflow.

Overall, the three-tier system architecture provides the technical foundation for the proposed production monitoring platform. The Client Layer enables users to interact with the system, the Application Layer transforms production entries into meaningful analytical information, and the Database Layer maintains the centralized production records. The coordinated operation of these layers allows AlDi Foods Inc. to monitor yield performance, extraction efficiency, raw material utilization, production stability, historical trends, and periodic production results through a centralized and organized digital platform.

## 3.4 Tools and Technologies Used

The development of the proposed system, *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.,"* required the integration of several software technologies and development tools to create a functional, reliable, maintainable, and user-friendly web-based production monitoring platform. The selected technologies supported the major requirements of the proposed system, particularly the digitization of production records, automated calculation of production indicators, centralized storage of production information, dashboard visualization, historical record management, and automated report generation.

The selection of the development technologies was based on their suitability for the proposed system and their ability to work together as an integrated web-based platform. The researchers considered factors such as compatibility, accessibility, ease of implementation, processing capability, maintainability, security, and suitability for the organization's operational requirements.

The proposed system was developed to address the limitations of the existing manual production monitoring process of AlDi Foods Inc. Production information such as raw material quantities, extraction output, and packed yield was previously dependent on manual recording and calculation. The selected technologies therefore needed to support a digital workflow in which production personnel could enter information through structured forms, the application could automatically process the submitted information, and the resulting information could be stored and presented through dashboards and reports.

The selected technologies also supported the system's centralized information management approach. Instead of maintaining production information across paper logbooks, spreadsheets, or separate records, the proposed platform consolidated production information within a centralized database. This allowed authorized users to retrieve historical production records more efficiently and provided a consistent source of information for analytics and reporting.

Another consideration was the system's ability to provide immediate processing of newly encoded production information. When production personnel entered data, the system processed the information, calculated the applicable production indicators, stored the results, and updated the appropriate dashboard information. In this study, real-time monitoring referred to the immediate processing and presentation of newly entered production data within the web-based platform rather than the use of industrial IoT or automated sensor infrastructure.

The selected technologies also supported the development methodology used in the study. During the development process, the researchers used these tools to design, implement, test, debug, and refine the system modules according to the requirements gathered from the intended users and the operational processes of AlDi Foods Inc.

## Frontend Technologies

### HTML (HyperText Markup Language)

HTML served as the primary markup language for developing the structural components of the proposed web-based production monitoring system. It provided the basic structure for the system's pages, forms, navigation menus, tables, dashboard containers, buttons, input fields, and other interface elements.

HTML was particularly important because the proposed system required multiple user-facing modules for recording and monitoring production information. The researchers used HTML to structure the interfaces for user authentication, dashboard access, production batch management, raw material recording, extraction output recording, packed yield recording, historical records, analytics, reports, user profiles, and administrative functions.

For production personnel, HTML forms provided structured fields where production-related information could be entered. These fields included production batch information, raw material quantities, extraction output, packed yield, production dates, and other information required by the system. Organizing these fields into standardized forms helped establish a consistent process for encoding production information.

HTML was also used to organize dashboard elements that displayed production information after it had been processed by the application. Summary cards, tables, charts, status indicators, and other interface components were structured using HTML so that users could view important production indicators in an organized manner.

The use of HTML also supported the role-based structure of the system. Different pages and interface components could be presented according to the user's assigned access permissions. Production personnel could access production recording functions, while authorized supervisors and administrators could access additional monitoring, historical, analytical, or administrative functions.

Because HTML is supported by modern web browsers, users could access the proposed platform through standard computers without requiring specialized client-side software. This supported the project's objective of developing a practical and accessible web-based production monitoring platform for AlDi Foods Inc.

HTML also worked together with CSS, PHP, JavaScript-based visualization components, and MySQL to create the complete web application. Therefore, HTML provided the structural foundation upon which the other technologies operated.

### CSS (Cascading Style Sheets)

CSS was used to control the visual presentation, layout, organization, and usability of the proposed system. While

HTML provided the structural components of the application, CSS controlled how these components were displayed to users.

The researchers used CSS to organize the appearance of navigation menus, buttons, forms, tables, dashboard cards, charts, report sections, and other interface elements. Consistent styling helped users understand the relationship between different system functions and navigate the application more efficiently.

The design of the proposed system considered the needs of production personnel who may interact with the application regularly during production activities. Forms were therefore organized to make information entry straightforward, while buttons and navigation elements were presented consistently across the different modules.

CSS also supported the visual presentation of analytical information. Since the proposed system provides yield percentages, extraction efficiency results, raw material utilization information, production stability classifications, and historical trends, appropriate spacing, alignment, typography, and layout were used to make these indicators easier to interpret.

Responsive design techniques were also applied to allow the interface to adapt to different screen sizes and resolutions. This helped maintain the readability and usability of the system when accessed through available computers or other compatible devices.

CSS further supported the dashboard by creating an organized visual environment for production statistics and graphical information. A properly structured dashboard allows supervisors and management personnel to identify important production information without having to examine individual production records manually.

Overall, CSS contributed to the usability and consistency of the proposed system by ensuring that production recording, monitoring, analytics, historical records, and reporting functions were presented through a clear and organized interface.

## Backend Technology

## PHP (Hypertext Preprocessor)

PHP served as the primary server-side programming language for implementing the core functionality and business logic of the proposed production monitoring system. It acted as an important processing component between the user interface and the MySQL database.

PHP processed requests submitted through the web interface, validated production information, communicated with the database, performed production calculations, retrieved historical records, prepared dashboard information, and supported automated report generation.

The major system functions implemented through PHP included:

- User authentication and login verification
- User account management
- Role-based access control
- Production batch processing
- Raw material recording
- Extraction output recording
- Packed yield recording

- Yield percentage calculation
- Extraction efficiency calculation
- Production stability interpretation
- Historical record retrieval
- Dashboard data processing
- Analytics processing
- Automated report generation

One of the primary responsibilities of PHP was processing production information after it was submitted by authorized users. For example, when production personnel entered the raw material quantity, extraction output, and packed yield for a production batch, PHP processed the submitted information according to the rules and calculations implemented in the system.

PHP also supported the automated calculation of production indicators. Instead of requiring users to manually calculate yield percentages and extraction efficiency using calculators or spreadsheets, the system processed these values automatically. This reduced repetitive calculations and helped maintain consistency in the computation process.

PHP was also responsible for validating submitted information before the records were stored. Validation helped prevent incomplete or improperly formatted information from being processed as valid production records.

The application also used PHP to implement the Production Stability Interpretation Module. Based on the predefined thresholds and conditions established for the system, PHP processed the relevant production indicators and assigned the corresponding Stable or Not Stable status.

PHP also handled requests for historical information. When a user selected a particular production period or requested specific records, PHP communicated with MySQL to retrieve the appropriate information and prepared it for presentation through the system interface.

The integration between PHP and MySQL was particularly important because the proposed system relied on centralized production information. PHP acted as the processing component while MySQL served as the primary storage system. This allowed production information to move between the user interface and database in an organized manner.

Through these functions, PHP supported the core objective of the proposed system: transforming manually encoded production information into automatically processed and organized production analytics.

## Analytics Processing Technology

### Chart.js

Chart.js served as the primary data visualization library used to present production analytics through graphical representations. The proposed system generated numerical production information such as yield percentages, extraction efficiency, raw material usage, production output, and stability classifications. Chart.js helped transform these numerical values into visual information that users could interpret more efficiently.

The system used Chart.js to support the visualization of:

- Current yield percentage

- Extraction efficiency performance
- Stable and Not Stable production trends
- Raw material consumption
- Production output trends
- Weekly raw material usage
- Monthly production performance
- Quarterly extraction efficiency
- Yearly production analytics
- Historical production trends

The use of graphical visualization was important because the proposed system was designed not only to store production records but also to provide production monitoring and analytical support. A table of numerical values may require users to examine several records individually, while a properly designed chart can provide a more direct view of changes and trends across production periods.

For example, production personnel or supervisors could use graphical representations to observe changes in yield performance across production batches. Similarly, extraction efficiency charts could help users identify periods or batches where extraction performance differed from other production records.

Chart.js also supported the dashboard's purpose as a centralized monitoring interface. Instead of requiring management personnel to manually consolidate information from different production records, the system could retrieve the relevant information from MySQL, process it through PHP, and present the resulting data through Chart.js.

The visualization functions therefore connected directly to the study's goal of improving production visibility. By presenting production information visually, the system allowed users to review operational performance more efficiently and supported data-informed production monitoring.

## Database Technology

## MySQL

MySQL served as the centralized relational database management system for the proposed production monitoring platform. It was responsible for storing, organizing, retrieving, updating, and maintaining the production information generated through system operations.

The use of MySQL was directly connected to one of the major problems addressed by the capstone: the dependence on manual and distributed production records. Rather than maintaining production information in separate paper documents or spreadsheet files, the proposed system stored the relevant information within a centralized database.

The database stored different categories of information, including:

- User accounts
- User roles and permissions
- Production batches

- Raw material input records
- Extraction output records
- Packed yield records
- Yield percentage results
- Extraction efficiency results
- Production stability status
- Historical production records
- Dashboard-related information
- Generated report information

The User Accounts records supported authentication and access control. User roles and permissions determined the system functions that could be accessed by different users.

The Production Batch records served as an organizational reference for individual production activities. Raw material inputs, extraction outputs, packed yields, and corresponding analytical results could be associated with the appropriate production batch.

The database also stored the production values required for analytical processing. Raw material quantities, extraction outputs, and packed yields provided the information needed for the system to calculate production indicators such as yield percentage and extraction efficiency.

Computed results could also be stored and associated with the corresponding production records. This allowed authorized users to retrieve previous production information and review historical analytical results without reconstructing the records manually.

Historical data stored in MySQL supported the system's reporting and analytics functions. Users could retrieve production information according to relevant periods and use the resulting information to review production performance over time.

The centralized database also helped address common problems associated with manual record keeping, such as misplaced documents, duplicated entries, inconsistent records, and delayed retrieval. By maintaining production information in a structured digital repository, the proposed system provided a more organized method for managing production records.

MySQL also worked with PHP to support dashboard and report generation. PHP retrieved the required records from MySQL, processed the information, and supplied the resulting data to the appropriate system components. Chart.js could then use the processed information to present graphical analytics through the dashboard.

Therefore, MySQL served not merely as a storage technology but as an important component of the centralized information-management approach of the proposed system.

## Development Environment

### Visual Studio Code

Visual Studio Code (VS Code) served as the primary Integrated Development Environment used by the researchers

during the development of the proposed production monitoring system. It provided a centralized environment for creating, editing, organizing, debugging, and maintaining the source code of the application.

VS Code supported the development of both frontend and backend components of the proposed system. The researchers used the environment to work with HTML, CSS, PHP, JavaScript, and other project files required by the application.

Its development features included:

- Syntax highlighting
- Intelligent code completion
- Error identification
- Debugging support
- Extension support
- Integrated terminal
- File management
- Version control integration

These features supported the researchers throughout the development process by helping them organize the different files and components of the proposed system.

The project contained multiple modules, including authentication, dashboard, raw material management, extraction recording, packed yield recording, historical records, analytics, reports, and administrative functions. VS Code provided a centralized environment where these components could be developed and maintained systematically.

The use of VS Code also supported debugging during the testing phase. When a system function did not produce the expected result, the researchers could inspect the relevant source code, identify potential errors, make corrections, and test the function again.

Therefore, Visual Studio Code contributed to the maintainability and organization of the source code throughout the system development process.

## Local Development Server

## XAMPP

XAMPP served as the local development and testing environment for the proposed web-based production monitoring system. It provided the necessary server-side components required to execute PHP applications and manage the MySQL database on the researchers' local machines.

XAMPP provided several components used during system development, including:

- Apache Web Server
- MySQL Database Server
- PHP Interpreter
- phpMyAdmin

The Apache Web Server allowed the researchers to run the PHP-based web application locally. This enabled them to access and test the system through a web browser before prototype deployment.

The MySQL Database Server provided the local database environment where production information, user accounts, production batches, raw material records, extraction records, packed yield information, analytical results, and other system information could be stored during development and testing.

The PHP Interpreter allowed the PHP scripts used by the system to execute within the local development environment.

The phpMyAdmin tool provided a graphical interface for managing the MySQL database. The researchers could use it to create and manage database structures, inspect stored information, and support database-related testing activities.

XAMPP was particularly useful during the development and testing stages because it allowed the researchers to test system functions before implementing the prototype in its intended operational environment.

For example, production recording forms could be tested by entering sample production information. The researchers could then verify whether PHP correctly processed the submitted information, whether MySQL stored the records properly, and whether the dashboard displayed the expected results.

This local development environment supported iterative development because changes to the application could be implemented and tested repeatedly before the system was presented as a prototype.

**Technology Integration** The technologies used in the proposed system were integrated to form a single web-based production monitoring platform. Each technology performed a specific role while communicating with the other components to support the overall operation of the system.

**HTML** provided the structural foundation of the web pages and production forms. **CSS** controlled the visual presentation, layout, responsiveness, and usability of the interface. **PHP** served as the primary backend technology responsible for processing user requests, validating production data, performing calculations, managing access control, retrieving records, and generating system outputs.

**MySQL** provided the centralized database where user information, production batches, raw material records, extraction outputs, packed yields, analytical results, historical records, and report-related information were stored.

**Chart.js** transformed processed production information into graphical representations used by the dashboard and analytical components. This allowed users to view production trends, yield performance, extraction efficiency, raw material utilization, and other relevant indicators more efficiently.

**Visual Studio Code** supported the development and maintenance of the application's source code, while **XAMPP** provided the local environment necessary for running the PHP application and MySQL database during development and testing.

The integration of these technologies followed the three-tier architecture presented in the preceding section. HTML and CSS primarily supported the **Client Layer**, PHP supported the major processing responsibilities of the **Application Layer**, and MySQL supported the **Database Layer**. Chart.js connected the processed information from the application and database with the visual analytics presented to users.

The overall technology interaction can be represented as:

**User → HTML/CSS Interface → PHP Processing → MySQL Database → PHP Processing → Chart.js/HTML Dashboard → User**

For a production transaction, the process begins when production personnel enter information through an HTML-based form. CSS provides the visual organization of the form and interface. Once the information is submitted, PHP validates and processes the data. The application performs the required calculations, including yield percentage and extraction efficiency, and applies the predefined rules for production stability classification.

After processing, the production information is stored in MySQL. When users access the dashboard, PHP retrieves the required information from the database and prepares it for presentation. Chart.js then displays applicable analytical information through graphical representations, while HTML and CSS organize the overall dashboard interface.

The same technology integration supports historical records and automated reports. Production information stored in MySQL can be retrieved by PHP according to the requested date, production batch, or reporting period. The resulting information can then be presented through the appropriate system interface.

This integrated approach allows the selected technologies to directly support the major modules and objectives of the proposed capstone. The technologies collectively enable the system to:

1. Digitally record production information.
2. Centralize raw material, extraction, and packed yield records.
3. Automatically calculate yield percentage and extraction efficiency.
4. Classify production performance as Stable or Not Stable based on predefined rules.
5. Provide updated production information through the dashboard.
6. Maintain historical production records.
7. Visualize production trends and performance indicators.
8. Generate periodic production reports.
9. Provide role-based access to authorized users.
10. Reduce dependence on manual calculations and scattered production records.

Overall, the selected tools and technologies provided the technical foundation required to develop the proposed production monitoring platform for AlDi Foods Inc. Their integration supported the transition from manual production recording toward a centralized digital process while remaining consistent with the scope of the study. The system focused on manually encoded production information, automated data processing, centralized storage, analytical visualization, and reporting rather than requiring advanced industrial automation, IoT devices, or sophisticated sensor infrastructure.

Through the coordinated use of HTML, CSS, PHP, Chart.js, MySQL, Visual Studio Code, and XAMPP, the proposed system was able to support the major production-monitoring functions identified in the study, including raw material recording, extraction output monitoring, packed yield recording, yield analytics, extraction efficiency monitoring, production stability interpretation, historical record management, dashboard visualization, and automated reporting.

## 3.5 Data Gathering Techniques

The researchers used three data gathering techniques to obtain the information needed for the development of the proposed system, namely interviews, observation, and document review. These techniques were conducted to understand the existing production monitoring process of AlDi Foods Inc. and to identify the requirements of the proposed system, “Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.”

The data gathering process focused on the existing procedures for recording raw material quantities, production batch information, extraction outputs, packed yields, production dates, and other production-related information. It also examined how production personnel performed calculations, monitored production performance, maintained historical records, and prepared production reports.

The information obtained from these techniques served as the basis for identifying the system requirements and designing the proposed web-based production monitoring platform. The gathered information was used to determine the necessary data fields, system functions, calculations, dashboard information, historical records, and reports that would be included in the system. The process focused on the actual production monitoring needs of AlDi Foods Inc. and did not require advanced industrial automation, IoT devices, robotics, or sophisticated sensor-based technologies.

## Interviews

The researchers conducted interviews with selected personnel of AlDi Foods Inc., particularly production supervisors, production personnel, and other authorized representatives who were familiar with the company's production monitoring activities. The interviews were conducted to obtain direct information regarding the current procedures used in recording, processing, checking, storing, and reporting production data.

The interview questions focused on the existing procedures for recording raw material quantities, production batch information, extraction outputs, and packed yields. The researchers also gathered information about how production personnel performed yield calculations, monitored extraction performance, identified production losses, and determined whether production results were within the expected range.

The researchers also asked about the problems encountered when using manual recording methods such as logbooks, paper forms, and spreadsheets. The identified concerns included repeated data entry, manual calculations, inconsistent records, delayed access to information, difficulty retrieving previous production records, and the possibility of errors during recording and computation.

The interviews also focused on the information that production personnel and supervisors needed when monitoring production activities. The participants were asked about the production indicators that were important for monitoring performance and the types of information that should be presented in a centralized system. These discussions helped the researchers identify the need for information such as raw material usage, extraction output, packed yield, yield percentage, extraction efficiency, production stability, and historical production performance.

The information obtained from the interviews helped the researchers identify the required system functions. These included user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, yield analytics, extraction efficiency monitoring, production stability monitoring, dashboard visualization, historical record retrieval, and automated reporting.

The interview results were also used during the requirements analysis and system design stages. The information provided by the participants helped the researchers determine how users would interact with the proposed system and what information should be available to authorized users. The identified requirements were then considered during the development of the system and its corresponding database, interfaces, calculations, and reporting functions.

## Observation

The researchers conducted direct observation to understand the actual production monitoring activities performed at AlDi Foods Inc. This technique allowed the researchers to examine the workflow followed by production personnel during the recording and monitoring of production information.

The observation focused on activities such as recording raw material quantities, preparing production batch information, monitoring extraction outputs, recording packed yields, performing production calculations, checking recorded information, and preparing production reports. The researchers also observed how production information was transferred from one recording activity to another and how personnel accessed previous production records when needed.

Through observation, the researchers were able to identify the actual sequence of activities involved in the existing production monitoring process. The observed process generally involved collecting production information, recording the data manually, performing calculations, checking the results, and preparing or reviewing production records. This helped the researchers determine which activities could be supported by the proposed digital system.

Observation also allowed the researchers to verify information obtained during the interviews. The researchers compared the actual workflow with the information provided by the production personnel to identify common procedures, repetitive tasks, and areas where manual recording could result in delays or inconsistencies.

The observation results supported the design of the system's data entry pages and workflow. The production process was considered when organizing the system pages for production batch management, raw material recording, extraction output recording, and packed yield recording. The observed workflow also supported the development of the dashboard, historical records, analytics, and reporting functions.

The observation further helped the researchers determine the usability requirements of the proposed system. Since production personnel would be responsible for entering production information, the system was designed to provide an organized and understandable data entry process. The system also provides immediate processing and presentation of newly encoded production information to support production monitoring.

However, the proposed system remains dependent on the data entered by authorized users. The system was designed as a web-based manual data entry platform and therefore does not automatically collect production information through IoT devices or industrial sensors.

## Document Review

The researchers conducted a review of available production-related documents used by AlDi Foods Inc. The document review was performed to determine the types of information recorded during production and to understand how production records were organized, maintained, calculated, and reported.

The documents reviewed included available production logbooks, daily production reports, raw material usage records, extraction output records, packed yield records, inventory records, monitoring forms, and historical production data. These documents were examined to identify the common data fields, recording formats, production dates, batch information, raw material quantities, extraction outputs, packed yields, and other production-related information.

The researchers also reviewed the existing methods used to calculate and monitor production performance. Particular attention was given to information related to yield percentage, extraction efficiency, raw material utilization, production output, and production losses. The review helped the researchers determine which calculations could be performed automatically by the proposed system after production data were entered.

The document review also helped identify possible difficulties associated with the existing manual recordkeeping process. These included inconsistent recording formats, incomplete information, repeated entries, difficulty locating historical records, and additional time required to prepare production summaries. These findings supported the need

for a centralized database where production information could be stored in an organized manner and retrieved when required.

The reviewed documents were also used as references when designing the database structure of the proposed system. The information helped the researchers determine the necessary data fields and relationships for production batches, raw materials, extraction records, packed yields, users, analytics, and reports.

The document review also supported the design of the dashboard and reporting functions. The production information found in the existing records was considered in determining the indicators that should be presented to authorized users. These indicators included raw material usage, extraction output, packed yield, yield percentage, extraction efficiency, and production stability.

In addition, the existing production records helped guide the development of historical reporting functions. The proposed system was designed to allow authorized users to retrieve and review production information according to different reporting periods, including weekly, monthly, quarterly, and yearly records. This supports easier review of production performance compared with relying only on manually maintained records.

The findings from the document review were compared with the information gathered through interviews and observation. This comparison helped the researchers verify whether the identified system requirements reflected the actual production monitoring activities of AlDi Foods Inc. The three data gathering techniques were therefore used together to provide a more complete understanding of the existing process and the requirements of the proposed system.

## Summary of Data Gathering Process

The data gathering process combined interviews, observation, and document review to obtain information about the existing production monitoring practices of AlDi Foods Inc. Interviews provided information about the experiences, requirements, and concerns of production personnel and supervisors. Observation allowed the researchers to examine the actual workflow and identify the activities involved in recording and monitoring production information. Document review provided information about the existing production records, data fields, calculations, and reporting formats.

The information gathered from these techniques was used to determine the requirements of the proposed production monitoring system. The researchers identified the need for a centralized platform that could record production batches, raw material quantities, extraction outputs, and packed yields. The system was also designed to automatically process production data and calculate yield percentage and extraction efficiency based on the encoded information.

The gathered information supported the development of production stability monitoring, which presents production results based on predefined criteria established for the system. The data gathering process also supported the development of a dashboard that presents important production indicators and allows authorized users to monitor current production information.

The information obtained from the existing records also guided the development of historical records and automated reports. The system was designed to organize production information so that authorized users could retrieve and review previous records and generate production summaries according to applicable reporting periods.

The gathered requirements also influenced the organization of the system interface and database. The system was developed using HTML and CSS for the user interface, JavaScript and Chart.js for interactive presentation and analytics, PHP for server-side processing, and MySQL for centralized data storage. XAMPP was used as the local development environment.

Overall, the data gathering techniques provided the researchers with the information needed to transform the existing manual production monitoring process into a centralized web-based system. The findings helped establish the system's required functions, data fields, calculations, dashboard indicators, historical records, and reporting features.

In the proposed system, the term real-time refers to the immediate processing and presentation of production information after it has been entered by an authorized user. It does not refer to automatic data collection through IoT devices, industrial sensors, or advanced automation technologies. The system therefore remains focused on manually encoded production data while providing faster processing, organized storage, and easier access to production information for monitoring and decision-making.

## 3.6 Testing Procedures

The proposed system underwent a series of structured testing procedures to verify that its functions operated according to the identified requirements of AlDi Foods Inc. The testing process was conducted after the development of the system components and was used to identify functional errors, inconsistencies, calculation problems, and usability issues before the prototype was presented for actual user evaluation.

The testing procedures were aligned with the objectives of the study, particularly the development of a web-based production monitoring platform capable of recording production information, calculating yield and extraction efficiency, monitoring production stability, presenting production analytics, maintaining historical records, and generating periodic reports. The testing activities focused on determining whether the developed functions performed their intended tasks based on the system requirements identified through interviews, observation, and document review.

The testing process also considered the three-tier architecture of the proposed system. The Client Layer was tested to determine whether users could properly access and interact with the system interfaces. The Application Layer was tested to verify data validation, calculations, production stability interpretation, and processing functions. The Database Layer was tested to determine whether production records could be properly stored, retrieved, and presented to authorized users.

Testing was performed using production scenarios based on the actual workflow identified during the data gathering process. These scenarios included entering production batch information, recording raw material quantities, entering extraction outputs and packed yields, processing production results, viewing dashboard information, retrieving historical records, and generating production reports. The use of these scenarios allowed the researchers to determine whether the system could support the production monitoring activities of AlDi Foods Inc.

The testing procedures also covered the major analytical functions of the proposed system. These included yield percentage computation, extraction efficiency calculation, raw material utilization, and production stability interpretation. The researchers verified whether the system produced the expected results based on the encoded production values and predefined criteria.

The testing process was not intended to introduce advanced industrial automation or IoT-based monitoring. Instead, it focused on verifying the functionality of the web-based system using manually encoded production information. In this study, real-time processing referred to the immediate processing and presentation of newly entered production information within the system.

The testing activities consisted of Unit Testing, System Testing, and User Acceptance Testing (UAT). Unit Testing examined individual system components, System Testing evaluated the integrated system, and UAT assessed the system from the perspective of intended users. The results of these testing activities were used to identify necessary corrections and improvements before the prototype was considered ready for presentation and evaluation.

## Unit Testing

Unit Testing was conducted to evaluate individual modules and functions of the proposed system independently. Each component was tested according to its intended purpose before and during its integration with other system components. This testing procedure allowed the researchers to identify errors within specific functions and correct them before they affected the operation of the complete system.

The following major functions were subjected to unit testing:

- User authentication and login;
- Production batch management;
- Raw material recording;
- Extraction output recording;
- Packed yield recording;
- Yield percentage computation;
- Extraction efficiency calculation;
- Production stability interpretation;
- Dashboard data retrieval;
- Historical record retrieval; and
- Automated report generation for weekly, monthly, quarterly, and yearly records.

The User Authentication function was tested to determine whether valid users could successfully log in and whether invalid credentials were properly rejected. The researchers also checked whether authenticated users could access the functions appropriate to their assigned role.

The Production Batch Management function was tested to verify whether production batch information could be properly entered, saved, updated, and retrieved. The researchers checked the required fields and verified that the entered information was properly associated with the corresponding production records.

The Raw Material Recording function was tested by entering sample raw material quantities and checking whether the values were correctly stored in the database. The Extraction Output and Packed Yield functions were similarly tested to determine whether production output information could be entered and associated with the correct production batch.

The Yield Percentage computation was tested using sample production values with expected results. The researchers compared the system-generated yield percentage with manually computed results to determine whether the implemented formula produced the correct output.

The Extraction Efficiency calculation was also tested using predetermined sample values. The researchers verified whether the system correctly processed the required production data and displayed the expected extraction efficiency result.

The Production Stability Interpretation function was tested using production results that represented different conditions. The researchers checked whether the system correctly classified the production result as “Stable” or “Not Stable” based on the predefined criteria established for the system.

Dashboard data retrieval was tested to determine whether newly recorded production information was properly reflected in the displayed indicators and visualizations. The researchers also checked whether the dashboard presented the appropriate raw material, extraction, packed yield, yield, efficiency, and stability information.

Historical record retrieval was tested by entering and saving several production records and then searching for previously stored information. The researchers verified whether the system could retrieve the appropriate records based on the available search or filtering functions.

The report generation function was tested for the applicable reporting periods, including weekly, monthly, quarterly, and yearly summaries. The researchers checked whether the system correctly collected the appropriate production records and presented the corresponding summary information.

Through Unit Testing, each major component was checked individually before being relied upon as part of the complete production monitoring workflow. This helped reduce the possibility of errors in individual functions and provided a basis for proceeding to integrated system testing.

## System Testing

System Testing was conducted to evaluate the proposed system as an integrated platform. Unlike Unit Testing, which focused on individual functions, System Testing examined how the different modules worked together during an entire production monitoring process.

The testing considered the interaction among the Client, Application, and Database layers of the proposed system architecture. Data entered through the user interface were processed by the application layer and stored in the database. The stored information was then retrieved for dashboard presentation, analytics, historical records, and report generation.

System testing was performed using simulated production scenarios based on the actual workflow identified during interviews, observation, and document review. These scenarios included the following activities:

- User login and access to authorized system functions;
- Creation and management of production batch information;
- Entry of raw material quantities;
- Entry of extraction output information;
- Entry of packed yield information;
- Processing of yield percentage;
- Processing of extraction efficiency;
- Interpretation of production stability;
- Updating of dashboard information;
- Retrieval of historical production records; and

- Generation of periodic production reports.

The researchers first tested the complete workflow from user authentication to production data entry. After entering the required production information, the researchers verified whether the system accepted the data, applied the appropriate validation, and stored the information correctly.

The researchers then examined whether the entered production data were correctly processed by the application. The calculated yield percentage and extraction efficiency were checked against expected values. The resulting production stability status was also verified against the predefined criteria.

The dashboard was tested to determine whether the information stored in the database was correctly retrieved and displayed. The researchers checked whether the dashboard indicators and visualizations corresponded to the encoded production records. The purpose of this testing was to verify that updated production information was reflected in the system after the data were processed.

The historical records function was tested by retrieving previously stored production information and comparing the displayed records with the original encoded values. This ensured that users could access production information for monitoring and review without relying solely on physical or manually maintained records.

The reporting function was also tested by generating reports for the available reporting periods. The researchers verified whether the appropriate production records were included and whether the generated summaries corresponded to the stored production information.

System Testing also included checking how the system responded to incomplete, invalid, or inappropriate data entries. The researchers examined whether the system provided appropriate validation and prevented incorrect or incomplete information from being processed when required fields were not properly entered.

The results of System Testing were used to identify integration issues between the user interface, application processing, and database. Corrections were applied when necessary so that the complete production monitoring workflow could operate consistently before the system was presented to the intended users for acceptance testing.

## User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to determine whether the proposed system was acceptable and suitable for the intended users of AlDi Foods Inc. This testing involved selected personnel who were familiar with the production monitoring process, such as production managers, supervisors, and production staff.

UAT focused on the users' experience while performing production monitoring activities using the developed system. The participants were given appropriate production scenarios and were asked to interact with the system functions relevant to their roles. Their feedback was used to determine whether the system was understandable, usable, accurate, and suitable for the intended production monitoring activities.

The respondents evaluated the system based on the following areas:

- Ease of use of the system interface;
- Functionality of the production monitoring features;
- Accuracy of yield and extraction efficiency calculations;
- Clarity of dashboard information and visualizations;
- Appropriateness of the Stable/Not Stable production interpretation;

- Usefulness of historical production records;
- Usefulness of weekly, monthly, quarterly, and yearly reports;
- Efficiency of recording and retrieving production information;
- Reliability of the system during use; and
- Overall user satisfaction.

During UAT, the participants were able to examine the process of entering production batch information, recording raw materials, entering extraction output and packed yield, viewing calculated production indicators, checking dashboard information, retrieving historical records, and reviewing generated reports.

The participants' feedback was important because the technical testing performed by the researchers could not fully determine whether the system was convenient and understandable for actual users. Their evaluation provided information about the practical usability of the system and whether the developed functions corresponded to the production monitoring activities of AlDi Foods Inc.

Any issues or suggestions identified during UAT were documented and considered for system refinement. These could include improvements to field labels, navigation, information presentation, data entry procedures, dashboard organization, or report presentation. Corrections were applied when appropriate and when they were consistent with the scope and objectives of the study.

The UAT results also supported the evaluation of the system according to the identified quality criteria of functionality, usability, accuracy, efficiency, reliability, and user satisfaction. These criteria were consistent with the evaluation approach used in the study and provided a basis for determining how well the developed prototype addressed the identified production monitoring requirements.

Overall, the testing procedures provided a systematic method for verifying the proposed production monitoring system before its prototype deployment and evaluation. Unit Testing verified individual functions, System Testing verified the interaction of the integrated modules, and User Acceptance Testing evaluated the system from the perspective of its intended users. The combined results provided evidence that the developed system was functioning according to its defined requirements and supported the production monitoring objectives of AlDi Foods Inc.

## 3.7 Evaluation Criteria

The developed production monitoring system was evaluated based on criteria that were aligned with the objectives and requirements of the study. The evaluation was conducted to determine whether the proposed system could support the production monitoring activities of AlDi Foods Inc. by providing functions for recording production data, automatically processing production measurements, presenting production analytics, maintaining historical records, and generating production reports.

The evaluation criteria were developed based on the identified problems in the existing production monitoring process and the intended functions of the proposed system. In the current process, production information was manually recorded and calculations were performed manually, which could result in delayed information processing, inconsistent records, difficulty in retrieving historical information, and increased possibility of calculation errors. Therefore, the evaluation focused on determining whether the proposed web-based system could provide an organized and usable platform for managing production information.

The evaluation also considered the findings from the related studies and literature discussed in Chapter 2. The reviewed studies highlighted the importance of organized production data, digital record management, production monitoring, automated calculations, and accessible production information for supporting operational

decision-making. The proposed system addressed these identified needs within the scope of AlDi Foods Inc. by providing a centralized web-based platform without requiring IoT devices or advanced industrial automation technologies.

The evaluation was also connected to the system requirements and development activities discussed in Chapter 3. The functions evaluated included user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, yield percentage computation, extraction efficiency calculation, production stability interpretation, dashboard presentation, historical record retrieval, and automated report generation. These functions were first examined through the testing procedures presented in Section 3.6 before the system was evaluated by the intended users.

The system was evaluated according to six criteria: functionality, usability, accuracy, efficiency, reliability, and user satisfaction. These criteria were used to determine whether the developed prototype performed its intended functions and supported the production monitoring requirements of AlDi Foods Inc. The evaluation results were intended to provide evidence for the findings presented in Chapter 4 and to serve as a basis for the conclusions and recommendations presented in Chapter 5.

## Functionality

Functionality was evaluated to determine whether the proposed system provided the required features and performed the intended production monitoring tasks. This criterion was directly related to the objectives of the study, particularly the development of a centralized web-based platform for recording, processing, monitoring, and reporting production information.

The evaluation covered the major functions of the system, including user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, yield percentage computation, extraction efficiency calculation, production stability classification, dashboard presentation, historical record retrieval, and automated report generation.

The researchers evaluated whether authorized users could successfully access the system and perform the functions assigned to them. The production recording functions were examined to determine whether users could enter the required production information and associate the data with the appropriate production batch.

The analytical functions were also evaluated to determine whether the system could automatically process the entered information. This included the computation of yield percentage and extraction efficiency, as well as the interpretation of production stability based on the predefined criteria of the system.

The dashboard function was evaluated based on its ability to present relevant production information in an organized manner. The evaluation considered whether production indicators, charts, and summarized information could help users monitor production performance without manually consolidating records from separate sources.

Historical records were evaluated to determine whether previously recorded production information could be retrieved when needed. The reporting function was also assessed to determine whether the system could generate production summaries for weekly, monthly, quarterly, and yearly periods.

Functionality was therefore used to determine whether the developed system addressed the functional requirements identified during interviews, observation, and document review. The results of this criterion were also used to determine whether the system functions tested in Section 3.6 operated according to the intended requirements.

## Usability

Usability was evaluated to determine whether the proposed system could be used by the intended users in a clear, understandable, and convenient manner. Since the system was designed to support production personnel and

authorized users of AlDi Foods Inc., the evaluation considered the users' ability to navigate the system and perform common production monitoring tasks.

The evaluation included the clarity of the user interface, organization of menus and modules, readability of information, ease of entering production data, and ease of accessing dashboard information. The researchers also considered whether users could understand the information presented through tables, indicators, charts, and production summaries.

The production data entry functions were evaluated based on whether users could understand the required fields for raw materials, extraction output, packed yield, and production batch information. The system's validation messages and prompts were also considered to determine whether they provided sufficient guidance when users entered incomplete or inappropriate information.

The dashboard and analytics functions were evaluated based on whether users could easily interpret the displayed yield, extraction efficiency, raw material utilization, and production stability information. The historical records and reporting functions were also evaluated based on the ease of locating and reviewing previously recorded production information.

Usability evaluation was important because a system may provide the required functions but still be difficult to use if its interface is unclear or its information is poorly organized. Therefore, this criterion helped determine whether the proposed system could be practically used within the production monitoring workflow of AlDi Foods Inc.

## Accuracy

Accuracy was evaluated to determine whether the system produced correct and consistent results based on the production information entered by authorized users. This criterion was particularly important because one of the identified problems in the existing process was the possibility of errors caused by manual calculations and inconsistent recording.

The evaluation focused on the system's automated calculations for yield percentage and extraction efficiency. The researchers compared system-generated results with manually computed expected values using sample production data. This comparison helped determine whether the formulas implemented in the application produced the expected results.

Production stability classification was also evaluated. The researchers used production values representing different conditions and checked whether the system correctly classified the results as “Stable” or “Not Stable” based on the predefined thresholds established for the system.

Accuracy was also considered across different parts of the system. The researchers checked whether information entered in the production recording modules was correctly reflected in the dashboard, historical records, analytics, and reports. This helped determine whether the system maintained consistency as production information moved from data entry to storage, processing, presentation, and reporting.

The evaluation of accuracy was limited to the correctness of system processing based on the data entered by users. Since the proposed system uses manually encoded production information rather than automatic sensor collection, the system cannot independently verify whether the original production values entered by users are physically correct. Therefore, accuracy evaluation focused on correct recording, processing, computation, classification, and presentation of the encoded information.

## Efficiency

Efficiency was evaluated to determine whether the proposed system could process and present production information within a reasonable amount of time and with less dependence on repetitive manual calculations and record consolidation.

The evaluation considered the time required to enter production information, process yield and extraction efficiency calculations, update dashboard information, retrieve historical records, and generate production reports. The researchers also examined whether the system could process newly entered information without requiring users to manually perform the same calculations separately.

In the context of this study, real-time processing referred to the immediate processing and presentation of production information after it was manually encoded into the system. It did not refer to automatic data collection through IoT devices, sensors, or industrial monitoring equipment.

Dashboard responsiveness was also considered because production personnel need to access updated information after production data have been entered. The researchers evaluated whether newly encoded information was reflected in the appropriate indicators and visualizations.

Historical record retrieval was evaluated based on the ability of the system to provide previously stored information without requiring users to manually search through physical logbooks or separate files. Report generation was also considered because the system was designed to organize production information into weekly, monthly, quarterly, and yearly summaries.

The efficiency criterion therefore focused on how the system supported faster organization, processing, retrieval, and presentation of production information within the defined prototype scope.

## Reliability

Reliability was evaluated to determine whether the proposed system could consistently perform its intended functions and maintain production information during repeated use. This criterion focused on the system's ability to provide consistent results when users performed similar production monitoring tasks under similar conditions.

The researchers evaluated whether production records could be consistently saved and retrieved after being entered. The database functions were checked to determine whether stored information remained available for dashboard presentation, historical record retrieval, and report generation.

The system's response to incomplete or invalid information was also considered. The researchers examined whether validation mechanisms could prevent inappropriate entries from being processed when required information was missing or incorrectly entered.

Reliability was also evaluated through repeated testing of the major production monitoring functions. The researchers checked whether the same production values produced consistent yield percentages, extraction efficiency results, and production stability classifications when processed under the same predefined conditions.

The reporting and dashboard functions were also examined for consistency. Information displayed in the dashboard, historical records, and reports was compared with the stored production information to determine whether the system consistently presented the appropriate records.

Within the scope of this study, reliability referred to the consistent operation of the developed prototype during testing and user evaluation. It did not represent enterprise-level system availability, long-term server uptime, or large-scale industrial reliability because these were outside the scope of the project.

## User Satisfaction

User satisfaction was evaluated to determine the overall perception of the intended users toward the developed production monitoring system. This criterion was included because the effectiveness of the proposed system depended not only on its technical functions but also on whether the intended users considered it useful and appropriate for their production monitoring activities.

The evaluation involved the intended users of the system, particularly personnel who were familiar with the production monitoring activities of AlDi Foods Inc. Their assessment provided feedback on their experience in using the system for entering production information, viewing production analytics, checking historical records, and generating reports.

The researchers considered the users' perception of the system's usefulness, ease of use, clarity of information, and suitability for the production monitoring workflow. Users were also given an opportunity to evaluate whether the automated calculations and dashboard information could support their understanding of production performance.

The user satisfaction evaluation also considered the usefulness of the system's historical records and automated reports. These functions were included because the existing process relied heavily on manually maintained records, making historical retrieval and consolidation more difficult.

Feedback from the users was used to identify areas that could still be improved in the prototype. Comments and evaluation results could be considered when refining the user interface, improving system functions, clarifying information presentation, and determining possible enhancements for future versions of the system.

User satisfaction therefore provided a user-centered perspective that complemented the technical testing procedures described in Section 3.6. While functionality, accuracy, efficiency, and reliability focused on system performance, user satisfaction focused on the users' overall experience and perception of the system's usefulness within the AlDi Foods Inc. production monitoring environment.

## Summary of Evaluation Approach

The evaluation of the proposed production monitoring system was conducted using the six identified criteria: functionality, usability, accuracy, efficiency, reliability, and user satisfaction. These criteria provided a structured basis for determining whether the developed prototype addressed the production monitoring requirements identified during the earlier stages of the study.

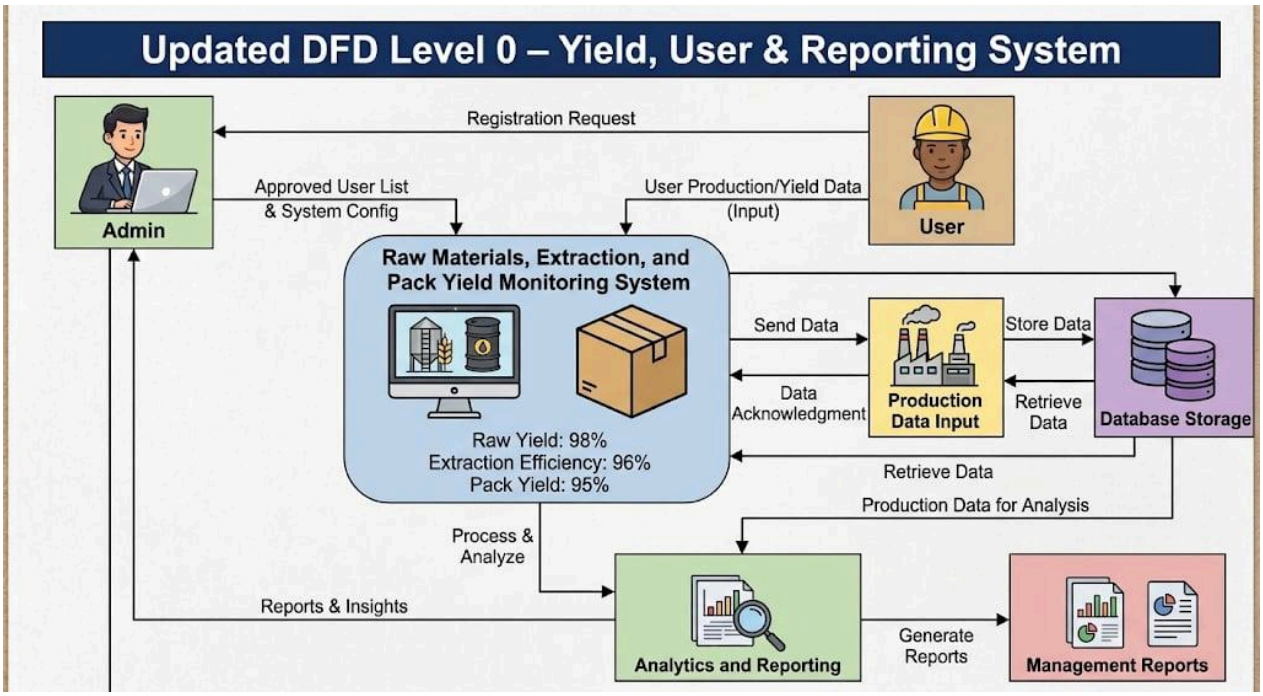
The evaluation process was connected to the overall research workflow. The problems identified in Chapter 1 served as the basis for the system requirements. The related studies and literature discussed in Chapter 2 provided supporting concepts related to digital production monitoring, organized production data, automated calculations, and accessible production information. The requirements, system architecture, development methodology, data gathering techniques, and testing procedures presented in Chapter 3 provided the foundation for developing and evaluating the prototype.

The system was first tested through Unit Testing and System Testing to verify individual functions and integrated system operations. User Acceptance Testing was then used to obtain feedback from the intended users. The evaluation criteria provided a structured means of assessing the prototype after these testing activities.

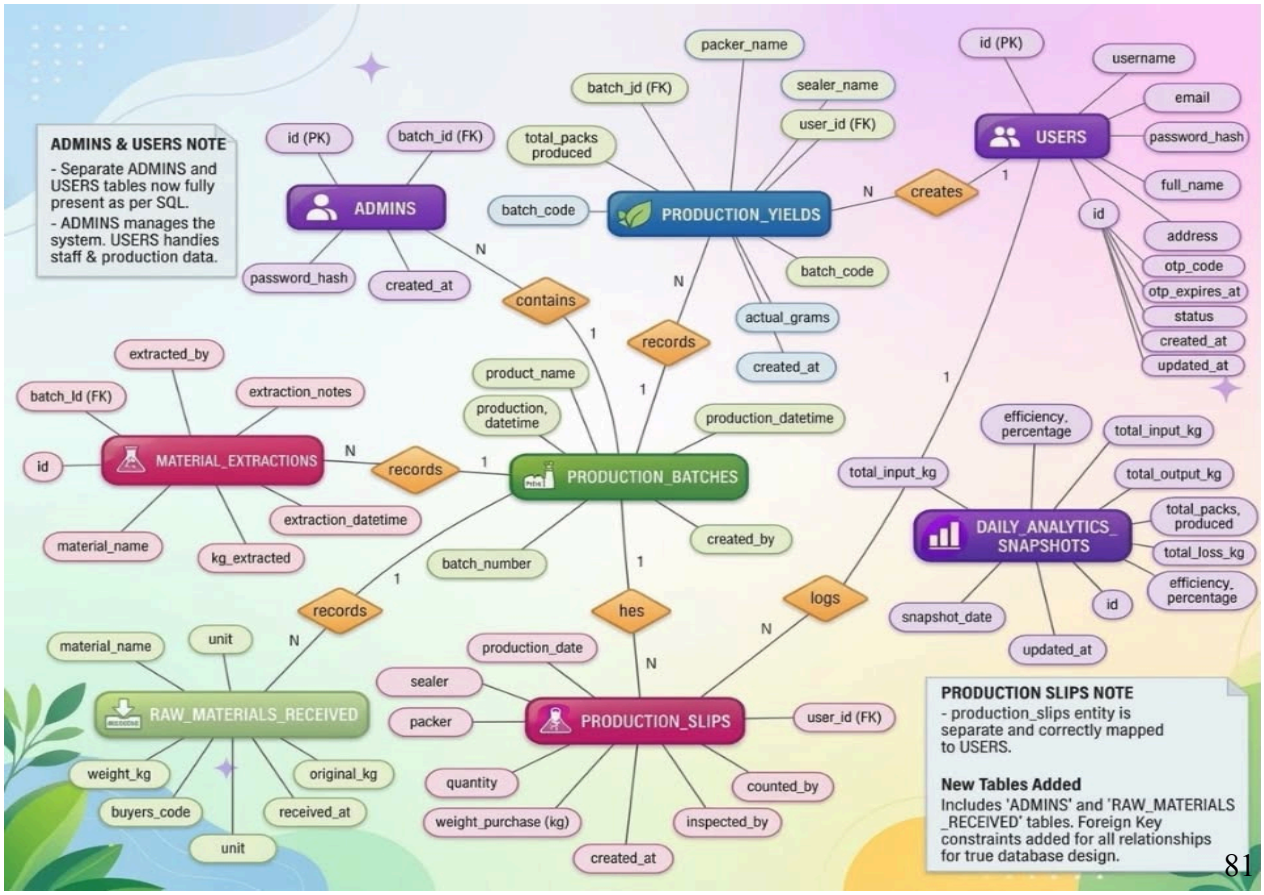
The results obtained from the evaluation were intended to be presented and analyzed in Chapter 4. The findings would indicate the extent to which the developed system satisfied the identified criteria and supported the objectives of the study. These findings would then serve as a basis for the conclusions, recommendations, implementation considerations, system impact, limitations, and future enhancements discussed in Chapter 5.

Overall, the evaluation criteria were designed to determine whether the proposed system could provide a functional, usable, accurate, efficient, reliable, and user-acceptable approach to digital production monitoring within the defined scope of AlDi Foods Inc. The evaluation remained focused on manually encoded production information and did not require IoT devices, advanced industrial automation, or other technologies outside the scope of the study.

## DFD (Data Flow Diagram)




## ERD (Entity Relationship Diagram)



**Users**
The Users entity represents production personnel who interact with the system. Users are responsible for creating production batches, encoding raw materials received, encoding raw material data, and recording packed yield output, production slip. Users play a central role in the system because they provide the primary production data used for real-time analytics and efficiency monitoring.

***Relationships:***

- A User could create multiple Production Batches.
- A User could record multiple Pack Yield entries under different batches.

Raw MATERIALS Management

**Production Batches**
The Production Batch entity represents a single production run or cycle within AlDi Foods Inc. Each batch contains detailed information about the production process, including raw materials used and outputs produced. This entity serves as the main container of production data used for yield computation and extraction efficiency analysis.

***Relationships:***

- A Production Batch belonged to one User.
- A Production Batch contained multiple Batch Materials records. A
- Production Batch could generate multiple Pack Yield records.

**Raw Materials**
The Raw Materials entity stores information about raw materials used in each production batch. This includes the quantity of raw inputs such as fruits or raw juice materials used during the extraction process. This entity is essential for calculating production efficiency and yield percentage.

***Relationships:***

- Batch Materials belonged to a specific Production Batch.
- Each Production Batch could have multiple material entries depending on production requirements.

**Extraction Records**
The Extraction Records entity captures intermediate output data obtained during the processing or extraction phase of a production batch. It logs extracted quantities, timestamps, and operator remarks to track step-by-step progress before final packaging. This entity helps monitor inline operational yield and spot losses occurring mid-process.

***Relationships:***

- Multiple Extraction Records can be logged for one Production Batch.

**Pack Yields**
The Pack Yield entity represents the final output of the production process. It records the amount of finished products (e.g., bottled or packed juice) produced from a specific production batch. This data is crucial for real-time yield analytics and extraction efficiency computation. ***Relationships:***

- Pack Yield belonged to a Production Batch. A Production Batch could produce multiple

- Pack Yield records depending on product variations or packaging types.

**Production Slip**

**Yield Analytics**
The Analytics entity is a standalone system module responsible for processing production data into meaningful insights. It handles real-time computation and visualization of production performance. This entity is central to the system's objective of providing digital production intelligence.

***Functions of Analytics:***

- Computed real-time yield percentage
- Analyzed extraction efficiency performance
- Generated production trends and visual reports
- Classified production status (Stable / Not Stable)
- Generated periodic reports (weekly, monthly, quarterly, yearly)

***Relationships:***

- Analytics was independent but used data from Production Batch, Batch Materials, and Pack Yield.

**Efficiency Records**
The Efficiency Records entity tracks performance efficiency rates for individual production runs. It evaluates expected output standard rates against actual packed output to identify operational bottlenecks and measure worker/machine throughput.

***Relationships:***

- An Efficiency Record belongs to one Production Batch.
- Efficiency Records data can be compiled into multiple Reports.

**Production Stability**
The Production Stability entity logs batch consistency metrics and quality status updates throughout the production cycle. It ensures that outputs meet safety, stability, and quality standards before final distribution.

***Relationships:***

- A Production Stability record belongs to one Production Batch.
- Production Stability data can be compiled into multiple Reports.

**Reports**
The Reports entity aggregates operational, yield, efficiency, and stability metrics into formal documents. It tracks report types and generation dates to support management oversight, compliance audits, and long-term decision-making.

***Relationships:***

- A Report compiles analytical data originating from a Production Batch (including Yield Analytics, Efficiency Records, and Production Stability).

**Summary of Relationships**

**Users**

- **Creates** multiple Production Batches.
- **Records** multiple Pack Yield entries across various batches.

**Production Batches**

- **Belongs to** one User.
- Raw materials received
- **Contains** multiple Raw Materials (Batch Materials) entries.
- **Generates** multiple Extraction Records.
- **Produces** multiple Pack Yield records (for different variations or packaging types).
- Production slip
- **Has** one Efficiency Record.
- **Has** one Production Stability record.

**Raw Materials Received**

**Raw Materials (Batch Materials)**

- **Belongs to** one specific Production Batch.

**Extraction Records**

- **Belongs to** one specific Production Batch (logs mid-process intermediate step data).

**Pack Yields**

- **Belongs to** one Production Batch.
- **Recorded by** a User.

**Production Slip**

**Yield Analytics**

- **Operates independently** while drawing source data directly from Production Batches, Batch Materials, and Pack Yields.

**Efficiency Records**

- **Belongs to** one Production Batch.

- **Compiles into** multiple Reports.

**Production Stability**

- **Belongs to** one Production Batch.
- **Compiles into** multiple Reports.

**Reports**

- **Aggregates** analytical data originating from Production Batches, including Yield Analytics, Efficiency Records, and Production Stability metrics.

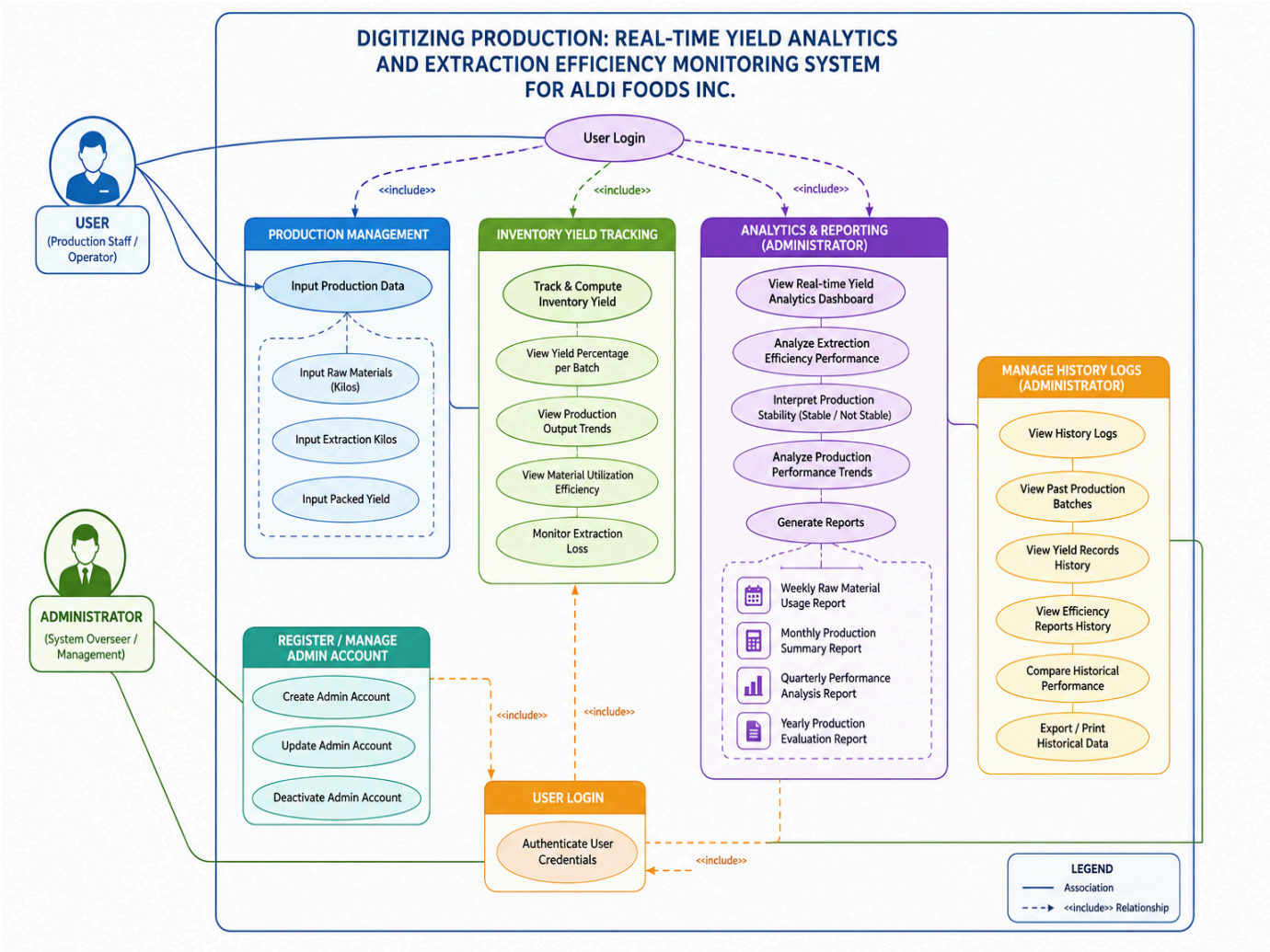
**System Role in Production Monitoring**
This ERD structure supports the main goal of the system, which is to digitize production monitoring at AlDi Foods Inc. It enables:

- **Real-time tracking of raw material usage** by logging specific inputs, quantities, and units mapped directly to each active production batch.
- **Automated yield computation** through the continuous integration of raw material inputs and packed yield outputs to determine yield percentages.
- **Extraction efficiency monitoring** by capturing mid-process operational logs, timestamps, and output volumes during intermediate extraction stages.
- **Production stability classification** by logging batch consistency metrics to automatically categorize quality and operational status.
- **Centralized reporting and analytics** by consolidating yield metrics, efficiency records, and stability status into downloadable management reports.

By structuring the system in this way, the platform ensured accurate, efficient, and real-time monitoring of production activities, replacing manual record-keeping processes and improving decision-making capabilities.

**Use Case Diagram**
The Use Case Diagram illustrates the interactions between the system and its primary actors within the Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc. It clearly defines the functional requirements of the platform and demonstrates how each actor engages with the system to facilitate production monitoring, data analytics, and operational reporting.

The system replaces manual production tracking by establishing a centralized digital platform. This transformation enables the real-time computation of yield metrics, extraction efficiency analysis, production stability interpretations, and automated report generation across the entire manufacturing workflow.




**Actors**

**The system consisted of two primary actors:**

**User**

The User represents production staff or operators responsible for encoding production data into the system. Users are directly involved in recording real-time production information such as raw material usage, extraction output, and packed yield data. The User plays a critical role in ensuring that production data is accurate, complete, and updated in real time, which is essential for generating reliable analytics and efficiency reports.

**Administrator**

The Administrator is responsible for overseeing the entire system. This actor manages user accounts, monitors production performance, reviews analytics dashboards, and generates reports for management decision-making. The Administrator has full access to system data and is responsible for evaluating production performance trends, monitoring efficiency levels, and ensuring data integrity across all modules.

**Register / Manage Admin Account**

This use case allows the Administrator to create, update, and manage admin accounts within the system. It ensures that only authorized personnel have access to sensitive production analytics and system configuration features.

**User Login**
Both Users and Administrators have to authenticate through the login system to access their respective functionalities. This ensures system security and controlled access to production data and analytics.

**Production Management**
This is the core module of the system where Users input all production-related data. It supports real-time monitoring and serves as the foundation for yield analytics and extraction efficiency computation.

Sub-functions included:

***Input Raw Materials Received***

***Input Raw Materials (Kilos)***

Users encode the quantity of raw materials used in production batches. This data is essential for computing yield and efficiency.

***Input Extraction Kilos***

Users record the amount of juice or extracted output obtained from the raw materials during production.

***Input Packed Yield***

Users enter the final packaged product output, which is used to evaluate production losses and efficiency.

***Inventory Yield Tracking***

This use case allows the system to automatically track and compute production yield based on raw material input and output data. It provides real-time updates on:

- Yield percentage per batch
- Production output trends
- Material utilization efficiency
- Extraction loss monitoring

This feature supports continuous monitoring of production performance and helps identify inefficiencies early.

***Input of Production Slip***

**Analytics & Reporting (Administrator)**

This use case enables the Administrator to access and analyze system-generated production data. It includes real-time dashboards and structured reports that support decision-making.

Key functionalities included:

- Real-time yield analytics visualization
- Extraction efficiency performance analysis

- Production stability interpretation (Stable / Not Stable)
- Trend analysis of production performance
- Generation of periodic reports:
    - Weekly raw material usage report
    - Monthly production summary report
    - Quarterly performance analysis
    - Yearly production evaluation report

This module transforms raw production data into meaningful insights for management and operational improvement.

***Manage History Logs***

This use case **allows** the Administrator to view and manage historical production records stored in the system database. It **provides** access to past production batches, yield records, and efficiency reports.

This supports:

- Audit trail of production activities
- Historical comparison of yield performance
- Detection of recurring inefficiencies
- Long-term production analysis and reporting

## System Interaction Summary

The proposed production monitoring system operates through a structured interaction between authorized users and the Administrator to support the production monitoring requirements of AlDi Foods Inc. The interaction was designed based on the existing production monitoring problems identified in the study, particularly the manual recording of raw material usage, extraction output, packed yield, and other production-related information. By transferring these activities into a centralized web-based platform, the system provides an organized process for entering, processing, storing, retrieving, and reviewing production information.

The system begins with the interaction of authorized users with the system interface. Production personnel are primarily responsible for encoding production information based on the actual production activities performed at AlDi Foods Inc. The information entered may include production batch details, raw material quantities, extraction output, and packed yield. These data serve as the primary inputs of the system and provide the basis for the succeeding calculations and production analytics.

The user interaction is supported by the system's authentication and access control functions. Authorized users must log in using their assigned credentials before accessing the production monitoring functions. This provides a controlled environment for managing production information and helps ensure that system functions are accessed only by users with appropriate permissions.

After authentication, users can access the production recording modules. Production batch information is first established to provide an organized reference for the related production data. Raw material quantities are then

recorded and associated with the appropriate production batch. Extraction output and packed yield information are also entered based on the production records available to the personnel. Through this process, the system creates a centralized digital record of production activities instead of relying solely on separate logbooks, paper records, or manually maintained spreadsheets.

Once the required production information has been entered, the Application Layer processes the data according to the programmed system rules and formulas. The system automatically calculates the required production indicators, including yield percentage and extraction efficiency. These calculations reduce the need for users to repeatedly perform manual computations and provide a consistent method for processing the encoded production values.

The system also evaluates production stability using the predefined criteria established during system development. Based on the processed production information, the system categorizes the production result as “Stable” or “Not Stable.” This classification provides users with an immediate indication of the production condition based on the defined thresholds. The stability classification is intended to support production monitoring and review rather than replace the judgment of production personnel or management.

In this study, the term real-time refers to the immediate processing and presentation of information after production data have been manually encoded into the system. The proposed system does not depend on IoT devices, automated sensors, or continuous industrial data streaming. Instead, production personnel manually provide the required information, after which the system immediately processes the data and updates the corresponding production information within the web-based platform.

The processed information is stored in the system database and becomes available for different monitoring and reporting functions. The Dashboard presents summarized production information through indicators, tables, and visualizations. These may include raw material utilization, extraction output, packed yield, yield percentage, extraction efficiency, and production stability. The dashboard allows authorized users and management personnel to review production information in a more organized manner compared with manually consolidating information from separate records.

The system also maintains historical production records. Users with appropriate access can retrieve previously encoded production information for monitoring, comparison, verification, and review. This function addresses the difficulty of locating historical production information from physical logbooks and separate files. By maintaining centralized records, the system provides a more organized source of production information that can support production monitoring and reporting activities.

The Administrator has a broader role in managing and overseeing the system environment. The Administrator can manage authorized users and access the available production information and analytical functions. Through the dashboard, historical records, and reports, the Administrator can review production performance and examine production information based on the records entered by production personnel.

The reporting function further supports management review by organizing production information according to specified periods. The system can provide weekly, monthly, quarterly, and yearly production summaries based on the available records. These reports allow management to review raw material utilization, extraction performance, yield information, and other production indicators without manually consolidating the same information from individual records.

The interaction between users, the application, and the database follows the three-tier architecture described in the system design. The Client Layer provides the interface through which authorized users enter and view information. The Application Layer processes the entered data, performs calculations, applies production stability criteria, and manages system functions. The Database Layer stores production records, user information, and other system data required for retrieval, analytics, and reporting. This structure supports the organized movement of information from data entry to processing, storage, presentation, and reporting.

The system interaction is also connected to the development and testing procedures presented in Chapter 3. The production workflow identified through interviews, observation, and document review served as the basis for designing the interaction among the users and system modules. The major functions were subsequently examined through Unit Testing and System Testing to verify that individual components and integrated workflows operated as intended. User Acceptance Testing (UAT) was then used to obtain feedback from the intended users regarding the system's functionality, usability, accuracy, efficiency, reliability, and overall satisfaction.

The interaction among the system components also supports the evaluation criteria presented in the study. Functionality is demonstrated through the ability of users to record production information and access the required monitoring functions. Usability is supported through the organized interface and production workflow. Accuracy is addressed through automated yield and extraction efficiency calculations and consistent presentation of stored information. Efficiency is supported by reducing repetitive manual calculations and simplifying information retrieval and reporting. Reliability is considered through consistent storage, processing, and retrieval of production records. User satisfaction is assessed through the intended users' experience and perception of the developed system.

The system interaction provides a structured workflow from production data entry to management review. Production personnel provide the manually encoded production information, the application processes and analyzes the information, the database stores the records, and the dashboard and reports present the processed results for monitoring and review. This workflow allows the developed system to serve as a centralized production monitoring platform for AlDi Foods Inc.

The interaction also supports the analytical requirements of the study by connecting production data with the system's calculated indicators. Raw material quantities, extraction output, and packed yield serve as the basis for determining production performance measures. The resulting yield percentage and extraction efficiency provide summarized information that can be reviewed through the dashboard and historical records. Production stability classification further provides a predefined interpretation of the production results based on the criteria implemented in the system.

The centralized storage of production information also supports historical analysis. Instead of requiring personnel to manually search through different production records, the system provides a structured database where previously encoded information can be retrieved. This supports the generation of periodic reports and provides a basis for reviewing production performance across different reporting periods.

The system interaction further addresses the research gap identified in the study by applying digital production monitoring within the specific operational context of AlDi Foods Inc. While related studies discussed in Chapter 2 demonstrate the value of digital tools, production monitoring, data organization, and automated analysis, the proposed system applies these concepts through a practical web-based platform that relies on manually encoded production data. This approach allows the project to remain within its defined scope while addressing the identified need for centralized production records and accessible production analytics.

The interaction also provides the foundation for the results and discussion presented in Chapter 4. The production data recorded during system operation, calculated yield and extraction efficiency values, production stability results, dashboard outputs, historical records, generated reports, and user evaluation responses provide the outputs that can be examined during the evaluation of the prototype. The results can then be interpreted according to the evaluation criteria established in Chapter 3.

The findings from the testing and evaluation activities subsequently provide a basis for Chapter 5. The conclusions can be based on the extent to which the developed system addressed the identified production monitoring problems and achieved the stated objectives. The recommendations can focus on improvements to system functionality, usability, reporting, data management, and other areas identified during testing and user evaluation. Future

enhancements can also be proposed based on the limitations of the current prototype and the needs identified during the implementation and evaluation process.

Overall, the System Interaction Summary demonstrates how the proposed platform connects production personnel, administrative users, production data, automated calculations, database storage, analytics, historical records, and reports into one centralized workflow. This interaction supports the primary purpose of the capstone: to digitize the production monitoring process of AlDi Foods Inc. and provide timely, organized, and accessible production information for operational monitoring and management review.

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CHAPTER 4

# SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULT, AND DISCUSSION

## 4.2 Overview of the Developed System

The Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc. is a web-based production monitoring application developed to support the digitization of the company's existing production monitoring process. The system was designed in response to the problems identified in the study, particularly the reliance on manually recorded production information, manual calculations, separate records, and difficulties in retrieving and reviewing historical production data. Through a centralized digital platform, the developed system provides a structured approach for recording, processing, analyzing, storing, retrieving, and reporting production information related to the juice extraction process of AlDi Foods Inc.

The development of the system was guided by the findings and requirements established throughout the earlier chapters of the study. Chapter 1 identified the operational difficulties associated with the existing production monitoring process, including the possibility of recording inconsistencies, delays in processing information, manual calculation requirements, difficulty in retrieving historical records, and limited visibility of production performance. These conditions created a need for a more organized method of managing production information. The proposed system therefore focused on digitizing the existing workflow rather than introducing complex industrial automation technologies.

The findings from Chapter 2 also provided a basis for the development of the system. The reviewed foreign and local studies and related literature demonstrated the value of digital production records, centralized information management, production monitoring, automated calculations, data visualization, and organized reporting. However, the proposed system was developed specifically for the operational context of AlDi Foods Inc. and within the defined scope of the study. It does not require IoT devices, continuous sensor data collection, advanced machine learning, robotics, or sophisticated industrial automation. Instead, it uses manually encoded production information and processes the data immediately after entry to provide timely production information.

The developed application provides an integrated platform for managing production batch information, recording raw materials, recording extraction output, recording packed yield, calculating production yield, determining extraction efficiency, interpreting production stability, displaying production analytics, maintaining historical records, and generating periodic reports. These functions are organized into interconnected modules so that information entered during production can proceed through data processing, storage, analysis, visualization, and reporting within the same system.

The overall operation of the system follows the three-tier architecture discussed in Chapter 3. The Client Layer provides the user interface through which authorized users interact with the application. The Application Layer handles data validation, business rules, calculations, production stability interpretation, and other system processes. The Database Layer stores user information, production batches, raw material records, extraction information, packed yield records, and other data required by the system. The interaction among these layers provides a structured flow of information from user input to processing and storage and eventually to dashboard presentation, historical retrieval, and reporting.

The system begins with the authentication process. Authorized users must provide their assigned login credentials before accessing the available production monitoring functions. The authentication mechanism establishes controlled access to the system and helps protect production information from unauthorized users. Based on the user's assigned access, the system provides the appropriate functions for recording, viewing, managing, and analyzing production information.

One of the primary components of the developed system is the Production Batch Management Module. This module provides an organized reference for individual production activities. Users can create and manage production batch information and associate the corresponding production records with the appropriate batch. Establishing production batches provides a structured way of organizing raw material, extraction, and packed yield information and supports the retrieval of production records during later monitoring and reporting activities.

The Raw Material Recording Module allows authorized users to encode the quantity of raw materials used for a production batch. This information serves as one of the primary inputs for the system's production calculations and analysis. By storing raw material information digitally, the system provides a centralized record that can be reviewed together with the corresponding extraction output and packed yield information.

The Extraction Output Module provides a means of recording the output obtained from the extraction process. The recorded extraction information is associated with the appropriate production batch so that the system can use it in the succeeding analytical processes. This module supports the study's objective of monitoring extraction performance by providing organized digital records of extraction-related production information.

The Packed Yield Module allows users to record the final packed output associated with a production batch. Together with raw material and extraction information, packed yield provides the production data required by the system to determine relevant yield indicators. The organization of these values within the same production record also helps maintain consistency between the original production information and the resulting analytical outputs.

The Yield Analytics Module processes the encoded production information to calculate the applicable yield percentage. The calculation is performed by the application based on the formula implemented during system development. Instead of requiring users to manually perform the calculation for every production batch, the system processes the values after they have been entered and presents the resulting indicator. This supports more consistent calculation and reduces the repetitive work associated with manual computation.

The system also includes an Extraction Efficiency Monitoring Module. This module calculates the extraction efficiency based on the production information encoded for the applicable batch. The resulting value provides an additional production performance indicator that can be reviewed alongside yield information. By presenting extraction efficiency in the system, production personnel and management can examine variations in extraction performance across recorded production activities.

Another component is the Production Stability Module, which interprets production results using the predefined criteria established during system development. Based on the applicable production values, the system classifies the result as “Stable” or “Not Stable.” This provides a simplified interpretation of production performance that can assist users in identifying production results that require closer review. The classification is based on the programmed criteria and is intended to support, rather than replace, the assessment of production personnel and management.

The use of the term real-time in the system refers to the immediate processing and presentation of newly encoded production information. When authorized users enter the required production data and submit the information, the system processes the data and makes the resulting information available within the applicable system functions. This definition is consistent with the scope established in the study because the system does not automatically collect production information through IoT devices, sensors, or industrial equipment.

The Dashboard Module provides a summarized view of production information and analytical results. It presents relevant production indicators through organized numerical information, tables, and graphical visualizations. Depending on the available records, the dashboard can present information related to raw material utilization, extraction output, packed yield, yield percentage, extraction efficiency, and production stability. The dashboard is intended to make production information easier to review compared with manually consolidating information from separate production records.

The dashboard also provides a connection between production data entry and management monitoring. Once information has been manually encoded and processed by the system, the resulting information can be reflected in the appropriate dashboard components. This allows authorized users to review updated production information without manually recalculating the corresponding indicators. The dashboard therefore serves as one of the primary interfaces for reviewing production performance within the developed prototype.

The Historical Records Module provides access to previously stored production information. Authorized users can retrieve production records for monitoring, verification, comparison, and review. This function addresses one of the limitations identified in the existing process, where historical information may be distributed across physical records, spreadsheets, or other manually maintained sources. Centralizing the records in the system provides a more organized method of accessing previously encoded production information.

The system also includes an Automated Reporting Module that organizes production information according to selected reporting periods. The developed system supports weekly, monthly, quarterly, and yearly production summaries based on the available records. These reports can include relevant production information and analytical results, allowing authorized users to review production performance without manually consolidating information from individual production records.

The centralized database serves as the foundation for the system's information management functions. MySQL is used to store user information, production batch records, raw material data, extraction output, packed yield, calculated or associated production information, and other system records. Centralized storage provides a common location for the production information used by the dashboard, historical records, analytics, and reports. It also supports more organized retrieval and reduces dependence on separate physical records for reviewing information stored within the system.

Data validation is incorporated into the system to support the quality of encoded information. Required fields and applicable input restrictions are used to reduce incomplete or inappropriate entries. The validation functions are particularly important because the proposed system relies on manually encoded production information. The system can process and analyze the information provided by authorized users, but it does not independently verify whether the original physical production measurements are correct. Therefore, the accuracy of the analytical results depends partly on the correctness of the information entered into the system.

The system was developed using PHP for server-side processing, HTML and CSS for the structure and presentation of the user interface, JavaScript for client-side interactions and system behavior, Chart.js for graphical data visualization, and MySQL for database management. The system was developed and tested using the XAMPP environment, which provided the local server and database environment required for the prototype. These technologies were selected because they support the development of a web-based information system within the technical and resource limitations of the study.

The use of a web-based architecture allows the system functions to be accessed through a standard web browser within the environment in which the prototype is deployed. This approach supports the objective of providing a centralized production monitoring platform without requiring specialized industrial monitoring equipment. The system therefore focuses on improving the organization and processing of production information using technologies appropriate to the defined project scope.

The Administrator Module provides management-related functions within the system. Authorized administrators can manage user access and review production information through the available monitoring and reporting functions. Administrator access helps establish control over the system environment and supports the management review of production information. Through the dashboard, historical records, analytics, and reports, administrators can examine the information recorded by production personnel and use the available outputs as references for production monitoring.

The interaction among production personnel, administrators, and the system follows the workflow described in the System Interaction Summary. Production personnel encode the required production information, the application processes the encoded values, the database stores the records, and the dashboard and reporting functions present the resulting information. This creates a structured flow from production data entry to production analysis and management review.

The developed system was also subjected to the testing procedures described in Chapter 3. Unit Testing was used to examine individual system functions, including authentication, production batch management, raw material recording, extraction output, packed yield, calculations, production stability, dashboard retrieval, historical records, and reporting. System Testing was then used to evaluate the interaction of these functions as an integrated production monitoring platform. User Acceptance Testing provided an opportunity for intended users to assess the system based on their experience with its functions and workflow.

The evaluation of the developed system was based on the criteria presented in Chapter 3, namely functionality, usability, accuracy, efficiency, reliability, and user satisfaction. Functionality focused on whether the system performed the required production monitoring tasks. Usability considered the clarity and ease of interaction with the system. Accuracy examined the correctness and consistency of system calculations and information presentation. Efficiency considered the processing, retrieval, and reporting of production information. Reliability considered the consistency of system operations and stored information. User satisfaction considered the users' overall perception of the system's usefulness and suitability for the production monitoring process.

The developed system also provides a connection between the technical outputs of the prototype and the research objectives established in Chapter 1. The ability to record production information digitally addresses the need for more organized production records. Automated yield and extraction efficiency calculations address the need to reduce repetitive manual calculations. Dashboard visualization provides a more accessible view of production performance. Historical records support easier retrieval of previous production information, while automated reports support the preparation of production summaries for management review.

The system further addresses the research gap identified through the related studies and literature in Chapter 2 by applying digital production monitoring concepts within the specific context of AlDi Foods Inc. The system does not attempt to reproduce large-scale industrial automation systems. Instead, it provides a practical web-based solution focused on manually encoded production data, centralized information management, automated calculations, production analytics, historical records, and reporting. This makes the developed prototype consistent with the scope and objectives of the study.

The outputs generated by the developed system provide the basis for the presentation and analysis of results in the succeeding sections of Chapter 4. The researchers can examine the operation of the individual modules, the processing of production information, the displayed analytics, the historical records, the generated reports, and the

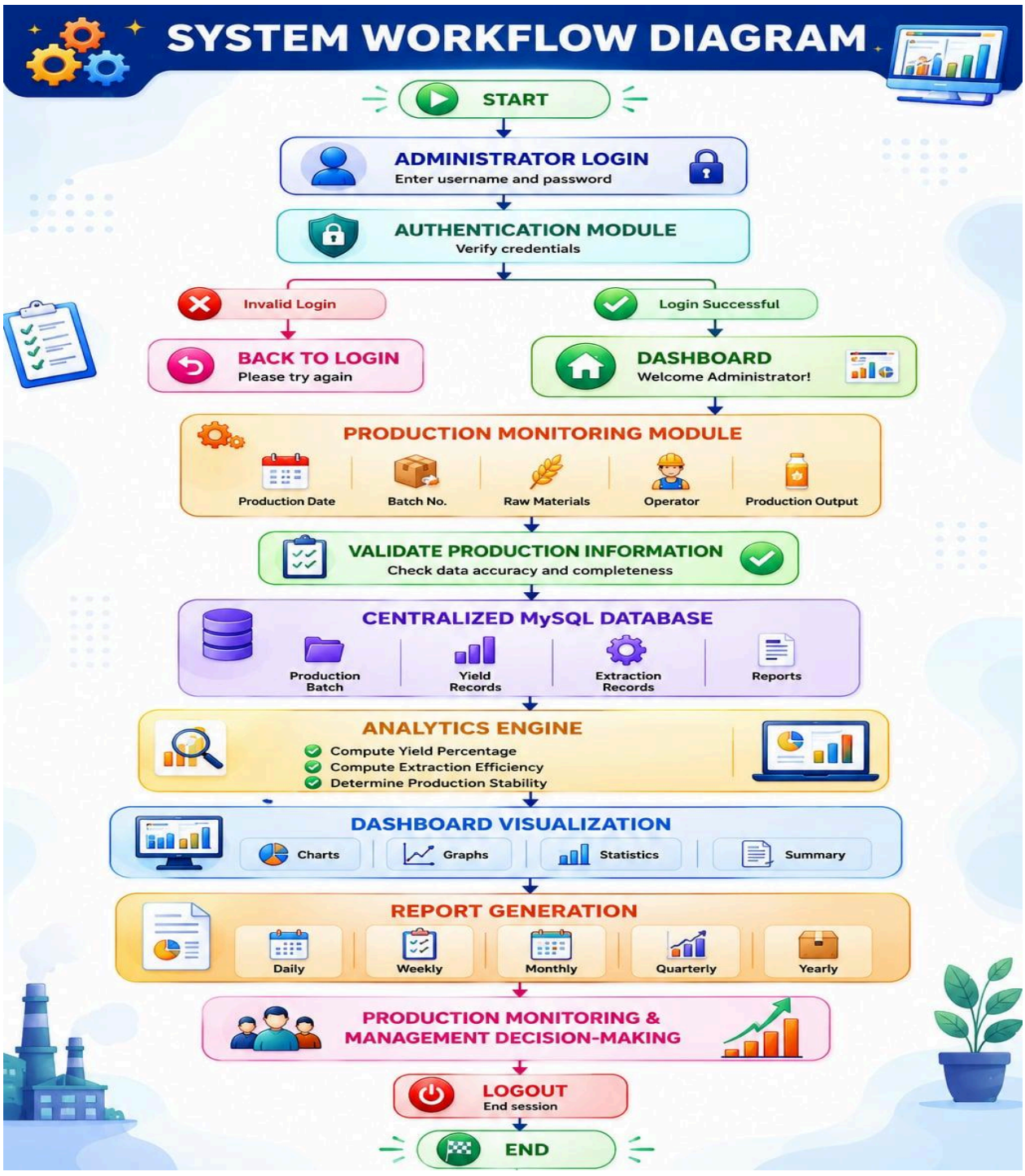
results of system evaluation. These outputs allow the researchers to determine whether the developed prototype performed according to the requirements and objectives established earlier in the study.

The findings from system testing and evaluation also provide the basis for the discussions in Chapter 5. The identified strengths, limitations, user feedback, and observed system performance can be used in developing the conclusions and recommendations of the study. Areas that may require further improvement can also be identified as possible future enhancements, particularly in relation to interface improvements, reporting capabilities, data management, system scalability, and other functions that may be considered beyond the current prototype scope.

Overall, the developed system provides AlDi Foods Inc. with a centralized web-based approach for managing production monitoring information. It connects production batch recording, raw material entry, extraction output, packed yield recording, automated yield computation, extraction efficiency monitoring, production stability interpretation, dashboard visualization, historical record management, and report generation within a single platform. By digitizing these activities, the system supports more organized production information management and provides timely analytical information after manual data entry.

The developed prototype therefore demonstrates the application of a practical and cost-conscious digital solution to the production monitoring requirements of AlDi Foods Inc. Its primary contribution is not the introduction of industrial automation or IoT-based monitoring, but the digitization and organization of existing production monitoring activities through centralized data management, automated calculations, analytical visualization, historical record retrieval, and structured reporting. In this manner, the system supports the overall objective of the study to improve the availability, organization, and usability of production information for operational monitoring and management review.

***Figure 4.1*** *illustrates the* ***System Workflow Diagram*** *of the proposed system.*




The workflow begins with the Administrator logging in and passing authentication to access the dashboard, where production details such as batch numbers, raw materials, and output are encoded and validated before being stored in the centralized MySQL database. Once stored, the system processes this data through an analytics engine to calculate yield and efficiency, presents visual insights on the dashboard, and generates periodic reports that empower management decision-making prior to session logout.

## 4.3 System Development Methodology

The development of the Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc. followed the Agile Software Development Methodology, an iterative and incremental approach that emphasizes continuous development, testing, evaluation, and refinement. The methodology was selected because the proposed system required the researchers to progressively develop several interconnected functions based on the production monitoring requirements identified at AlDi Foods Inc.

The use of Agile was appropriate for the developmental nature of the study because the project involved designing, developing, testing, and evaluating a web-based production monitoring prototype. Instead of developing the entire application at once, the researchers organized the development activities into manageable phases and modules. Each phase provided information that was used in succeeding development activities, allowing the researchers to identify requirements, design the system, implement the functions, test the developed components, and incorporate necessary improvements.

The Agile approach also supported the researchers in maintaining alignment between the developed system and the operational requirements identified during data gathering. Interviews, observation, and document review provided information regarding the existing production monitoring process, including the recording of raw material usage, extraction output, packed yield, production batch information, calculations, historical records, and reporting requirements. These findings were translated into system requirements and subsequently into the different modules of the proposed application.

The development process consisted of seven major phases: Planning, Requirement Analysis, System Design, Development, Testing, Deployment (Prototype), and Review and Feedback. Although the phases are presented sequentially, the development process allowed the researchers to return to earlier activities when testing results or user feedback indicated that modifications were necessary. This iterative process helped ensure that the prototype remained consistent with the objectives, scope, and requirements of the study.

## Planning

The Planning phase served as the initial stage of the system development process. During this phase, the researchers established the direction and scope of the project based on the production monitoring problems identified at AlDi Foods Inc. The researchers defined the objectives, expected system outputs, project limitations, development resources, technologies, and general development activities required to produce the proposed prototype.

The researchers examined the existing production monitoring practices to understand how production information was currently recorded, calculated, stored, and reviewed. The existing process relied primarily on manually recorded information, including raw material usage, extraction output, packed yield, and other production-related data. The researchers considered the difficulties associated with this process, such as possible recording inconsistencies, delayed processing, repetitive manual calculations, difficulty in retrieving historical information, and limited visibility of production performance.

The identified problems provided the initial basis for determining the scope of the proposed system. Rather than replacing the entire industrial production process, the project focused specifically on digitizing the production monitoring and information management activities within the defined scope of the study. The system was therefore planned as a web-based platform capable of accepting manually encoded production information and immediately processing the entered data.

The Planning phase also considered the limitations established in the study. The proposed system was not designed as an IoT or IIoT platform and did not require automatic sensor data collection, robotics, advanced machine learning, or sophisticated industrial automation. The system was planned around manually encoded production information so that the prototype could provide a practical and cost-conscious approach to production monitoring.

The researchers also identified the technologies required for development. PHP was selected for server-side processing, HTML and CSS for the structure and presentation of the interface, JavaScript for client-side interaction, Chart.js for graphical data visualization, MySQL for centralized database management, and XAMPP as the local development and server environment. These technologies provided the technical foundation required to implement the proposed web-based production monitoring system.

The output of the Planning phase was a defined project direction that established the system scope, objectives, technologies, development requirements, and expected functions. These outputs served as the foundation for the subsequent Requirement Analysis phase.

## Requirement Analysis

The Requirement Analysis phase focused on identifying and documenting the functional and non-functional requirements of the proposed production monitoring system. The requirements were derived from the actual production monitoring activities and information needs identified during the data gathering process.

As discussed in Section 3.5, the researchers used interviews, observation, and document review to gather the information required for system development. Interviews were conducted with relevant production personnel and authorized representatives to determine how production information was currently recorded, calculated, reviewed, and reported. Observation was used to examine the actual production monitoring workflow, while document review provided additional information from existing production records, logbooks, daily reports, raw material records, extraction output records, packed yield information, inventory-related documents, and other applicable monitoring forms.

The gathered information was analyzed to determine the data that needed to be recorded and the functions that needed to be provided by the system. The major functional requirements included user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, yield percentage computation, extraction efficiency calculation, production stability classification, dashboard presentation, historical record retrieval, and automated report generation.

The system also required functions for organizing production information according to production batches. This requirement allowed raw material, extraction output, and packed yield information to be associated with the appropriate production activity. Such organization provided a foundation for subsequent calculations, dashboard presentation, historical record retrieval, and reporting.

The analytical requirements were also identified during this phase. The system needed to calculate yield percentage and extraction efficiency using the production information entered by authorized users. It also needed to classify production results as “Stable” or “Not Stable” according to the predefined criteria established for the system.

Non-functional requirements were also considered. The system was expected to provide an understandable interface, appropriate access control, consistent data processing, organized information storage, efficient retrieval of production records, and reliable presentation of production information. These requirements were later reflected in the evaluation criteria presented in Section 3.7, including functionality, usability, accuracy, efficiency, reliability, and user satisfaction.

The Requirement Analysis phase therefore established the information and system functions that the prototype needed to support. These requirements became the basis for the architecture, database, interface, and module designs developed in the next phase.

## System Design

After the system requirements were identified, the researchers proceeded to the System Design phase. This phase focused on translating the identified requirements into a structured design that could guide the actual development of

the application. The researchers designed the system architecture, database structure, process flows, user interfaces, and relationships among the different system components.

The system architecture followed a three-tier structure consisting of the Client Layer, Application Layer, and Database Layer. The Client Layer provides the interface through which authorized users interact with the system. The Application Layer manages system processing, validation, calculations, production stability interpretation, and other application functions. The Database Layer stores the production information and other system records required for processing, retrieval, analytics, and reporting.

The researchers also prepared system diagrams to represent the planned operation and structure of the application. The System Workflow Diagram illustrates the major development and operational processes. The System Architecture Diagram presents the interaction among users, application functions, and data storage. The Data Flow Diagram (DFD) illustrates how production information moves from the users through the system and database and eventually becomes available for monitoring and reporting. The Entity Relationship Diagram (ERD) defines the relationships among the database entities required by the application.

The database design was developed to support the centralized storage of production information. The database structure accommodates information associated with users, production batches, raw materials, extraction records, packed yields, analytical results, and reports, consistent with the final implemented database structure of the prototype. The relationships among these records allow the system to connect production inputs with the resulting analytical information.

The user interface was also designed during this phase. The researchers emphasized a simple and organized interface to support the production monitoring workflow. The design included interfaces for authentication, dashboard monitoring, raw material management, production batch information, extraction recording, packed yield recording, historical records, analytics, reports, and user-related functions according to the implemented system modules.

The design phase also considered the user workflow identified during the interviews and observations. The interface was therefore structured so that users could enter production information in an organized sequence, while the system could process the entered values and present the resulting information through the dashboard, historical records, and reports.

The outputs of the System Design phase served as the blueprint for the Development phase. By establishing the architecture, database structure, data flow, and interface design before implementation, the researchers had a defined structure to guide the coding and integration of the system components.

## Development

The Development phase involved translating the approved system design into the functional web-based prototype. The researchers implemented the system using the technologies identified during the Planning phase. PHP was used for server-side processing and system logic, while HTML and CSS were used to structure and present the user interface. JavaScript was used to support client-side interactions and dynamic system behavior. Chart.js was integrated for graphical presentation of production information, while MySQL was used for centralized storage and retrieval of system records. XAMPP provided the local server and database environment during development and testing.

The development process followed the Agile principle of incremental implementation. Rather than implementing all functions simultaneously, the researchers developed the system components progressively and verified their operation before integrating them into the larger production monitoring workflow.

The User Authentication Module was developed to provide controlled access to the system. This module allows authorized users to log in using their assigned credentials and access the functions permitted by their role. The

authentication function provides the initial security mechanism for protecting production information and controlling access to the system.

The Production Batch Management Module was then developed to organize individual production activities. The module allows production batch information to be recorded and associated with other production data. This provides a reference through which raw material, extraction output, packed yield, and related analytical information can be organized.

The Raw Material Recording Module was implemented to allow users to encode the quantity of raw material used for a production batch. The Extraction Output Module and Packed Yield Module were subsequently developed to record the corresponding production output information. These modules form the primary data-entry components of the production monitoring workflow.

The Yield Analytics Module was implemented to process the encoded production information and calculate the applicable yield percentage. The Extraction Efficiency Module was also developed to calculate extraction efficiency based on the applicable production values. These automated calculations were designed to reduce repetitive manual computation and provide consistent processing of the encoded information.

The Production Stability Module was developed to classify production results according to the predefined criteria established for the prototype. The system produces a “Stable” or “Not Stable” classification based on the applicable production conditions and programmed thresholds.

The Dashboard Module was implemented to present summarized production information through indicators, tables, and graphical visualizations. The dashboard allows users to review production information after it has been encoded and processed by the system. The information presented may include raw material utilization, extraction output, packed yield, yield percentage, extraction efficiency, and production stability.

The Historical Records Module was developed to provide access to previously stored production information. This module supports the retrieval and review of records that may be needed for monitoring, verification, comparison, or reporting purposes.

The Automated Reporting Module was also implemented to organize production information into applicable reporting periods, including weekly, monthly, quarterly, and yearly summaries. The module uses the stored production information to present organized production reports without requiring users to manually consolidate the same information from separate records.

During development, the researchers performed module-level verification to determine whether newly implemented functions operated according to their intended requirements. When problems or inconsistencies were identified, the affected functions were modified before proceeding with further integration. This iterative process allowed the researchers to refine the prototype throughout development.

The Development phase therefore transformed the requirements and system designs into an operational prototype that could proceed to formal testing. The resulting application provided the core functions required to support the digitization of production monitoring activities at AlDi Foods Inc.

## Testing

After the major system functions were developed, the prototype underwent the structured Testing phase described in Section 3.6. Testing was conducted to verify whether the individual components and integrated system operated according to the requirements established during the Requirement Analysis phase.

The first level of testing involved Unit Testing. Individual functions were examined to determine whether they performed their intended tasks. The researchers tested user authentication, production batch management, raw

material recording, extraction output recording, packed yield recording, yield percentage computation, extraction efficiency calculation, production stability classification, dashboard data retrieval, historical record retrieval, and automated report generation.

For the analytical functions, sample production values were entered and the resulting system calculations were compared with expected values. This allowed the researchers to verify the implementation of the yield percentage and extraction efficiency calculations. Production stability classification was also examined using production values representing different conditions to determine whether the system correctly applied the predefined criteria.

After individual functions were verified, System Testing was performed to examine the system as an integrated platform. System Testing considered the interaction among the Client, Application, and Database layers. The researchers tested complete production monitoring scenarios, beginning with user authentication and production batch creation and continuing through raw material entry, extraction output, packed yield entry, automated processing, dashboard updating, historical record retrieval, and report generation.

System Testing also included the examination of invalid, incomplete, or inappropriate data entries. The researchers checked whether the applicable validation mechanisms responded appropriately when required information was missing or when entered information did not meet the required conditions.

Integration among the different functions was evaluated as part of System Testing rather than treated as a separate testing phase. The researchers verified whether data entered through the interface were properly processed by the application, stored in the database, retrieved for dashboard presentation, and included in historical records and reports.

Finally, User Acceptance Testing (UAT) was conducted with the intended users to obtain feedback regarding the developed prototype. The UAT focused on the users' experience with the production monitoring functions and their assessment of the system's functionality, usability, accuracy, efficiency, reliability, and overall satisfaction.

The testing results were used to identify functions that required correction or refinement. Necessary adjustments were made before the prototype proceeded to the Deployment phase. This iterative testing process helped reduce the possibility of functional errors and ensured that the system was sufficiently prepared for prototype evaluation.

## Deployment (Prototype)

After the completion of the required testing and corrections, the project proceeded to the Deployment (Prototype) phase. This phase involved preparing the developed application for demonstration and evaluation within the defined environment of the study.

The prototype was configured using the XAMPP environment, with Apache providing the local web server functionality and MySQL supporting the system database. The required database tables and system records were configured according to the developed database design. User accounts and appropriate access permissions were also established for the intended evaluation activities.

The prototype deployment allowed the researchers and intended users to interact with the developed system using the same general workflow implemented during development and testing. Sample or applicable production information could be encoded to demonstrate the operation of production batch management, raw material recording, extraction output recording, packed yield recording, yield and extraction efficiency calculations, production stability classification, dashboard visualization, historical records, and report generation.

The intended users were also oriented regarding the basic operation of the prototype. The orientation covered system login, navigation, production data entry, viewing of analytical results, dashboard interpretation, historical record retrieval, and report generation according to the functions available in the implemented system.

The Deployment phase in this study refers specifically to prototype deployment and evaluation. It does not represent a claim that the system was fully deployed as a permanent enterprise-wide production system of AlDi Foods Inc. Full-scale implementation, long-term infrastructure management, enterprise-level security administration, and large-scale production deployment were outside the defined scope of the study.

## Review and Feedback

The final phase of the Agile development process involved Review and Feedback. This phase provided an opportunity for the researchers to evaluate the developed prototype based on the observations and responses obtained during testing and user evaluation.

The intended users provided feedback regarding the system interface, navigation, production data entry, dashboard presentation, analytical results, historical records, report generation, and general system operation. Their feedback was important because the system was designed to support actual production monitoring activities and therefore needed to be understandable and practical for the intended users.

The researchers reviewed the feedback in relation to the evaluation criteria established in Section 3.7. Functionality was considered in terms of whether the system performed the required production monitoring tasks. Usability focused on the clarity and ease of using the system. Accuracy considered the correctness and consistency of the system's processing and calculations. Efficiency considered the processing, retrieval, and presentation of production information. Reliability considered the consistency of system operations and stored records. User satisfaction considered the users' overall perception of the system's usefulness and suitability.

The feedback obtained during this phase was also considered in identifying areas for refinement. Issues related to interface organization, data entry, information presentation, reporting, or other system functions could be addressed through additional development iterations when necessary and within the scope of the prototype.

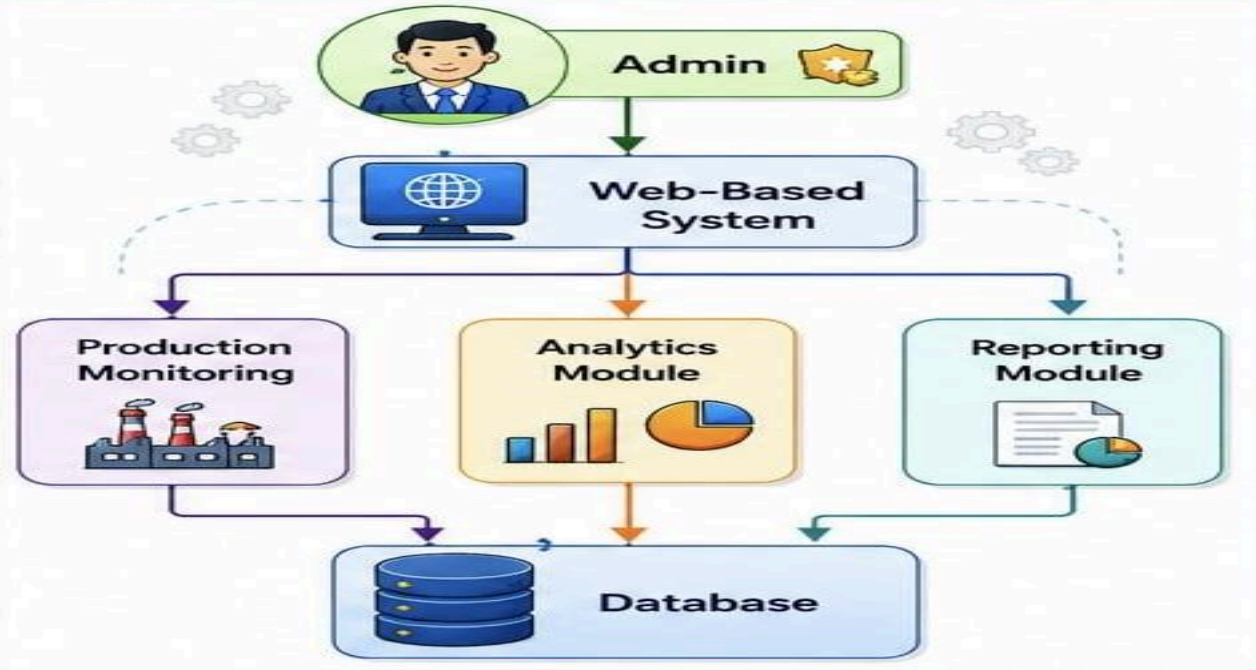
The Review and Feedback phase therefore completed the iterative development cycle by connecting system development with actual user evaluation. The information obtained from this phase contributed to the evaluation results presented in Chapter 4 and provided supporting information for the conclusions and recommendations discussed in Chapter 5.

Overall, the Agile Software Development Methodology provided the researchers with a structured yet flexible approach for developing the proposed production monitoring system. The seven phases—Planning, Requirement Analysis, System Design, Development, Testing, Deployment (Prototype), and Review and Feedback—provided a logical progression from identifying the production monitoring problems to developing, testing, and evaluating the proposed solution.

The methodology also maintained a direct connection with the overall research process. The production monitoring problems identified in Chapter 1 established the need for the system; the related studies and literature in Chapter 2 provided supporting concepts and identified relevant research gaps; the data gathering, architecture, design, development, testing, and evaluation procedures in Chapter 3 guided the construction of the prototype; and the resulting system outputs and evaluation findings provided the basis for the discussions in Chapter 4 and the conclusions and recommendations in Chapter 5.

Through this iterative approach, the researchers developed a web-based production monitoring prototype that remained within the defined scope of the study. The resulting system focused on manually encoded production information, immediate processing after data entry, automated yield and extraction efficiency calculations, production stability interpretation, dashboard analytics, historical record management, and periodic reporting. This approach supported the primary objective of the capstone to digitize and organize production monitoring activities at AlDi Foods Inc. without requiring IoT devices or advanced industrial automation technologies.

*Figure 4.2* presents the interaction between the Administrator and the Web-Based System, highlighting the major functional modules of the proposed application.




As *illustrated in Figure 4.2,* the Administrator serves as the primary user of the web-based application and is responsible for managing all production monitoring activities. After successfully logging into the system, the administrator interacts with the Web-Based System, which serves as the central platform for recording, processing, storing, and presenting production information.

The application is composed of three major functional modules: the Production Monitoring Module, the Analytics Module, and the Reporting Module. The Production Monitoring Module enables administrators to encode production batch information, including the production date, raw material quantity, operator information, and production output. These production records are stored securely in the centralized MySQL database, ensuring data consistency and efficient retrieval of historical production information.

The Analytics Module retrieves production data from the database and automatically computes the production yield percentage and extraction efficiency for each production batch. The computed results are displayed through graphical dashboards, charts, and summary tables, allowing administrators to monitor production performance, evaluate extraction efficiency, identify operational trends, and assess overall production stability.

The Reporting Module generates comprehensive production reports based on the processed information stored in the database. These reports provide summarized production information, yield analytics, extraction efficiency results, and other relevant operational data that support production monitoring, managerial evaluation, and data-driven decision-making.

The interaction illustrated in Figure 4.2 demonstrates how the proposed system integrates production monitoring, analytics, database management, and reporting into a single web-based platform. Throughout the Agile development process, feedback obtained from administrators and intended users is continuously incorporated into succeeding iterations, ensuring that the developed application remains reliable, user-friendly, efficient, and aligned with the production monitoring requirements of AlDi Foods Inc.

## 4.4 System Architecture

- The system architecture of the proposed production monitoring system for AlDi Foods Inc. follows a three-tier architecture composed of the Client Layer, Application Layer, and Database Layer. The architecture was designed to provide a structured separation between the user interface, system processing functions, and data storage. This approach supports the objectives of the study by providing a centralized web-based platform for recording, processing, monitoring, and reviewing production information related to raw material utilization, extraction output, and packed yield.

- The architecture is connected to the production monitoring problems identified in Chapter 1, particularly the reliance on manual logbooks, spreadsheets, and paper-based records. Instead of requiring production personnel to perform repeated manual calculations and search through separate records, the proposed system allows authorized users to encode production information through a web-based interface. The submitted information is then processed by the Application Layer and stored in the centralized database. The resulting information can be immediately presented through dashboards, historical records, and reports.

- The system consists of the following three core layers:

- Client Layer (Presentation Layer). The Client Layer serves as the front-end or presentation component of the system. It provides the web-based interface through which authorized users, including production personnel, supervisors, and administrators, interact with the application. Users first access the authentication interface where they provide their account credentials before accessing the functions assigned to their roles.

- After successful authentication, authorized users can access the appropriate production monitoring functions. These functions include production batch management, raw material recording, extraction output recording, packed yield recording, dashboard viewing, historical record access, and analytics. The interface is designed to organize production information into understandable forms, tables, dashboard elements, and reports so that users can review production performance without relying solely on handwritten records or separate spreadsheets.

- The Client Layer also serves as the main point for manual data encoding. Production personnel enter relevant production information, such as the quantity of raw materials used, extraction output, and packed yield. The data entered by the user is submitted to the Application Layer for validation and processing. Therefore, the system does not depend on IoT devices, automated industrial sensors, or continuous sensor streaming. In this study, the term real-time refers to the immediate processing and presentation of newly encoded production information after it has been submitted to the system.

- The Client Layer also provides access to visual information generated from the processed production data. Through the dashboard and analytics functions, users can review production-related indicators such as raw material utilization, extraction output, packed yield, yield percentage, extraction efficiency, and production stability status. These interfaces support supervisors and authorized personnel in reviewing production information in a more organized and accessible manner.

- Application Layer (Logic Layer / Central Processing Engine). The Application Layer serves as the central processing component of the system. It receives requests and data submitted through the Client Layer, applies the appropriate validation and business rules, performs calculations, and communicates with the Database Layer when information needs to be stored or retrieved. This separation allows the system's processing functions to operate independently from the presentation interface and database storage.

- One of the primary responsibilities of this layer is to validate production information before it is stored. The system checks submitted data based on the defined requirements and validation rules to help prevent

incomplete or inappropriate entries. After valid production information is received, the Application Layer processes the data according to the calculations implemented in the system.

- The Application Layer performs the automated calculation of yield percentage and extraction efficiency based on the production data entered by authorized users. These calculations reduce the need for users to repeatedly perform manual computations and allow the calculated results to be presented consistently throughout the system. The layer also applies the predefined rules used to classify production stability into categories such as “Stable” and “Not Stable.” The classification is based on the thresholds and conditions defined for the proposed system rather than on automated machine or sensor readings.

- In addition to calculations and classifications, the Application Layer manages the system's production monitoring functions. These include user authentication, production batch processing, raw material recording, extraction output processing, packed yield processing, dashboard data preparation, historical record retrieval, and automated report generation. When users request historical production information or reports, the Application Layer retrieves the necessary records from the Database Layer, processes them as required, and prepares the information for presentation through the Client Layer.

- The Application Layer also supports the real-time aspect of the proposed system. Once an authorized user submits valid production information, the application processes the data and makes the corresponding calculated and summarized information available through the system interface. This provides more immediate visibility of production information compared with a process where calculations and summaries are performed only after manual consolidation of separate records.

- Database Layer (Data Management Layer / Centralized Repository). The Database Layer serves as the centralized data storage component of the system. It is responsible for storing, organizing, retrieving, and maintaining the production and user information required by the application. A relational database is used to provide structured storage of records and to establish relationships among the different data entities used by the system.

- The database stores information associated with user accounts and roles, production batches, raw material records, extraction outputs, packed yields, calculated production indicators, and production stability information. By maintaining these records in a centralized repository, the system provides a common location for production information instead of relying on multiple logbooks, spreadsheets, or paper-based records.

- The Database Layer also supports historical monitoring. Previously encoded production records can be retrieved by the application when authorized users need to review past batches, compare production information, or generate reports. This supports the historical records and analytics functions described in Chapter 4 and addresses the difficulty of retrieving and consolidating historical production information identified in the existing production process.

- Data consistency is also supported through the structured organization of database records and the validation performed by the Application Layer before information is stored. However, because the proposed system relies on manually encoded production data, the accuracy of the stored information also depends on the correctness of the information entered by authorized users. The architecture therefore provides a structured method for managing production information while recognizing the limitations of manual data entry.

- The interaction among the three layers follows a structured sequence. First, an authorized user accesses the system through the Client Layer and enters or requests production information. Second, the Application Layer receives the request, validates the data, performs the required calculations and business rules, and communicates with the Database Layer. Third, the Database Layer stores new information or retrieves existing records. The Application Layer then processes the retrieved information when necessary and sends

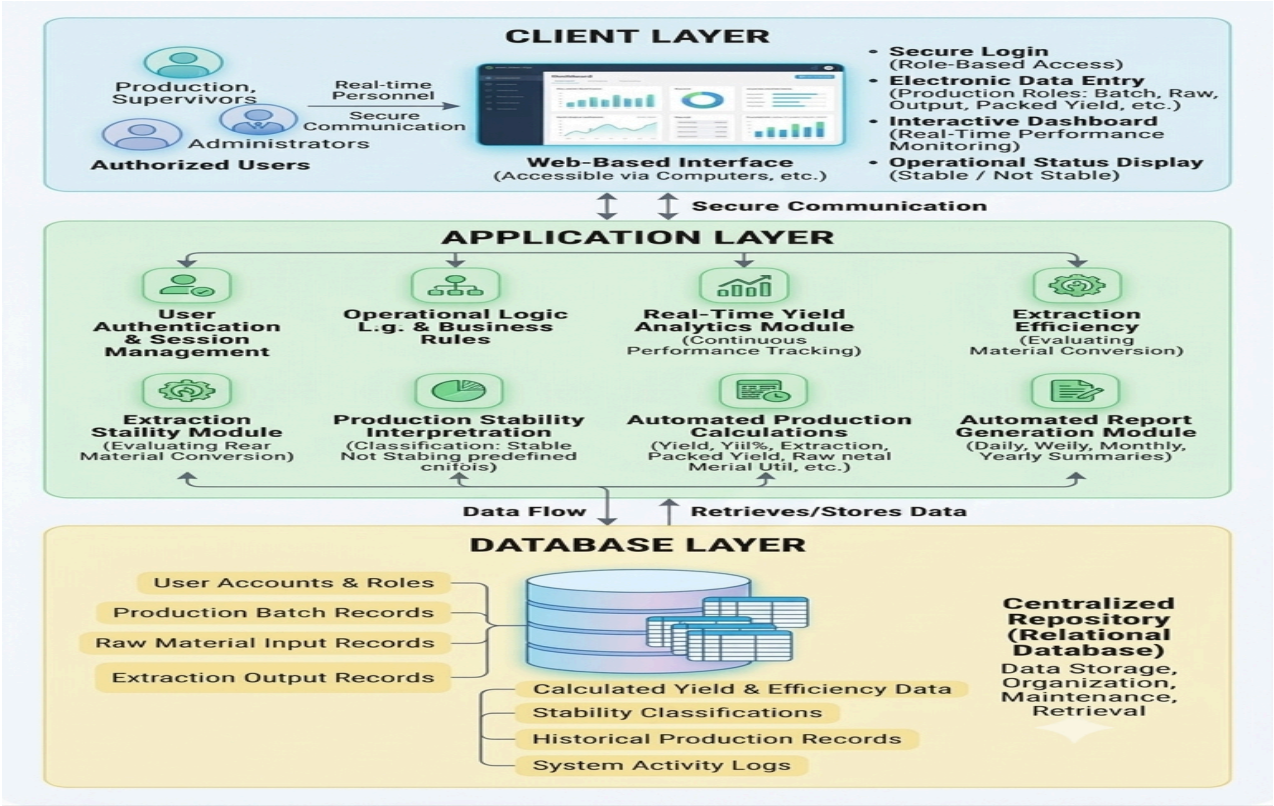
the appropriate result back to the Client Layer, where it is presented to the user through forms, tables, dashboards, analytics, historical records, or reports.

- This three-tier architecture supports the overall system objectives established in Chapter 1 and the requirements identified through the data-gathering techniques discussed in Chapter 3. The interviews, observations, and document review provided information regarding the existing production recording process and the data required for monitoring. These requirements were translated into the system functions implemented across the three architectural layers. The architecture also provides a basis for the testing procedures described in Section 3.6, where the individual functions and the interaction among the system components are evaluated.

- The architecture further supports the system's evaluation criteria, particularly functionality, usability, accuracy, efficiency, reliability, and user satisfaction. The Client Layer supports usability by providing interfaces for data entry and information viewing. The Application Layer supports functionality and calculation processing, while the Database Layer supports organized storage and retrieval of production records. These components work together to provide the functions evaluated during system testing and User Acceptance Testing (UAT).

- Overall, the three-tier architecture provides the structural foundation of the proposed AlDi Foods production monitoring system. It separates user interaction, application processing, and data management while allowing the three components to communicate as part of one centralized system. Through this architecture, manually encoded production information can be converted into processed and organized production indicators, including yield percentage, extraction efficiency, and production stability. The resulting information can then be presented through dashboards, historical records, analytics, and reports to support the monitoring and review of production performance.

*Figure 4.3 illustrated* the overall System Architecture of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.*



## 4.5 Hardware and Software Requirements

The successful implementation of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* requires appropriate hardware and software resources to ensure the efficient operation of the proposed web-based application. Since the system is responsible for recording production batch information, automatically computing production yield and extraction efficiency, generating graphical dashboards, and producing management reports, reliable computing resources are necessary to maintain system performance, data accuracy, and operational efficiency.

The hardware and software requirements are carefully selected based on the functional requirements identified during the system analysis and design phases. These resources provide the necessary environment for the installation, development, testing, deployment, and daily operation of the proposed application. The selected technologies are widely available, cost-effective, and compatible with the existing information technology infrastructure of AlDi Foods Inc., making the proposed system practical for implementation without requiring expensive specialized equipment.

The hardware specifications ensure that the application can efficiently process production data, perform automated computations, retrieve historical records, generate graphical visualizations, and support report generation with minimal processing delays. Likewise, the software requirements provide the development framework, programming environment, database management capabilities, and web technologies needed to build and operate the application successfully.

Since the proposed system utilizes a centralized database, all production information is securely stored and managed within a single repository. This centralized approach improves data consistency, minimizes data redundancy, and enables administrators to retrieve production records quickly whenever monitoring, analysis, or report generation is required. The recommended hardware specifications also provide sufficient processing power and storage capacity to accommodate increasing production records as the company continues to expand its operations.

The software technologies selected for the development of the system are based on open-source and industry-standard applications that are widely used in web development. These technologies support the implementation of the system's major functionalities, including secure administrator authentication, production batch management, automated yield computation, extraction efficiency monitoring, dashboard visualization, centralized database management, and automated report generation. The use of open-source software also minimizes implementation costs while ensuring that the system remains maintainable, scalable, and easy to update in the future.

Furthermore, the combination of the recommended hardware and software provides a stable and reliable environment for the continuous operation of the proposed application. The selected components ensure that the system can perform real-time production monitoring, efficiently process production information, maintain accurate historical records, and provide administrators with timely analytical reports that support operational planning and decision-making.

The following sections present the minimum and recommended hardware specifications as well as the software requirements necessary for the successful development, deployment, and implementation of the proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* These requirements serve as the technical foundation that enables the application to operate efficiently while achieving the objectives of improving production monitoring, minimizing manual computation, enhancing data accuracy, and supporting data-driven decision-making within the company.

### 4.5.1 Hardware Requirements

The hardware requirements specify the minimum and recommended computer specifications necessary for the successful implementation and operation of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* These hardware components provide the computing resources needed to support the installation, execution, and maintenance of the proposed web-based application. Selecting appropriate hardware ensures that the system can perform efficiently while processing production information, executing automated computations, storing production records, and generating analytical reports without experiencing significant delays or performance issues.

Since the proposed system operates as a web-based application, the hardware must be capable of supporting server-side processing, database management, and real-time interaction between the administrator and the application. The recommended specifications are selected based on the expected workload of the system, including encoding production batch information, computing yield percentage and extraction efficiency, generating graphical dashboards through Chart.js, retrieving historical production records, and producing management reports.

Adequate processing power is essential because the application performs multiple operations simultaneously, including validating user inputs, processing production data, executing automated computations, communicating with the MySQL database, and displaying analytical results through graphical visualizations. A computer with a higher-performance processor enables the application to execute these operations more efficiently, resulting in faster response times and an improved user experience.

Memory (RAM) also plays an important role in ensuring smooth system performance. Sufficient memory allows the web browser, web server, database server, and other supporting applications to run simultaneously without affecting system responsiveness. As production records continue to increase over time, additional memory helps maintain stable system performance while retrieving historical production information and generating reports.

Storage capacity is another important hardware consideration because the application stores production batch information, yield records, extraction efficiency records, generated reports, and other related system files within the database. The use of Solid-State Drive (SSD) storage is recommended because it provides significantly faster data access, shorter application loading times, quicker database transactions, and improved overall system performance compared to traditional hard disk drives (HDD). Faster storage also enhances the retrieval of historical production records and improves the efficiency of report generation.

The monitor specification ensures that administrators can comfortably view the system's user interface, production forms, dashboards, charts, graphs, and generated reports. A higher display resolution provides clearer visualization of production analytics, allowing administrators to analyze production performance more effectively. Standard USB keyboard and mouse devices provide the necessary input tools for encoding production information, navigating the system, and performing administrative tasks efficiently.

Although the proposed system can operate within a local network environment, a reliable internet or network connection is recommended to support system accessibility, software updates, and future deployment in a networked production environment. A stable connection also facilitates faster communication between the web server and client devices when the application is accessed through multiple workstations.

*Table 4.1* presents the minimum and recommended hardware specifications required for the implementation of the proposed system.

**Table 4.1 Hardware Requirements of the Proposed System**

| Hardware Component | Minimum Specification | Recommended Specification |
|---|---|---|
| Processor | Intel Core i3 or AMD Ryzen 3 | Intel Core i5 or AMD Ryzen 5 and above |
| Memory (RAM) | 4 GB | 8 GB or higher |
| Storage | 256 GB SSD | 512 GB SSD or higher |
| Monitor | 1366 × 768 Resolution | Full HD (1920 × 1080) |
| Keyboard and Mouse | Standard USB Keyboard and Mouse | Standard USB Keyboard and Mouse |
| Internet Connection | 10 Mbps | 50 Mbps or higher |

The minimum hardware specifications are sufficient to install and operate the proposed web-based application under normal production monitoring conditions. These specifications allow administrators to encode production information, perform automated yield and extraction efficiency computations, access production records, and generate reports with acceptable system performance.

However, the recommended hardware specifications provide improved processing capability, faster database operations, greater storage capacity, and enhanced responsiveness, particularly when handling larger volumes of production data and generating graphical dashboards and management reports. The additional computing resources also improve the overall reliability of the application during continuous daily operation and reduce the likelihood of performance bottlenecks.

Overall, the recommended hardware configuration provides a stable and efficient computing environment for the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* By utilizing these hardware resources, the proposed system can efficiently process production information, maintain accurate production records, generate real-time analytical results, and support effective production monitoring and decision-making. Furthermore, the selected hardware specifications provide sufficient capacity to accommodate future expansion of the system as additional production data and functional requirements are introduced.

## 4.5.2 Software Requirements

The proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* is developed using widely available, reliable, and open-source software technologies to ensure cost-effectiveness, compatibility, ease of maintenance, and long-term scalability. The selected software components

provide the necessary development environment, web server, database management system, programming languages, and development tools required to build, deploy, and operate the proposed web-based application successfully.

The software requirements are carefully selected based on the functional and technical requirements identified during the system analysis and design phases of the study. Each software component performs a specific role in supporting the overall operation of the application, from developing the user interface and implementing the system's business logic to storing production information and generating graphical dashboards and management reports.

The use of open-source technologies significantly reduces implementation and maintenance costs for AlDi Foods Inc. while ensuring that the application remains flexible, reliable, and easy to upgrade. Since the proposed system is intended for small- and medium-sized manufacturing operations, the selected software provides a practical and sustainable solution without requiring expensive proprietary software licenses.

The operating environment of the application is based on Microsoft Windows 10 or Windows 11, which provides a stable platform for system development, deployment, and daily operation. The operating system supports all required development tools and server applications while providing compatibility with the selected programming languages and database management system.

For local development and deployment, the researchers utilize XAMPP, an integrated web server package that includes Apache, PHP, and MySQL. XAMPP simplifies the installation and configuration of the development environment by providing all essential server components within a single package. This environment enables the application to run locally during software development, testing, debugging, and demonstration before deployment in the production environment.

The Apache Web Server serves as the application's web server by processing client requests and delivering web pages to users through a web browser. Apache manages communication between the administrator and the application, allowing production information to be entered, processed, retrieved, and displayed efficiently. Its stability and widespread adoption make it suitable for supporting the daily operation of the proposed production monitoring system.

The core business logic of the application is developed using PHP (Hypertext Preprocessor). PHP is responsible for processing administrator requests, validating user inputs, communicating with the MySQL database, performing automated yield and extraction efficiency computations, and generating production reports. Through PHP, the application integrates all functional modules into a unified web-based platform that supports production monitoring and management decision-making.

The system interface is developed using HTML5, CSS3, and JavaScript, which work together to create a responsive, organized, and user-friendly web application. HTML5 provides the structural foundation of the web pages, CSS3 enhances the visual appearance and layout of the interface, while JavaScript adds interactivity and dynamic functionality. These technologies allow administrators to navigate the application efficiently, encode production information, view analytical dashboards, and generate reports with ease.

To improve data interpretation and production monitoring, the proposed system utilizes Chart.js for data visualization. This JavaScript library enables the application to display production information using charts, graphs, and graphical dashboards. Through visual representations of production yield and extraction efficiency, administrators can quickly identify production trends, monitor operational performance, compare production batches, and evaluate the effectiveness of the juice extraction process. The graphical dashboards provide management with meaningful insights that support timely and data-driven decision-making.

The proposed system uses MySQL as its Database Management System (DBMS). MySQL stores all production-related information, including administrator accounts, production batches, yield records, extraction records, and generated reports. The centralized database ensures that production information is stored securely,

organized efficiently, and easily retrievable whenever monitoring, analysis, or report generation is required. The relational database structure also maintains data integrity through the use of primary keys and foreign key relationships, minimizing data redundancy while ensuring consistency across all production records.

During system development, Visual Studio Code serves as the primary source code editor. This lightweight yet powerful development environment enables the researchers to write, organize, debug, and maintain the application's source code efficiently. Its support for multiple programming languages, extensions, and integrated development features significantly improves coding productivity and software quality throughout the implementation phase.

The application is accessed using modern web browsers such as Google Chrome and Microsoft Edge, both of which fully support HTML5, CSS3, JavaScript, and Chart.js technologies. These browsers ensure proper rendering of the system interface, responsive navigation, accurate graphical visualization, and reliable execution of all application functions. Their compatibility with current web standards contributes to a consistent user experience during daily system operation.

To support collaborative software development and maintain version history, the researchers utilize Git as the version control system. Git enables the researchers to monitor code revisions, manage updates, restore previous versions when necessary, and maintain an organized development workflow throughout the implementation of the project. Version control also improves collaboration among the development team by ensuring that modifications to the source code are properly documented and managed.

*Table 4.2* presents the software requirements used in the development and implementation of the proposed system.

**Table 4.2 Software Requirements of the Proposed System**

| **Software** | **Purpose** |
|---|---|
| Windows 10/11 | Operating System |
| XAMPP | Local Web Server Environment |
| Apache | Web Server |
| PHP | Server-side Programming Language |
| HTML5 | Web Page Structure |
| CSS3 | User Interface Design |

| | |
|---|---|
| JavaScript | Client-side Scripting |
| Chart.js | Data Visualization and Graph Generation |
| MySQL | Database Management System |
| Visual Studio Code | Source Code Editor |
| Google Chrome / Microsoft Edge | Web Browser |
| Git | Version Control (Development) |

## Software Requirements

Each of these software components contributes to the successful operation of the proposed Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc. The integration of these technologies enables the application to automate production recording, compute yield percentage and extraction efficiency, manage production records within a centralized database, generate graphical dashboards, and produce management reports efficiently.

Furthermore, the selected software technologies provide a stable, secure, and scalable environment that supports the long-term implementation of the proposed system. Their compatibility with one another ensures seamless communication between the web interface, application logic, and database, resulting in reliable system performance and efficient data processing. As production requirements continue to evolve, these technologies also provide a flexible foundation for future system enhancements, including additional analytical features, expanded reporting capabilities, multiple user roles, and integration with other production management functions.

Overall, the software requirements presented in this section provide the technological foundation necessary for the successful development, deployment, and operation of the proposed web-based production monitoring system. By utilizing proven and widely adopted software technologies, the developed application delivers a reliable, user-friendly, and cost-effective solution that supports the production monitoring objectives of AlDi Foods Inc. while improving data accuracy, operational efficiency, and management decision-making.

## 4.6 System Design

The system design of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* was developed based on the functional and nonfunctional requirements identified during the requirements analysis, data-gathering, and system development activities discussed in Chapter 3. The design translates the identified limitations of the existing manual production-recording process into a structured web-based production monitoring system. The proposed design focuses on centralizing production information, reducing repetitive manual calculations, improving data organization, and providing timely analytical information for production monitoring and operational planning.

The proposed system follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. The presentation layer provides the web-based interface through which authorized production personnel, supervisors, and administrators interact with the system. It includes the login page, dashboard, production-recording forms, historical records, analytics, profile management, and administrative interfaces. The application layer contains the business logic responsible for authentication, data validation, production processing, automated calculations, production-stability classification, and report generation. The data layer provides centralized storage for user accounts, production batches, raw material information, extraction records, packed yields, analytical results, and reports.

The system design was structured to support the actual production workflow of AlDi Foods Inc. Rather than requiring production personnel to maintain separate paper records or spreadsheets, the system provides a centralized platform where production information can be encoded and associated with a specific production batch. The recorded information is subsequently processed by the application to calculate yield percentage and extraction efficiency and to determine the corresponding production stability status based on predefined criteria.

At the center of the system design is the production batch, which serves as the main reference for organizing production information. Each production batch contains the relevant raw material input, extraction output, and packed yield information. This structure allows the system to maintain a complete digital record of a production activity while ensuring that the information used for analytical calculations belongs to the appropriate batch.

The relational database model serves as the foundational data structure of the proposed system. It organizes operational information into related tables designed to minimize unnecessary data duplication and maintain data consistency. The USERS table manages user accounts, credentials, roles, and access information. It is connected to the PRODUCTION_BATCHES table to identify the authorized user responsible for creating or recording a production batch.

The PRODUCTION_BATCHES table functions as the central entity of the production-recording process. Each batch is assigned a unique identifier that allows its associated production information to be retrieved and processed. Through this relationship, the system can connect the raw materials used in a particular batch with the corresponding extraction output and final packed yield.

The RAW_MATERIALS table stores information regarding the raw material inputs recorded for production. The quantity of raw material entered by authorized personnel becomes one of the primary values used by the system in calculating production performance. The EXTRACTION_RECORDS table stores the output obtained during the extraction process, while the PACKED_YIELDS table records the final quantity of product that has been packed. These records are associated with the corresponding production batch to maintain traceability throughout the production process.

The system also incorporates analytical components that process the recorded production information. The YIELD_ANALYTICS component supports the automatic calculation and presentation of yield-related indicators based on the recorded production input and output. The EFFICIENCY_RECORDS component supports the computation and monitoring of extraction efficiency using the relevant raw material and extraction-output values. These automated computations reduce the need for production personnel or supervisors to manually perform calculations using separate spreadsheets or calculators.

The PRODUCTION_STABILITY component evaluates the calculated production indicators against predefined criteria established for the system. Based on the applicable thresholds, the system assigns a production status such as *Stable* or *Not Stable*. This classification provides an immediate summary of the recorded production condition and allows authorized users to identify production batches that may require further review.

The processed information is presented through the system's dashboard and analytics functions. The dashboard provides summarized production information and graphical visualizations that allow users to monitor production

performance. Through these features, authorized personnel can review yield results, extraction-efficiency values, raw material utilization, and production stability without manually consolidating information from separate records.

The system design also includes a historical-record function that allows users with appropriate permissions to retrieve previously recorded production batches. Historical records provide access to production information that can be reviewed according to dates, batches, and other available criteria. This improves information retrieval compared with the existing manual process, where personnel may need to search through physical logbooks, spreadsheets, or separate production documents.

The reporting component is designed to organize stored production information into useful summaries. The system can generate production reports for defined periods, including daily, weekly, monthly, quarterly, and yearly records, depending on the reporting functions implemented in the system. These reports may contain raw material utilization, extraction outputs, packed yields, yield percentages, extraction-efficiency results, and production-stability classifications. By consolidating these indicators, the system provides management with accessible production information for monitoring and operational planning.

Data validation is incorporated into the system design to improve the accuracy and reliability of manually encoded production information. Required fields are validated before records are submitted, while numerical production fields are checked to ensure that appropriate values are entered. Database relationships and constraints further help maintain the integrity of production records by ensuring that related information is associated with the correct production batch and authorized user.

The system design also incorporates user authentication and role-based access to protect production information and administrative functions. Users must provide valid credentials before accessing protected areas of the system. Different user roles can be assigned appropriate permissions based on their responsibilities. This allows production personnel to perform necessary recording activities while administrators can manage users and access broader system information.

The proposed system's real-time monitoring capability is based on the immediate processing of manually encoded production information. When an authorized user submits production data, the application processes the information and updates the corresponding analytical results without requiring manual calculation. Therefore, real-time monitoring in this project refers to the timely availability and processing of encoded production data rather than continuous automated data collection through IoT or industrial sensors.

The overall system workflow can be represented as:

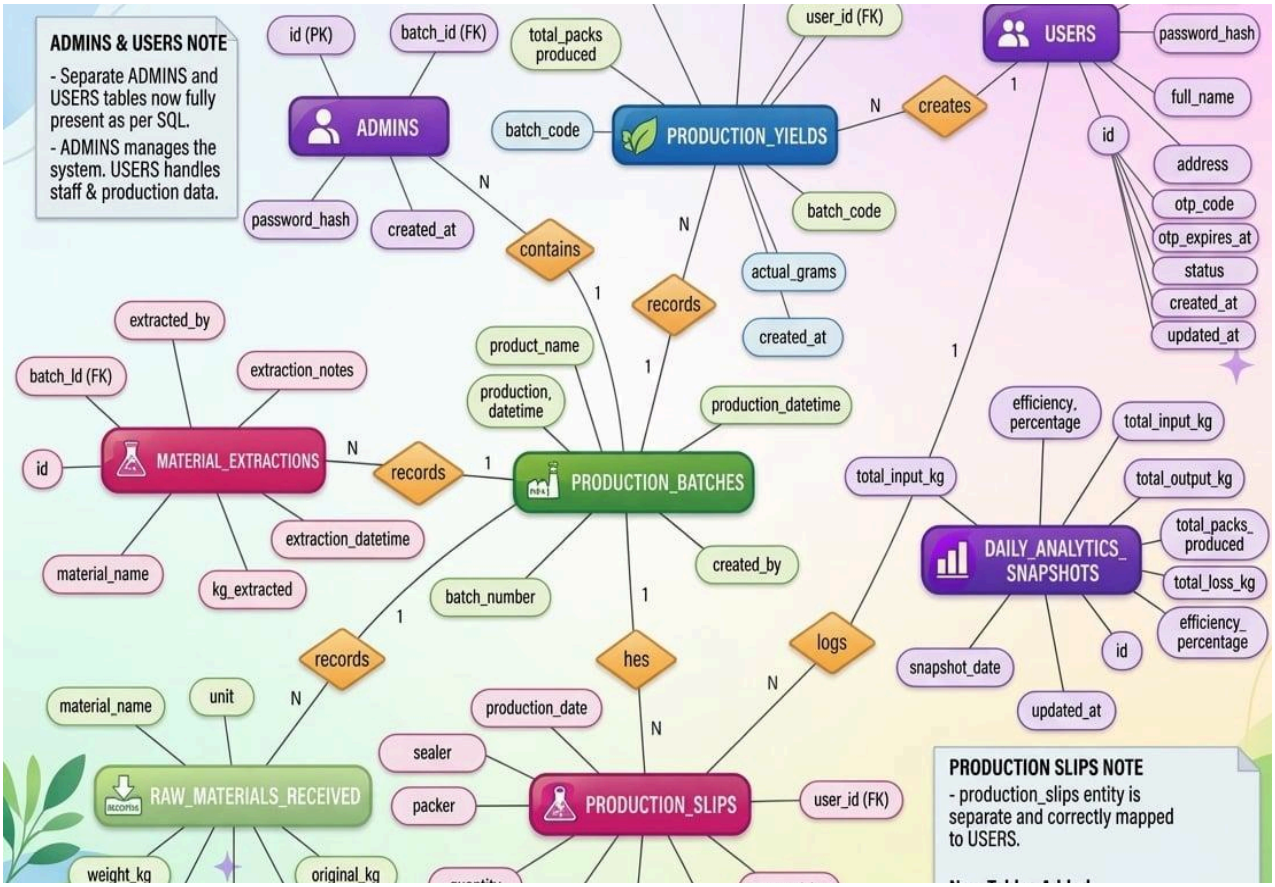
User Authentication → Production Batch Creation → Raw Material Recording → Extraction Output Recording → Packed Yield Recording → Automated Yield and Efficiency Computation → Production Stability Evaluation → Dashboard and Analytics → Historical Records → Report Generation

This workflow establishes a continuous digital process from production-data entry to information analysis and reporting. It ensures that the information entered during production can be processed and reused by different system components without requiring users to repeatedly encode the same information.

The design also supports the maintainability and future scalability of the application. The separation of the presentation, application, and data layers allows individual components to be modified or enhanced without requiring major changes to the entire system. Additional analytical indicators, reporting functions, user roles, or production-record fields can be incorporated in the future as the operational requirements of AlDi Foods Inc. change.

Overall, the system design provides an integrated structure for digitizing the production-monitoring activities of AlDi Foods Inc. By combining a three-tier architecture, centralized relational database, production-batch management, automated calculations, production-stability classification, dashboard visualization, historical record management,

and automated reporting, the proposed design directly addresses the limitations of manual production recording identified in the study. The design enables manually encoded production information to be transformed into organized, accessible, and timely analytical information while maintaining the project's practical, cost-effective, and




non-IoT-based scope.

*Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.*

The primary entities of the database include:

- Users
- Raw Material Received
- Raw Materials
- Production Batches
- Extraction Records
- Packed Yields
- Production Slips
- Yield Analytics

- Efficiency Records
- Production Stability
- Reports

**Users Table**
This table stores account credentials and access permissions for system users.

| Field | Description |
|---|---|
| user_id | Primary Key |
| name | Complete Name |
| username | Login Username |
| password | Account Password |
| role | User Access Level |

**Production Batches Table**
This table acts as the central hub tracking production batch information.

| Field | Description |
|---|---|
| batch_id | Primary Key |
| user_id | Foreign Key (Created by User) |
| production_date | Date of Production |

| | |
|---|---|
| operator_name | Name of Assigned Operator |
| product_name | Name of Product |
| remarks | Additional Operational Notes |

**Raw Materials Received Table**
This table records the raw materials that are received for production.

| Field | Description |
|---|---|
| raw_material_received_id | Primary Key |
| raw_material_received_name | Name of Raw Material Received |
| weight_kg | Measurement Unit (e.g., kg, L) |
| buyers_code | Code of the Suppliers |
| received_at | Timestamp |

**Raw Materials Table**
This table records the raw materials utilized in each production batch.

| Field | Description |
|---|---|
| raw_id | Primary Key |
| batch_id | Foreign Key |
| material_name | Name of Raw Material |
| quantity_input | Amount of Material Used |
| unit | Measurement Unit (e.g., kg, L) |

**Extraction Records Table**
This table tracks the volume and details of extracted outputs from production.

| Field | Description |
|---|---|
| extraction_id | Primary Key |
| batch_id | Foreign Key |
| extracted_output | Volume of Extracted Output |
| extraction_time | Time of Extraction Process |

| | |
|---|---|
| remarks | Extraction Notes |

**Packed Yields Table**

This table stores the final packaged yield measurements derived from production batches.

| Field | Description |
|---|---|
| packed_id | Primary Key |
| batch_id | Foreign Key |
| packed_output | Total Packaged Volume/Units |
| packed_date | Date Packaging Completed |

**Yield Analytics Table**
This table stores calculated yield percentages to measure overall output performance.

| Field | Description |
|---|---|
| yield_id | Primary Key |
| batch_id | Foreign Key |
| yield_percent | Calculated Yield Percentage |

**Efficiency Records Table**
This table records resource-to-output conversion efficiency metrics.

| Field | Description |
|---|---|
| efficiency_id | Primary Key |
| batch_id | Foreign Key |
| efficiency_% | Extraction Efficiency Metric (%) |

**Production Stability Table**

This table evaluates batch performance stability against organizational criteria.

| Field | Description |
|---|---|
| stability_id | Primary Key |
| batch_id | Foreign Key |
| status | Operational Status (e.g., Stable / Not Stable) |

**Reports Table**
This table logs generated production summaries derived from analytical and stability metrics.

| Field | Description |
|---|---|
| report_id | Primary Key |

| | |
|---|---|
| batch_id | Foreign Key |
| report_type | Type of Summary (e.g., Daily, Monthly) |
| generated_date | Date Report Created |

The relationships among these tables allow the application to track raw material consumption, measure extraction performance, calculate packaged yields, evaluate production stability, and generate comprehensive management reports. Utilizing proper indexing and foreign key constraints helps maintain referential integrity while optimizing query performance across all operational and analytical workflows.

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# CHAPTER 5

# CONCLUSION AND RECOMMENDATIONS

## 5.1 Introduction

This chapter presents the conclusion and recommendations of the study entitled “Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.” It summarizes the major findings of the study, evaluates the extent to which the objectives established in Chapter 1 are achieved, and explains how the developed system addresses the operational challenges identified in the existing production-monitoring process of AlDi Foods Inc. The chapter also presents recommendations for system implementation, maintenance, improvement, and possible future research.

The study is conducted in response to the existing manual production-monitoring process of AlDi Foods Inc., particularly in recording and monitoring juice extraction activities. Before the development of the proposed system, production information such as raw material usage, extraction output, and packed yield was recorded using logbooks, spreadsheets, paper-based records, and other manual documentation methods. Although these methods provide a means of documenting production activities, they require personnel to perform repetitive data-entry and calculation tasks. The manual process also makes it more difficult to consolidate production information, monitor performance promptly, retrieve historical records, and generate organized reports for production analysis.

The identified problems are consistent with the concerns presented in Chapter 1, particularly regarding data-entry errors, inconsistent records, delayed computation, limited visibility of production performance, and difficulty in retrieving historical information. These limitations can affect the ability of production personnel and supervisors to obtain timely information regarding raw material utilization, extraction performance, packed yield, and overall production stability. Consequently, there is a need for a centralized system that can organize production information and provide analytical results without requiring extensive manual computation.

To address these concerns, the researchers develop a web-based production-monitoring system designed specifically for the production-monitoring requirements of AlDi Foods Inc. The system provides a centralized platform where authorized users can encode production information and access processed results through a web-based interface. The system records raw material usage, extraction output, and packed yield information and uses the recorded values to automatically calculate yield percentage and extraction efficiency. The resulting information is presented through dashboards, analytics, historical records, and reports to support production monitoring and operational planning.

The development of the system is guided by the requirements identified through the data-gathering activities discussed in Chapter 3. Interviews, observations, and examination of existing production records provide the researchers with information regarding the current production workflow, data requirements, user responsibilities, and reporting needs. These requirements serve as the basis for designing the system's functional modules, database structure, user interface, and analytical functions.

The researchers use Agile Software Development Methodology to organize the system development process. The development activities are divided into requirement analysis, system design, development, testing, and deployment of the prototype. The iterative nature of the methodology allows the researchers to review system requirements, develop functional components, identify issues, and refine the system according to the identified requirements and feedback obtained during development and testing.

The developed system integrates several interconnected modules that support the production-monitoring workflow of AlDi Foods Inc. These include user authentication, production batch management, raw material recording, extraction output recording, packed yield recording, yield analytics, extraction efficiency monitoring, production stability interpretation, dashboard visualization, historical record management, and automated reporting. The integration of these functions transforms production information from manually maintained records into structured digital records that can be processed and retrieved through a centralized system.

The system follows the three-tier architecture discussed in Chapter 4, consisting of the presentation layer, application layer, and data layer. The presentation layer provides the web interface through which authorized users interact with the system. The application layer handles business rules, validation, automated calculations, production processing, and system functions. The data layer provides centralized storage for production and user information through the relational database. This architectural structure supports the organization, processing, and retrieval of production information while allowing the different components of the system to work together.

The real-time capability of the developed system is defined within the scope of the study as the immediate processing and presentation of manually encoded production information. When authorized personnel enter and submit production data, the system processes the information and provides the corresponding analytical results without requiring separate manual calculations. Therefore, the system does not depend on continuous automated data collection, IoT devices, or advanced industrial automation technologies. Instead, it focuses on providing timely digital processing of production data entered by authorized users.

The findings presented in this chapter are based on the system development, implementation, and evaluation activities described in the preceding chapters. The conclusion considers the identified problems in Chapter 1, the concepts and findings from the related literature and studies in Chapter 2, the development methodology and technical approach in Chapter 3, and the implemented system components presented in Chapter 4. Through this connection, the chapter determines how the developed system responds to the identified production-monitoring requirements of AlDi Foods Inc.

Overall, the study demonstrates the application of a practical and cost-effective web-based information system for digitizing production-monitoring activities. By centralizing production records, automating calculations, providing analytical dashboards, maintaining historical information, and generating production reports, the developed system provides a structured approach to monitoring production performance. The recommendations in this chapter are therefore directed toward the continued use, maintenance, evaluation, and future enhancement of the system to support the changing production-monitoring requirements of AlDi Foods Inc.

## 5.2 Summary of Findings

The study successfully achieves its primary objective of designing and developing a web-based production-monitoring system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* The developed system is designed to address the limitations of the existing manual production-monitoring process of AlDi Foods Inc. by providing a centralized platform for recording, processing, analyzing, retrieving, and reporting production information. The findings presented in this section are based on the problems and objectives identified in Chapter 1, the related literature and studies reviewed in Chapter 2, the development methodology applied in Chapter 3, and the system implementation and functionalities presented in Chapter 4.

Based on the requirements gathered through interviews, direct observation, and examination of existing production records, the researchers identify several challenges in the company's existing production-monitoring process. These include the reliance on handwritten records and spreadsheets, repetitive manual data entry, manual computation of production indicators, delays in consolidating production information, difficulty in retrieving historical records, and limited visibility of production performance. These challenges establish the need for a centralized digital system capable of organizing production information and providing timely analytical results.

The findings show that the developed system successfully digitizes the recording of production activities. Authorized users can create production batches and record important production information, including raw material quantities, extraction outputs, and packed yields through structured web-based forms. The digitization of these records reduces dependence on separate paper-based documents and spreadsheets and provides a more organized method of storing production information. Each production record is associated with its corresponding production batch, allowing production information to be systematically organized and retrieved.

The system successfully utilizes a centralized MySQL database to store and manage production-related information. The database structure connects the different components of the production process, including users, production batches, raw materials, extraction records, and packed yields. This centralized storage approach improves the organization of production information and allows authorized users to retrieve previously recorded data through the system. It also reduces the need to manually consolidate information from different production documents when reviewing production activities.

One of the major findings of the study is the successful implementation of automated yield-percentage computation. The system processes the required production values entered by authorized users and automatically calculates the corresponding yield percentage using the predefined computation implemented during system development. This functionality reduces the need for repetitive manual calculations and promotes consistency in the processing of production information. The calculated yield results can subsequently be used by the dashboard, analytics, historical records, and reporting functions of the system.

The system also successfully implements extraction-efficiency monitoring. By using the recorded raw material input and extraction output, the application provides an extraction-efficiency result for each applicable production batch. This allows production personnel and supervisors to review the relationship between the amount of raw material used and the resulting extraction output. The availability of this information within the same production-monitoring platform provides a more organized basis for comparing production batches and identifying records that may require further examination.

Another significant finding is the implementation of production stability interpretation. The system evaluates relevant production indicators against predefined thresholds established during system development and assigns a corresponding status, such as **Stable** or **Not Stable**. This function provides users with an immediate interpretation of recorded production performance. Instead of manually comparing calculated production values with predefined criteria, the system performs the comparison automatically. A *Not Stable* classification serves as an indication that a production record may require further review; it does not independently determine the specific cause of the production condition.

The study further finds that the developed dashboard provides a centralized visual representation of production information. The dashboard incorporates summary information, charts, graphs, and production indicators based on the data recorded in the system. This allows authorized users to review production performance without manually examining individual records or consolidating information from separate documents. The dashboard therefore improves the visibility of production information and supports more convenient monitoring of yield, extraction efficiency, raw material utilization, and production stability.

The system's real-time analytics capability is also successfully implemented within the defined scope of the study. When authorized personnel enter and submit production information, the system processes the available data and provides the corresponding analytical results through the appropriate system interfaces. In this study, real-time monitoring refers to the immediate processing and presentation of manually encoded production information. The system does not depend on continuous IoT-based data collection, industrial sensors, robotics, or advanced industrial automation. This approach remains consistent with the scope and limitations established in Chapter 1 and provides a practical and cost-effective method of digitizing production monitoring.

The developed system also successfully provides historical production-record management. Production information entered into the system is stored in the centralized database and can be retrieved by authorized users. Historical records allow production personnel and supervisors to review previous batches, compare production information, and access records for documentation and analysis. This functionality addresses the difficulty of retrieving historical information from physical logbooks, spreadsheets, and separate production documents identified during the requirements-analysis stage.

The reporting functionality further contributes to the organization of production information. The system provides production summaries for applicable reporting periods, including weekly, monthly, quarterly, and yearly records, based on the implemented reporting functions. These reports can consolidate relevant production indicators such as raw material utilization, extraction output, packed yield, yield percentage, extraction efficiency, and production stability. By automating the organization of production information into reports, the system reduces the need for users to manually consolidate records when preparing production summaries.

The findings also demonstrate the successful integration of the major system modules. The user-authentication module controls access to the application, while production-batch management organizes individual production activities. Raw material recording, extraction-output recording, and packed-yield recording provide the production information required by the analytical functions. The yield and extraction-efficiency components process the recorded values, while the production-stability function interprets the results based on predefined criteria. The dashboard and analytics modules present the processed information, while historical records and reporting functions provide mechanisms for reviewing and summarizing previously recorded production data.

The overall workflow of the developed system can therefore be summarized as:

**User Authentication → Production Batch Creation → Raw Materials Received → Raw Material Recording → Extraction Output Recording → Packed Yield Recording → Automated Yield and Extraction-Efficiency Computation → Production Stability Evaluation → Dashboard and Analytics → Historical Records → Report Generation**

This integrated workflow establishes a continuous digital process from production-data entry to analysis and reporting. It minimizes the need to repeatedly enter or manually consolidate the same production information across different records. The centralized structure also allows information generated from one production activity to be utilized by several system functions, improving the overall organization of production data.

The findings are also consistent with the concepts presented in Chapter 2, particularly those concerning the digitization of operational records, computerized production monitoring, yield analysis, data organization, and the use of production information to support operational decisions. The reviewed studies provide a foundation for understanding how digital systems can improve the management and processing of production information. The present study applies these concepts specifically to the production-monitoring requirements of AlDi Foods Inc. by focusing on raw material usage, extraction output, packed yield, yield percentage, extraction efficiency, production stability, historical records, dashboards, and reports.

In relation to Chapter 3, the findings demonstrate that the Agile Software Development Methodology provides a structured approach for developing the proposed system. The requirements gathered through interviews, observation,

and existing production documents are translated into system functions and design components. The development process covers requirement analysis, system design, development, testing, and prototype deployment. The resulting system reflects the production-monitoring requirements identified during the data-gathering process and incorporates the system architecture, database design, interfaces, and functional modules discussed in Chapter 4.

The three-tier architecture implemented in Chapter 4 also supports the developed system by separating its presentation, application, and data functions. The presentation layer provides the web-based interface used by authorized personnel, the application layer processes business rules and automated computations, and the data layer manages centralized production information. This separation provides an organized structure for the system and supports the interaction among the different modules developed for production monitoring.

Overall, the findings demonstrate that the developed system addresses the major production-monitoring problems identified in Chapter 1. It provides a centralized platform for digitizing production records, automates yield and extraction-efficiency computations, interprets production stability using predefined criteria, presents production information through dashboards and analytics, maintains historical records, and supports automated reporting. These functions collectively reduce the dependence on manual calculations and fragmented production records while improving the accessibility and organization of production information.

The study therefore finds that the developed system fulfills the major functional objectives established for the project. It provides AlDi Foods Inc. with a practical web-based platform for transforming manually encoded production information into organized and timely analytical information. The system remains within the intended scope of the study by focusing on manual production-data encoding rather than requiring IoT devices, continuous sensor monitoring, advanced machine learning, robotics, or full industrial automation. Through its centralized database, automated computations, dashboard visualization, historical records, and reporting capabilities, the system provides a structured foundation for improving the production-monitoring process of AlDi Foods Inc.

## 5.3 Conclusion

The study, entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.,"* is conducted to address the limitations of the existing production-monitoring process of AlDi Foods Inc., particularly the reliance on manual recording, repetitive computation, delayed consolidation of production information, limited visibility of production performance, and difficulty in retrieving historical production records. Based on the requirements analysis, system development, implementation, and evaluation presented in Chapters 1 to 4, the researchers conclude that the developed web-based production-monitoring system provides a structured digital approach to managing production information and addresses the major requirements identified during the study.

The study begins with the identification of operational problems associated with the existing manual production-recording process. As discussed in Chapter 1, production personnel rely on handwritten records, spreadsheets, and other manual documentation methods to record raw material usage, extraction output, and packed yield. These methods require repetitive encoding and manual calculations and can make production information less accessible for immediate monitoring and historical analysis. The proposed system is therefore developed to provide a centralized digital platform that organizes production information and processes relevant production indicators automatically.

Based on the findings presented in Chapter 5.2, the researchers conclude that the developed system addresses the primary objective of digitizing the production-monitoring process of AlDi Foods Inc. Authorized users can record production batches, raw material quantities, extraction outputs, and packed yields through structured web-based interfaces. The recorded information is stored in a centralized relational database, allowing production records to be organized according to their corresponding production batches and retrieved by authorized users when required.

The centralized recording mechanism provides an organized alternative to maintaining production information through separate paper records and spreadsheets. By storing related production information within one database, the

system provides a common source of production records for its authorized users. This structure supports easier retrieval and reduces the need to repeatedly consolidate information from separate documents when reviewing production activities.

The researchers further conclude that the system addresses the objective of automating production-related calculations. The developed application automatically processes the required production values to calculate yield percentage and extraction efficiency. This reduces the need for users to manually perform repetitive mathematical calculations and provides consistent computational processing based on the formulas implemented in the system.

The automated calculation functions are particularly relevant to the problems identified in Chapter 1 because manual computation is one of the activities that can contribute to delays and computational inconsistencies. Through automated processing, production information entered into the system can be converted into corresponding production indicators without requiring users to perform the calculations separately. The calculated values can then be utilized by the dashboard, analytics, historical-record, and reporting functions.

The study also concludes that the developed system provides extraction-efficiency monitoring appropriate to the production-monitoring requirements of AlDi Foods Inc. The system uses the recorded raw material input and extraction output to provide an extraction-efficiency result for the applicable production batch. This enables authorized personnel to review extraction performance using organized production information and provides a basis for comparing results among production batches.

The extraction-efficiency function does not independently determine the physical or operational cause of production losses. Instead, it provides calculated information that can assist production personnel and supervisors in identifying records that may require further investigation. This distinction maintains the system within the intended scope of the study and avoids claiming capabilities that are not implemented in the proposed application.

The researchers also conclude that the system addresses the objective of providing timely production analytics through its dashboard and analytical functions. Production information entered by authorized users is processed by the application and presented through summary information, charts, graphs, and other implemented production indicators. These features allow users to review production information in a more organized visual format rather than manually examining individual records.

Within the context of this study, the term real-time refers to the immediate processing and presentation of manually encoded production information after it is submitted to the system. The system does not rely on continuous automated data collection through IoT devices, industrial sensors, robotics, or advanced industrial automation. This definition is consistent with the scope and limitations established in Chapter 1 and ensures that the real-time capability of the system is understood in relation to its actual implementation.

The production-stability interpretation function further supports the monitoring objective of the study. The application compares applicable production indicators with predefined thresholds established during system development and provides a corresponding status, such as Stable or Not Stable. This feature provides users with an immediate interpretation of the recorded production condition and can assist supervisors in identifying production records that require additional review.

The researchers emphasize that the Stable or Not Stable classification serves as an analytical indicator rather than an automatic diagnosis of the cause of a production problem. Further investigation remains the responsibility of authorized production personnel and management. This approach allows the system to support production monitoring while maintaining appropriate limitations regarding automated decision-making.

The study also concludes that the historical-record functionality addresses the difficulty of retrieving previously recorded production information identified during the initial assessment. The centralized database retains production records that can be accessed by authorized users for review, comparison, documentation, and analysis. The

availability of historical information provides a more organized approach to reviewing previous production batches compared with manually searching through physical logbooks or separate spreadsheets.

The reporting functionality further supports the production-management requirements of the organization. Based on the implemented reporting features, production information can be organized into applicable reporting periods and summarized using relevant indicators such as raw material utilization, extraction output, packed yield, yield percentage, extraction efficiency, and production stability. This reduces the amount of manual consolidation required when preparing production summaries and provides management with structured production information for review.

The researchers further conclude that the integration of the system modules creates a unified production-monitoring workflow. User authentication provides controlled access to the application, production-batch management organizes individual production activities, and raw material, extraction, and packed-yield modules capture the primary production information. The analytical functions process the recorded values, while the dashboard, historical records, and reporting components present and organize the resulting information.

The implemented workflow can be summarized as:

User Authentication → Production Batch Creation → Raw Material Recording → Extraction Output Recording → Packed Yield Recording → Automated Yield and Extraction-Efficiency Computation → Production Stability Evaluation → Dashboard and Analytics → Historical Records → Report Generation

This integrated workflow demonstrates how production information can move from data entry to processing, analysis, storage, and reporting within one centralized application. The integration reduces the need for users to maintain separate systems for production recording and computation and provides a consistent flow of information among the different system functions.

The three-tier architecture implemented in Chapter 4 also contributes to the organization of the developed application. The presentation layer provides the web-based interfaces through which authorized users interact with the system. The application layer manages business rules, data validation, automated computations, production-stability interpretation, and other processing functions. The data layer provides centralized storage for user and production information. The separation of these functions provides a structured foundation for the developed system and supports future modification of individual components.

The researchers conclude that the development methodology presented in Chapter 3 contributes to the systematic development of the proposed system. The use of Agile Software Development Methodology allows the researchers to progressively translate the identified requirements into system components through requirement analysis, system design, development, testing, and prototype deployment. The iterative development process provides opportunities to review system functions and address identified issues during development.

The findings are also aligned with the concepts discussed in Chapter 2 concerning the digitization of production records, computerized monitoring, production analytics, yield analysis, and the use of organized operational information to support decision-making. The related literature and studies provide a foundation for the importance of improving the management and processing of production information. The present study applies these concepts to the specific production-monitoring requirements of AlDi Foods Inc. through a web-based system focused on manually encoded production data.

The study further demonstrates that digital production monitoring does not necessarily require sophisticated industrial automation technologies to provide useful production information. Within the scope of this project, the combination of a web-based interface, centralized database, automated calculations, dashboard visualization, historical record management, and reporting functions provides a practical approach to digitizing production monitoring. This makes the proposed solution consistent with the project's cost-effective and non-IoT-based scope.

The researchers therefore conclude that the developed system provides a practical digital platform for improving the organization and processing of production information at AlDi Foods Inc. It addresses the major problems identified during the study by providing centralized production recording, automated yield and extraction-efficiency computation, production-stability interpretation, dashboard analytics, historical record management, and reporting functions.

Based on the system implementation and evaluation presented in Chapter 4, the developed application addresses the major functional objectives established in Chapter 1. It transforms manually encoded production information into structured digital records that can be processed, analyzed, visualized, retrieved, and reported through a centralized platform. The system consequently provides production personnel and authorized management users with more accessible production information for monitoring and operational review.

The researchers recognize, however, that the developed system is a web-based prototype designed within the defined scope, resources, and requirements of the study. Its effectiveness is dependent on the accuracy of the information entered by authorized users, the proper use of the system, the availability of the required computing environment, and the continued maintenance of the application and database. The system should therefore be regarded as a digital production-monitoring and analytical support tool rather than a replacement for production personnel, physical quality-control procedures, or managerial judgment.

Overall, the study concludes that *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* successfully provides a centralized and structured approach to digitizing the company's production-monitoring activities. Through the integration of production recording, automated computation, extraction-efficiency monitoring, production-stability interpretation, dashboard analytics, historical records, and reporting, the system addresses the operational requirements identified throughout the study.

The developed system demonstrates the application of Information Technology to a specific production-monitoring environment in the Philippine food-manufacturing context. It provides a foundation that AlDi Foods Inc. may further evaluate, maintain, and enhance according to its future operational requirements. The study consequently establishes that a properly designed web-based information system can provide organized, timely, and accessible production information while remaining within a practical, cost-effective, and non-IoT implementation scope.

## 5.4 Recommendations

Based on the findings, conclusions, and overall evaluation of the proposed system, the researchers recommend several improvements and future directions to further enhance the effectiveness, usability, scalability, and long-term sustainability of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* These recommendations are intended for AlDi Foods Inc., future researchers, and software developers who may continue the development and enhancement of the proposed web-based production monitoring system.

The researchers recommend that AlDi Foods Inc. fully implement the developed system in its daily production operations to maximize the benefits of digital production monitoring. The adoption of the system is expected to improve the accuracy of production recording, reduce manual computation, enhance production monitoring, and support faster and more informed managerial decision-making. The company is also encouraged to provide adequate training for administrators and production personnel to ensure the proper utilization and maintenance of the application.

The researchers also recommend the implementation of regular database backup and maintenance procedures to protect production records from accidental data loss and system failures. Routine software updates and preventive maintenance are likewise recommended to maintain the reliability, security, and efficiency of the application throughout its operational use.

For future system enhancement, the researchers recommend expanding the functionalities of the application by incorporating additional production management features such as inventory management, raw material tracking, supplier management, production scheduling, quality assurance monitoring, and employee performance monitoring. The integration of these modules would further improve the overall production management capabilities of the system.

The researchers further recommend the implementation of role-based access control by allowing multiple authorized users with different levels of system privileges. Future versions of the application can include separate user accounts for production supervisors, quality assurance personnel, warehouse staff, and management to improve collaboration while maintaining data security and accountability.

To improve accessibility and operational flexibility, the researchers recommend developing a mobile-responsive version of the application or a dedicated mobile application. This enhancement enables production personnel and management to monitor production activities, review reports, and access production information using smartphones or tablet devices.

The researchers also recommend the integration of advanced data analytics and predictive reporting features in future versions of the system. Additional analytical tools, such as production forecasting, trend analysis, and performance prediction, can assist management in

making more strategic operational decisions and identifying potential production issues before they occur.

Although the current study intentionally excludes Industrial Internet of Things (IIoT) technologies and advanced automation to maintain affordability for small and medium-sized enterprises, future researchers are encouraged to investigate the possible integration of sensor-based data collection, automated production monitoring, or machine learning techniques if organizational resources and operational requirements permit such enhancements.

Future researchers are also encouraged to conduct further evaluation of the system using larger groups of users and longer implementation periods. Additional studies can assess the long-term effects of the system on production efficiency, operational performance, user satisfaction, and organizational productivity using different evaluation methodologies and performance indicators.

Lastly, the researchers recommend that future studies continue exploring innovative information technology solutions that support digital transformation within the manufacturing industry. Similar production monitoring systems can be developed for other food processing companies and manufacturing organizations to improve production management, enhance operational efficiency, reduce manual workloads, and promote data-driven decision-making through modern web-based technologies.

Overall, these recommendations are intended to support the continuous improvement, sustainability, and future expansion of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.*, ensuring that the developed application remains relevant, adaptable, and capable of meeting the evolving operational requirements of the organization.

**System Improvements**

Although the developed system successfully achieves its primary objectives of digitizing production monitoring, automating yield computation, monitoring extraction efficiency, and generating production reports, there are still opportunities to improve its functionality and overall performance.

The researchers recommend incorporating an automated notification and alert feature into future versions of the application. This feature can notify administrators whenever production results fall below the expected yield percentage or extraction efficiency, or when production stability is classified as Not Stable. Such notifications allow production personnel to immediately review production records and take appropriate corrective actions.

Future versions of the system can also include a more comprehensive dashboard with additional visualization features. While the current system provides charts, graphs, and production summaries, future enhancements can include interactive filters, comparison of production batches by date, graphical trend analysis, and customizable dashboard layouts. These features provide management with clearer insights into production performance and support more effective decision-making.

Another recommended enhancement is the integration of barcode or QR code technology for production batch identification. This feature simplifies production batch recording, reduces manual encoding errors, improves the traceability of production records, and makes information retrieval more efficient. To strengthen system security, future versions should implement additional authentication and security features such as stronger password encryption, password recovery options, user activity logs, automatic session timeout, and role-based access control if additional user roles are introduced. These improvements help protect production information from unauthorized access while improving accountability among system users.

The researchers also recommend enhancing the report generation module by allowing reports to be exported in different formats such as PDF, Microsoft Excel, and CSV. Providing multiple export options makes it easier for management to archive production reports, prepare documentation, and perform further data analysis whenever necessary.

To ensure the safety of production records, future versions of the application should include automatic database backup and recovery functions. Regular backup procedures help protect valuable production information from accidental deletion, hardware failure, or unexpected system interruptions.

Finally, the researchers recommend improving the usability of the system by continuously refining the user interface based on user feedback. Enhancing navigation, simplifying data entry forms, and improving dashboard organization make the system easier to learn and use for production personnel and administrators.

**Future Research Directions**

This study focuses on developing a web-based production monitoring system for AlDi Foods Inc. that digitizes production recording, computes yield percentage and extraction efficiency, stores historical production records, and generates production reports. Although the proposed system successfully addresses the identified operational problems, the researchers recommend that future researchers can further improve the system by expanding its functionality while remaining aligned with the needs of manufacturing organizations.

The researchers recommend that future researchers can enhance the dashboard by incorporating more advanced statistical summaries, production comparisons, and additional graphical reports that provide deeper insights into production performance. These enhancements can help management evaluate long-term production trends and identify areas for operational improvement.

The researchers also recommend that future researchers can develop a mobile-responsive or dedicated mobile application version of the system to enable administrators and production supervisors to access production information using smartphones or tablets. This enhancement improves accessibility and allows authorized users to monitor production records even when they are away from the production office.

Another possible enhancement is the expansion of the system to include additional management modules such as inventory management, raw material inventory tracking, supplier information management, quality assurance monitoring, production scheduling, and employee management. Integrating these modules creates a more comprehensive manufacturing information system while maintaining the centralized database established in the current study.

The researchers also recommend that future studies can focus on improving report generation by incorporating more detailed production summaries, customized report templates, and comparative analyses of production performance over different reporting periods. Such improvements provide management with more comprehensive information for production planning and operational evaluation.

The researchers likewise encourage future researchers to conduct comparative studies involving other food manufacturing companies to determine whether the proposed production monitoring framework can be adapted and implemented in different production environments. These studies can help evaluate the flexibility and applicability of the developed system across various manufacturing processes.

Finally, the researchers recommend that future researchers evaluate the long-term impact of implementing digital production monitoring systems on operational efficiency, production consistency, data accuracy, reporting efficiency, and overall organizational productivity. Such studies provide additional evidence regarding the effectiveness of digital production monitoring systems in supporting continuous operational improvement.

Here is the present tense version of your Implementation Strategies section. Since this is under Recommendations, the recommendations are expressed as "recommend", while future actions remain as should/would where appropriate.

**Implementation Strategies**

For the successful implementation of the proposed system, the researchers recommend that AlDi Foods Inc. establish a structured deployment strategy that includes proper preparation, employee training, continuous monitoring, and regular system evaluation. Proper implementation helps maximize the benefits of the developed application while ensuring consistent and accurate production monitoring.

Before full implementation, the researchers recommend that production personnel and administrators undergo comprehensive user training covering system navigation, production data encoding, dashboard interpretation, report generation, and basic troubleshooting procedures. Adequate training improves user confidence, minimizes encoding errors, and promotes the proper utilization of the system.

The researchers also recommend that management establish standardized operating procedures (SOPs) that clearly define responsibilities for recording production information, validating encoded data, reviewing production reports, and maintaining database accuracy. Standardized procedures ensure consistency in production monitoring activities and improve the reliability of stored production information.

To maintain data integrity, the researchers recommend that regular database backups be performed according to an established schedule. Backup files are stored securely to minimize the risk of data loss caused by hardware failure, software malfunction, or other unforeseen circumstances.

The researchers further recommend that periodic system maintenance be conducted to ensure continuous system reliability and performance. This includes updating software components, correcting identified system errors, optimizing database performance, reviewing security settings, and monitoring overall system functionality.

The researchers also recommend that management regularly analyze the production reports and dashboard information generated by the system during production meetings and operational planning. Continuous evaluation of yield percentage, extraction efficiency,

production stability, and production output would support informed decision-making and encourage continuous process improvement.

Finally, the researchers recommended that user feedback should be collected regularly after system implementation. Suggestions and observations from administrators and production personnel should be documented and considered in

future system updates. Continuous improvement based on actual user experience would help ensure that the system remained effective, user-friendly, and responsive to the operational requirements of AlDi Foods Inc.

Here is the past tense version of Section 5.5 Impact of the Study. Since this section discusses the completed study, all present-tense descriptions have been converted to past tense while maintaining academic style.

## 5.5 Impact of the Study

The development of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* provides practical, technological, organizational, and academic contributions to the digitalization of production-monitoring activities within a food-manufacturing environment. The study was developed in response to the problems identified in Chapter 1, particularly the reliance of production personnel on manual methods for recording raw material usage, extraction outputs, packed yields, and other production information. The use of logbooks, paper-based records, spreadsheets, and manual calculations may result in human errors, inconsistent records, repetitive work, delayed consolidation of information, difficulty retrieving historical records, and limited visibility of production performance. These conditions may make it more difficult for supervisors and production managers to obtain organized and timely information for production monitoring and operational decision-making.

In response to these identified problems, the researchers developed a web-based and centralized production-monitoring system that provides a structured approach to recording, processing, storing, retrieving, and presenting production information. The developed system incorporates the major requirements identified during the data-gathering activities described in Chapter 3 and translates these requirements into functional modules presented in Chapter 4. These functions include user authentication, production batch management, raw material recording, extraction-output recording, packed-yield recording, automated yield-percentage calculation, extraction-efficiency monitoring, production-stability classification, dashboard visualization, historical record management, and report generation. Through these functions, the study demonstrates how information technology can be applied to address a specific operational problem while maintaining a practical and cost-effective system scope.

For AlDi Foods Inc., the developed system provides a centralized environment for managing production-monitoring information. Instead of depending entirely on separate physical records or manually maintained spreadsheets, authorized users can enter production information through designated system interfaces and store the resulting records in a centralized database. Raw material quantities, extraction outputs, packed yields, and other relevant production information can be associated with production batches, allowing the organization to maintain a more structured record of production activities. This centralized approach may make production information easier to access, review, and organize, particularly when supervisors or managers need to examine current or previous production records.

The system also provides an impact on the way production information is processed. In the existing manual process, personnel may need to perform calculations repeatedly when determining production indicators. The developed system automates selected calculations after the required information has been entered. Yield percentage and extraction efficiency can therefore be calculated according to the formulas implemented in the system. This reduces the need for repeated manual computation and provides a consistent method for processing the same types of production data. However, the accuracy of the resulting calculations remains dependent on the accuracy of the values entered by authorized users. The system can process the information consistently, but it cannot guarantee that manually entered production measurements are correct.

The system also contributes to improved production visibility by presenting relevant production information through dashboards, tables, graphical representations, and reports. Instead of reviewing individual production records separately, authorized users can view summarized information and calculated indicators through the system. Information regarding raw material utilization, yield percentage, extraction efficiency, and production stability can provide additional perspectives when reviewing production performance. This may help supervisors and production

managers identify production records that require further examination and provide a more organized basis for discussing production conditions.

The real-time capability of the developed system also contributes to its operational value. Within the context of this study, real-time does not refer to continuous automated collection of information from production machines, sensors, or IoT devices. Rather, it refers to the immediate processing and presentation of production information after authorized users manually encode and submit the required data. Once a production record is entered, the system can immediately process the applicable calculations and present the resulting information through the appropriate system interface. This provides more timely access to production information compared with processes that require separate manual calculation and consolidation before the results can be reviewed.

The production-stability classification further contributes to production monitoring by providing an indicator based on predefined thresholds. Production results can be classified as “Stable” or “Not Stable” according to the criteria configured within the system. This feature can help users identify production records that may require additional attention or review. However, the classification should not be interpreted as an automatic determination of the cause of a production problem. A “Not Stable” result indicates that a measured production condition does not meet the defined threshold, while the identification of the actual cause still requires further investigation and professional judgment by appropriate production personnel.

Another impact of the study is the improvement of historical production-record management. Manual records may be difficult to retrieve when information from previous production batches is required. Physical documents may need to be searched individually, while information stored in separate spreadsheets may require additional consolidation. Through the centralized database developed for the system, production information can be stored systematically and retrieved by authorized users. Historical records can therefore serve as a source of information for reviewing previous production activities, examining production trends, preparing reports, and supporting operational discussions.

The centralized storage of information may also contribute to greater consistency in production documentation. The system provides designated fields and modules for different categories of production information, allowing users to follow a more standardized recording process. Data validation mechanisms can also help prevent certain input-related problems, such as incomplete required fields or invalid numerical values. Although these features cannot completely eliminate human errors, they provide additional controls that can support more consistent data entry and processing.

The study may also influence the work of production personnel by providing a structured digital workflow for recording production information. Instead of using different recording methods for different production measurements, personnel can follow the system's defined sequence of production batch creation, raw material recording, extraction-output recording, and packed-yield recording. This organized workflow can make the responsibilities associated with production data entry clearer and provide a consistent process for documenting production activities. Proper training and standard operating procedures remain necessary, however, because the usefulness of the system depends on users entering complete and accurate information.

For supervisors and production managers, the system provides an additional source of information that can support data-driven production monitoring. The availability of calculated production indicators and historical records allows management to review production performance using stored data rather than relying solely on manually compiled records or individual observations. The dashboard and reporting functions can provide summarized information that may assist in identifying changes in production performance and determining which records require further investigation. The system therefore supports decision-making by providing organized information, while the responsibility for interpreting the information and determining appropriate actions remains with management and authorized production personnel.

The study also has an impact on the use of business analytics within the production environment. The system transforms manually encoded production measurements into processed indicators that can be presented through dashboards and reports. Raw material quantities, extraction outputs, and packed yields become structured production information that can be used to calculate yield percentage and extraction efficiency. The resulting indicators provide a basis for monitoring production performance and examining relationships between production inputs and outputs. This demonstrates how business analytics concepts can be incorporated into a focused production-monitoring system without requiring complex analytical infrastructure.

From an Information Technology perspective, the study demonstrates the practical application of concepts in systems analysis and design, database management, web application development, software engineering, data visualization, and business analytics. The requirements identified through interviews, observation, and examination of existing production records in Chapter 3 were translated into system requirements and design components. The development process used a Developmental Research Design and Agile Software Development approach, allowing the researchers to proceed through requirement analysis, system design, development, testing, and prototype deployment. This connection between the research process and the software-development process demonstrates how IT methodologies can be applied to address actual organizational requirements.

The system architecture presented in Chapter 4 also demonstrates the application of structured software design principles. The three-tier architecture separates the presentation layer, application layer, and data layer. The presentation layer provides the user interface through which authorized users interact with the system, the application layer manages authentication, validation, calculations, production processing, stability classification, and other business logic, while the data layer manages the centralized storage of production information. This separation provides an organized foundation for maintaining and potentially extending the system in the future.

The technologies used in the development of the system also contribute to its practical value. The use of HTML, CSS, JavaScript, PHP, MySQL, Chart.js, and XAMPP demonstrates that commonly available web-development technologies can be combined to create a production-monitoring platform. The system does not require expensive industrial automation equipment, robotics, or IoT/IIoT infrastructure to provide its intended production-monitoring functions. This supports the project's objective of developing a practical and cost-conscious digital solution appropriate to the identified needs and resources of AlDi Foods Inc.

The study may also contribute to the gradual digital transformation of AlDi Foods Inc. by demonstrating how an organization can begin digitalizing a specific operational process without attempting to automate its entire production environment. The developed platform focuses on the management of production information and addresses specific issues associated with manual recording and calculation. Through centralized data storage, automated processing, dashboards, historical records, and reporting, the system provides a foundation upon which additional digital capabilities may be developed in the future.

The study may have relevance beyond AlDi Foods Inc., particularly for small and medium-sized food-manufacturing organizations that continue to depend on manual production records. Organizations with limited financial and technological resources may find it difficult to immediately implement highly automated manufacturing systems. The developed project demonstrates an alternative approach in which an organization can begin with a focused web-based information system that addresses specific production-monitoring requirements. However, the system should not be assumed to be directly applicable to all organizations because production processes, formulas, thresholds, workflows, reporting requirements, and user responsibilities may differ. Any application to another organization would require appropriate requirements analysis and system modification.

From an academic perspective, the study contributes to the field of Information Technology by demonstrating how theoretical concepts can be applied to a real organizational setting. The project integrates concepts involving requirements analysis, system architecture, database design, web programming, user authentication, data validation, automated computation, dashboard development, reporting, testing, and system evaluation. It provides an example of

how an IT project can begin with an identified business problem and progress toward the development of a functional software solution.

The study may also serve as a useful reference for future BSIT students and researchers who intend to develop systems related to production monitoring, manufacturing information systems, yield analysis, database applications, dashboards, or business analytics. The development methodology, system architecture, database design, production workflow, interface design, and evaluation procedures documented in the study may provide a foundation for similar projects. Future researchers may adapt these concepts to other organizations or industries while conducting a new requirements analysis to ensure that the resulting system is appropriate for the intended users.

The study's impact should also be considered together with its limitations. The developed system is a prototype specifically designed according to the production-monitoring requirements of AlDi Foods Inc. Therefore, its functions are tailored to the identified workflow and may require modification before being applied to other production environments. The project focuses on production monitoring, yield analytics, extraction-efficiency monitoring, production stability, historical records, dashboards, and reporting. Other business functions such as purchasing, sales, accounting, supplier management, comprehensive inventory management, and enterprise-wide resource planning are outside the defined scope.

The system also depends on manually encoded production information. Although it provides immediate processing after data submission, it does not automatically collect production information from machines, sensors, or other industrial equipment. IoT and IIoT technologies are therefore not part of the current implementation. Similarly, advanced machine-learning or predictive analytics functions are not required for the current system. These technologies may be considered in future research if sufficient historical data, organizational requirements, technical resources, and appropriate evaluation methods become available.

Despite these limitations, the study provides a practical contribution by demonstrating that a focused digital system can address several information-management challenges associated with manual production monitoring. Its impact is not based on replacing the entire production process but on improving how production information is recorded, calculated, stored, retrieved, and presented. The system provides a structured environment in which production personnel can encode information, the application can process selected indicators, and authorized supervisors and managers can review the resulting information.

Overall, the study demonstrates the potential of a centralized web-based production-monitoring platform to support the digitalization of production information at AlDi Foods Inc. The problems identified in Chapter 1 provided the basis for the system requirements, the related studies and literature in Chapter 2 provided a foundation for understanding digital production monitoring and analytics, the methodology in Chapter 3 guided the requirements gathering and system development process, and the implementation presented in Chapter 4 translated these requirements into functional system components. The resulting platform provides a practical connection between information technology and production operations by bringing together production data recording, automated calculations, analytics, historical records, and reporting within a single system.

Ultimately, the impact of the study lies in demonstrating how an appropriately scoped information system can support the gradual digital transformation of a food-manufacturing operation. By centralizing manually encoded production information and providing immediate processing and accessible production indicators, the developed system can support more organized production monitoring and provide management with additional information for operational review. The project also establishes a foundation for future improvements, additional research, and possible integration with more advanced technologies when such developments become appropriate for the needs and resources of AlDi Foods Inc.

## 5.6 Summary

This chapter presents the overall summary of the study entitled “Digitizing Production: Real-Time Yield Analytics

and Extraction Efficiency Monitoring for AlDi Foods Inc." It consolidates the major findings, conclusions, recommendations, and contributions of the study based on the research objectives and the results obtained throughout the development of the proposed system. The summary connects the identified production monitoring problems discussed in Chapter 1, the related studies and literature presented in Chapter 2, the research design and system development methodology described in Chapter 3, and the system implementation, functionalities, testing, and evaluation presented in Chapter 4.

As established in Chapter 1, the study was conducted in response to the production monitoring challenges encountered in AlDi Foods Inc., particularly the use of manual methods for recording raw material usage, extraction output, and packed yield. The existing process relied on logbooks, spreadsheets, paper-based records, and manual calculations, which could contribute to inconsistent records, encoding errors, delayed consolidation of information, repetitive calculations, and difficulty in retrieving historical production data. These conditions limited the availability of organized and timely production information that could be used by production personnel, supervisors, and management for monitoring and decision-making. In response to these identified concerns, the study aimed to develop a web-based production monitoring system capable of centralizing production records and providing automated production analytics.

The review of related studies and literature in Chapter 2 provided the basis for the development of the proposed system. The reviewed literature demonstrated the importance of digital record management, production monitoring, yield measurement, centralized data storage, and analytical information in supporting production-related activities. The studies also showed that digital technologies can be applied to improve the organization and accessibility of operational information. Although some related studies utilized advanced technologies such as sensors, automation, or other digital manufacturing technologies, the present study focused on a more practical and cost-effective approach appropriate to the identified requirements and scope of AlDi Foods Inc. The developed system therefore concentrated on the digital processing of manually encoded production information rather than implementing a complete automated manufacturing or IoT-based environment.

Based on the requirements analysis presented in Chapter 3, the researchers gathered information through interviews, observations, and examination of existing production records, forms, and reports. These activities helped identify the information required for production monitoring, including raw material quantities, extraction outputs, packed yields, production batches, and related records. The gathered requirements served as the foundation for the system design and development. The study applied a Developmental Research Design and an Agile software development approach consisting of requirement analysis, system design, development, testing, and deployment of the prototype. This methodology enabled the researchers to develop the system according to the identified operational requirements and to evaluate its functionality during the development process.

The developed system provides a centralized web-based platform for managing production information. It allows authorized users to create and manage production batches and record the corresponding raw materials, extraction outputs, and packed yields. Through the centralized database, production information can be stored in a structured manner and retrieved when needed. This addresses the concerns identified in Chapter 1 regarding scattered records and difficulty in accessing historical production information. The system also provides user authentication and access control to ensure that system functions and production information are available according to the assigned user roles.

One of the major functionalities of the developed system is the automated computation of production yield percentage. The system processes the encoded production values and automatically calculates the corresponding yield indicator instead of requiring users to perform the calculation manually for every production record. This functionality addresses the repetitive manual computation identified during the requirements analysis and provides a consistent method for processing the recorded production values. However, the reliability of the calculated results remains dependent on the accuracy and completeness of the data entered by authorized users.

The system also provides extraction-efficiency monitoring based on the production information recorded by the users. By processing the relevant raw material and extraction output values, the system provides an indicator that can be used to review extraction performance. This functionality supports the objective of making production performance information more accessible to users. The system, however, does not independently determine the causes of low extraction efficiency because such determination may require additional operational information, equipment inspection, material assessment, or management analysis.

Another major functionality is the production stability classification. The system evaluates recorded production information using predefined thresholds and presents the resulting condition as “Stable” or “Not Stable.” This provides production personnel and management with an additional indicator for monitoring production conditions. The classification is intended as a monitoring aid and should not be interpreted as an automatic diagnosis of production problems. When a production record is classified as “Not Stable,” further investigation by the responsible personnel is still required to determine the possible operational or production-related causes.

The dashboard and analytics functions further support the objective of providing timely production information. The system presents processed production information through summary indicators, charts, graphs, and other analytical displays. This reduces the need for users to manually consolidate information from separate records before reviewing production performance. In the context of this study, real-time analytics refers to the immediate processing and presentation of information after manually encoded production data are submitted to the system. The system does not continuously collect production information through sensors and does not implement IoT, IIoT, robotics, or full industrial automation. This definition is consistent with the scope established by the researchers and ensures that the term “real-time” accurately describes the system's implemented functionality.

The historical records and reporting functions provide additional support for production monitoring and information management. Production records stored in the centralized database can be retrieved for review and analysis. These records can help users examine previous production batches, raw material usage, extraction performance, and packed yield. The availability of organized historical information also provides a basis for identifying production trends and reviewing previous records, subject to the quality and completeness of the information entered into the system. Where applicable, generated reports provide a more organized presentation of accumulated production information for production monitoring and management purposes.

The system architecture and technical implementation discussed in Chapter 4 were designed to support the identified requirements. The three-tier architecture separates the system into the presentation layer, application layer, and data layer. The presentation layer provides the interface through which authorized users interact with the system. The application layer handles authentication, data validation, production processing, automated calculations, stability classification, analytics, and reporting functions. The data layer provides centralized storage for user and production information through the database. This architectural structure provides an organized foundation for the developed web-based application and can support future maintenance and enhancement.

The system was developed using HTML, CSS, JavaScript, PHP, MySQL, Chart.js, and XAMPP, as discussed in Chapter 4. These technologies provided the necessary components for creating the user interface, implementing the application logic, managing the production database, presenting analytical visualizations, and supporting the local development and testing environment. The selected technologies also supported the study's objective of developing a practical and cost-effective production monitoring platform without requiring specialized industrial automation equipment or IoT devices.

The testing and evaluation conducted in Chapter 4 provided a basis for determining whether the developed system performed according to its intended functions and requirements. The evaluation considered software and user-related factors such as functionality, usability, accuracy, efficiency, reliability, and user satisfaction. The results indicated that the system was capable of performing its intended production monitoring functions for its target users within the conditions of the study. The findings support the use of the developed system as a digital alternative for organizing

and processing production monitoring information at AlDi Foods Inc. However, the results should be interpreted within the defined scope of the prototype, the available production data, the intended users, and the testing conditions.

The study's findings also demonstrate the relationship between information technology and business analytics in addressing an operational production problem. Rather than merely storing production records, the system transforms manually encoded data into calculated indicators and visual information that can be reviewed by authorized users. This allows production information to become more useful for monitoring and analysis. The system does not replace the judgment of production personnel or management; instead, it provides organized information that can support their existing production monitoring and decision-making processes.

The recommendations presented in Chapter 5 focus on the continuous improvement and future expansion of the developed system. Possible system improvements include enhanced data validation, additional reporting capabilities, improved dashboard visualizations, expanded batch traceability, stronger security and access-control mechanisms, backup and recovery features, improved user-interface design, and mobile or responsive accessibility. Additional production-management functions may also be considered as future system requirements if the operational needs of AlDi Foods Inc. expand.

Future research may explore the use of larger and longer-term production datasets to identify additional trends and patterns in raw material utilization, extraction performance, packed yield, and production stability. Statistical analysis, predictive analytics, Artificial Intelligence (AI), and Machine Learning (ML) may also be investigated in future studies when sufficient historical data are available. The possible integration of sensors, Internet of Things (IoT), or Industrial Internet of Things (IIoT) technologies may likewise be studied if the organization later requires automated or continuous data collection. These technologies are considered future research directions and are not part of the current implementation of the developed system.

The study also recommends proper implementation procedures to ensure that the system can be used effectively. Authorized users should receive appropriate training and orientation regarding account management, production data entry, record verification, dashboard interpretation, and report generation. Procedures for data validation, correction, database backup, security, and system maintenance should also be established. Since the present system relies on manually encoded information, users should verify the accuracy and completeness of production values before submitting records. Regular feedback and periodic system evaluation may further help identify additional requirements and areas for improvement.

Overall, the study demonstrates that the application of information technology can provide a practical approach to addressing the production monitoring requirements of AlDi Foods Inc. By centralizing production records and providing automated yield calculations, extraction-efficiency monitoring, production stability indicators, dashboard analytics, historical records, and reporting capabilities, the system provides authorized users with more organized and timely production information. These functions directly respond to the production monitoring concerns identified in Chapter 1 while applying the principles and findings supported by the related literature in Chapter 2.

The study further demonstrates how the research and development process presented in Chapter 3 was translated into an implemented system in Chapter 4. The requirements gathered from the intended production environment guided the system design, while the Agile development approach supported the progressive development and testing of the prototype. The resulting system therefore reflects the connection between the identified organizational problem, supporting literature, development methodology, technical implementation, and system evaluation.

In conclusion, *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* provides a web-based production monitoring platform aligned with the identified operational requirements of the organization. The study addresses the need for centralized production information by digitizing manually recorded production data and providing automated calculations and analytical displays. Although the

system does not provide complete automated production monitoring and does not eliminate the need for human judgment or accurate data entry, it establishes a practical foundation for improving the organization, accessibility, and processing of production information.

The developed system also provides a foundation for future digital transformation initiatives within the organization. Its current implementation demonstrates how a cost-effective web-based solution can be applied to production monitoring without requiring advanced industrial automation technologies. Future enhancements may build upon the existing system by introducing additional analytical capabilities, expanded production functions, stronger security, mobile accessibility, integration with other information systems, or automated data-collection technologies. In this way, the study contributes not only to the immediate production monitoring needs of AlDi Foods Inc. but also to future research and development involving digital production information systems for similar manufacturing environments.

## REFERENCES

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